

BIO 203- Methods in Molecular BIOLOGY

HANDOUTS (001- 096) MID + FINAL TERM

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BIO-203 METHODS IN MOLECULAR BIOLOGY

HAND OUTS Module # (001- 096)

MODULE# 001 Polymerase chain Reaction

Molecular technology has become a crucial tool for identifying new genes with importance in medicine, agriculture, animal production and health, environment and the industry related to these areas. Among the applications of molecular techniques is important to highlight the use of the Polymerase Chain Reaction (PCR) in the identification and characterization of viral, bacterial, parasitic and fungal agents. This technique was developed by Kary Mullis in the mid 80's and since then it has been considered as an essential tool in molecular biology which allows amplification of nucleic acid sequences (DNA and RNA) through repetitive cycles in vitro. The mechanisms involved in this methodology are similar to those occurring in vivo during DNA replication. The amplification of specific nucleic acid sequences, even in the presence of millions of other DNA molecules, is achieved by thermostable DNA polymerase enzyme (as the name of this technique suggests: "polymerase chain reaction") and specific primers. Primers are short sequences of DNA or RNA (oligonucleotides) that initiate DNA synthesis.

Reference: Polymerase Chain Reaction Edited by Dr. Patricia Hernandez-Rodriguez, Chapter # 8: Polymerase Chain Reaction: Types, Utilities and Limitations, Page 157-158.

MODULE# 002 Steps of PCR

PCR is a biochemical process capable of amplifying a single DNA molecule into millions of copies in a short time.

A typical PCR consists of:

Initial Denaturation: The reaction temperature is increased to 95 °C and the reaction is incubated for 2–5 min (up to 10 min depending on enzyme characteristics and template complexity) to ensure that all complex, double-stranded DNA (dsDNA) molecules are separated into single strands for amplification.

Cycling:

- 1. **Denaturation:** The reaction temperature is increased to 95 °C, which melts (disrupts the hydrogen bonds between complementary bases) all dsDNA into single-stranded DNA (ssDNA).
  2. **Annealing:** In annealing short DNA molecules called primers bind to flanking regions of the target DNA. The temperature is lowered to approximately 5 °C below the melting temperature (T_m) of the primers (often 45–60 °C) to promote primer binding to the template.
  3. **Extension:** The temperature is increased to 72 °C, which is optimum for DNA polymerase activity to allow the hybridized primers to be extended.DNA polymerase extends the 3′ end of each primer along the template strands.
- **Repeat:** Steps 1–3 repeated ("cycled") 25–35 times to exponentially produce exact copies of the target DNA.
- **Final elongation:** Single step is performed at a temperature of 70-74°C for 5-15 minutes after the last PCR cycle to ensure that any remaining single-stranded DNA is fully extended.
- **Final hold temperature:** This step at 4-15°C for an indefinite time may be employed for short-term storage of the reaction.

Figure: Three steps of PCR─denaturation, annealing, and extension─as shown in the first cycle, and the exponential amplification of target DNA with repeated cycling.

Reference:

https://www.thermofisher.com/pk/en/home/life-science/cloning/cloning-learning-center/invitrogen-school-of-molecular-biology/pcr-education/pcr-reagents-enzymes/pcr-basics.html

https://www.sigmaaldrich.com/PK/en/technical-documents/technical-article/genomics/pcr/polymerase-chain-reaction

MODULE# 003 Ingredients of PCR

- A PCR template for replication **can be of any DNA source, such as genomic DNA (gDNA), complementary DNA (cDNA), and plasmid DNA**. Contamination of template with other materials from the sample source can result in PCR inhibition and in extreme cases, complete failure of the assay.
- DNA polymerases are critical components in PCR, since they synthesize the new complementary strands from the single-stranded DNA templates. All DNA polymerases possess 5′→ 3′ polymerase activity, which is the incorporation of nucleotides to extend primers at their 3′ ends in the 5’ to 3’ direction.
- Oligonucleotide primers are typically 15–25 nucleotides in length with 40–60% GC and a T_m that is similar for each member of the pair. Primer pairs should have a Tm within 5°C of each other. A simple method to assess the T_m of short oligos (<25 bases) is to use the formula: $T_m = 4(G + C) + 2(A + T)$
- Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification. dNTPs consist of four basic nucleotides - dATP, dCTP, dGTP, and dTTP.The concentration of each dNTP (dATP, dCTP, dGTP, dTTP) in the PCR is usually 200 µM.
- Free Mg^{2+} ions are required as a co-factor for DNA polymerase activity; however, Mg^{2+} ions form complexes with dNTPs, primers and DNA templates.
- Buffer of the PCR reaction mixture serves as a chemical environment to maintain an activity and stability of the DNA polymerase.

Figure: Requirements and purpose of amplification cycles (denaturation, annealing and extension) in a polymerase chain reaction (PCR).

References:

https://www.thermofisher.com/pk/en/home/life-science/cloning/cloning-learning-center/invitrogen-school-of-molecular-biology/pcr-education/pcr-reagents-enzymes/pcr-basics.html

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MODULE# 004 Primers for PCR

A primer is a short single-stranded DNA fragment used in certain laboratory techniques, such as the polymerase chain reaction (PCR). In the PCR method, a pair of primers hybridizes with the sample DNA and defines the region that will be amplified, resulting in millions and millions of copies in a very short timeframe. Oligonucleotide primers are typically 15–25 nucleotides in length with 40–60% GC and a T_m that is similar for each member of the pair. A simple method to assess the T_m of short oligos (<25 bases) is to use the formula: $T_m = 4(G + C) + 2(A + T)$**.** The main property of primers is that they must correspond to sequences on the template molecule (must be complementary to template strand). The size of the primer is very important as well. Short primers are mainly used for amplifying a small, simple fragment of DNA. On the other hand, a long primer is used to amplify a eukaryotic genomic DNA sample. The structure of the primer should be relatively simple and contain no internal secondary structure to avoid internal folding. One also needs to avoid primer-primer annealing which creates primer dimers and disrupts the amplification process.

Primers should generally have the following properties:

- Length of 18-24 bases
- 40-60% G/C content
- Start and end with 1-2 G/C pairs
- Melting temperature (Tm) of 50-60°C
- Primer pairs should have a Tm within 5°C of each other
- Primer pairs should not have complementary regions

Figure: Forward and Reverse Primers.

References:

https://www.genome.gov/genetics-glossary/Primer

https://www.addgene.org/protocols/primer-design/

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MODULE# 005 DNA polymerases

The DNA polymerase typically used in PCR is called *Taq* polymerase, after the heat-tolerant bacterium from which it was isolated (***T****hermus* ***aq****uaticus*).The function of Taq polymerase is to replicate the target DNA. The DNA polymerases recognize primers as start tags.The availability of the thermostable DNA polymerase provided the means to automate the reaction. It was originally isolated from Thermophilus aquaticus that thrives in hot springs. Taq DNA polymerase has a temperature optimum of 75–80°C and survives prolonged exposure to temperatures as high as 96°C. So, it can stay active after each of the denaturation steps.

Taq polymerase lacks proofreading activity and unable to correct wrong incorporated nucleotide bases. On average, these errors occur approximately once per 9000 nucleotides.This might be important for some applications, such as cloning, because the product of the polymerase chain reaction may not be a completely precise copy of the original sequence. Another imperfection of Taq polymerase is that it can only efficiently amplify fragments of a few thousand base pairs. Amplification of very long fragments (up to 40 kb) requires special methods.These problems can be solved by using other thermostable polymerases, such as Pfu and Pwo DNA polymerases. These enzymes have proofreading activity.

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MODULE# 006 Standard PCR reaction

Standard PCR reagents include a set of appropriate primers for the desired target gene or DNA segment to be amplified, DNA polymerase, a buffer for the specific DNA polymerase, deoxynucleotides (dNTPs), DNA template, and sterile water. It also includes a negative control, and a positive control. The amplification parameters will vary depending on the primers and the thermal cycler used. It may be necessary to optimize the system for individual primers, template, and thermal cycler. 25-30 cycles of amplification are recommended.

Table: PCR reagents in the order they should be added.

Reagent	**Concentration of stock solutions**	**Volume**	**Final Concentration**
Sterile H2O		*Q.S.* to 50 µl	
PCR Buffer	10X	5 µl	1X
dNTP's	10 mM	1 µl	200 µM
MgCl2	25 mM	3 µl	1.5 mM
Forward Primer	20 µM = 20 pmol/µl	1 µl	20 pmol

Reverse Primer	20 µM = 20 pmol/µl	1 µl	20 pmol
Template DNA	Variable	Variable	~10^5 Molecules
Taq DNA Polymerase	5 Units/µl	0.5 µl	2.5 Units
			50 µl/Reaction

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https://www.sigmaaldrich.com/PK/en/technical-documents/protocol/genomics/pcr/standard-pcr

MODULE# 007 Interpretation of Results

- A primer dimer (PD) is a potential by-product in the polymerase chain reaction (PCR). As its name implies, a PD consists of two primer molecules that have attached (hybridized) to each other because of strings of complementary bases in the primers.
- The term "mispriming" can be used when PCR products are generated through primer annealing to partially complementary sequences. Mispriming is also of great concern in clinical molecular diagnostics, especially when the PCR assay must discriminate between closely related sequences
- Both positive and negative controls are used in PCR experiments. The positive control controls a known sample of DNA it shows that the primers have attached to the DNA strand. The negative control contains all reagents and a DNA template lacking the target sequence, shows if contamination of the PCR experiment with foreign DNA has occurred.
- The presence of blanks (samples that contain no cDNA or RNA) is as important as it is in any PCR experiment, in order to evaluate possible sample contamination. One blank per mastermix reaction (i.e. per primer pair) is recommended.

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MODULE# 008 Types of PCR

In recent years, modifications or variants have been developed from the basic PCR method to improve performance and specificity, and to achieve the amplification of other molecules of interest in research as RNA. Some of these variants are:

1. Multiplex PCR which simultaneously amplified several DNA sequences (usually exonic sequences) in one reaction.
2. Nested PCR increases the specificity of the amplified product for a second PCR with new primers that hybridize within the amplified fragment in the first PCR.
3. Semi quantitative PCR which allows an approximation to the relative amount of nucleic acids present in a sample.
4. Reverse Transcriptase PCR which generates amplification of RNA by synthesis of cDNA (DNA complementary to RNA) that is then amplified by PCR.
5. Real time PCR which performs absolute or relative quantification of nucleic acid copies obtained by PCR.

Reference:

Polymerase Chain Reaction Edited by Dr. Patricia Hernandez-Rodriguez, Chapter # 8: Polymerase Chain Reaction: Types, Utilities and Limitations, Page 157-158.

MODULE# 009 Nested PCR

Nested PCR is a modification of PCR that was designed to improve sensitivity and specificity. Nested PCR involves the use of two primer sets and two successive PCR reactions. The first set of primers is designed to anneal to sequences upstream from the second set of primers and are used in an initial PCR reaction. Amplicon resulting from the first PCR reaction is used as a template for a second set of primers and a second amplification step. Sensitivity and specificity of DNA amplification is significantly enhanced with this technique. The major disadvantage of nested PCR is the high rate of contamination that can occur during the transfer of first-round products to the second tube for the second round of amplification. This can be avoided either by physically separating the first- and second-round amplification mixtures with a layer of wax or oil or by designing closed-tube amplification and detection protocols.

References:

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Nolte, F. S., & Hill, C. E. (2011). Polymerase chain reaction and other nucleic acid amplification technology. *McPherson RA, Pincus MR. Henrys Clinical Diagnosis and Management by Laboratory Methods*, *21*, 1239-1249.

MODULE# 010 Multiplex PCR

The multiplex polymerase chain reaction is a widespread molecular technique utilized for the amplification of multiple targets in a single PCR experiment. In a multiplex assay, more than one target sequence can be amplified using multiple primer pairs in a reaction mixture. This process amplifies DNA in samples using multiple primers and a temperature-mediated DNA polymerase in a thermal cycler. It is preferred that primers have similar annealing temperatures and must be non-complementary to each other to avoid primer-dimers and inefficient reactions. Multiplex PCRs are more complicated to develop and are often less sensitive than single-primer-set PCR reactions, but they allow multiple targets to be detected from a single specimen in one reaction. It consists of multiple sets of primer within a single PCR mixture to produce amplicons of varying sizes that are specific to different DNA sequences. By targeting multiple sequences at once, additional information may be gained from single test run that otherwise would require several times the reagents and more time to perform. This technique requires two or more probes that can be distinguished from each other and detected simultaneously. There is a range of different probe technologies available, all using fluorophores. If amplicon sizes overlap, the different amplicons may be differentiated and visualised using primers that have been dyed with different color fluorescent dyes. Each Multiplex PCR assay must also have a detection step capable of identifying each amplicon. This can be done by gel electrophoresis with visual identification of separate amplicons of different size or by hybridization with specific DNA probes and detection using spectrophotometry, fluorometry, autoradiography, or chemiluminescence. Multiplex PCR has the potential to produce considerable savings of time and effort within the laboratory without compromising test utility. Since its introduction, multiplex PCR has been successfully applied in many areas of nucleic acid diagnostics, including gene deletion analysis, mutation and polymorphism analysis, quantitative analysis, and RNA detection. In the field of infectious diseases, the technique has been shown to be a valuable method for identification of viruses, bacteria, fungi, and/or parasites.

References:

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Wiedbrauk, D. L., & Farkas, D. H. (Eds.). (1995). *Molecular methods for virus detection*. Elsevier.

MODULE# 011 Reverse Transcriptase PCR (RT-PCR)

The discovery of retroviral reverse transcriptase in the early 1970s ultimately made RT-PCR possible. Reverse transcriptase is an RNA-dependent DNA polymerase, catalyzing DNA synthesis using RNA as the template. The end product is known as complementary DNA (cDNA). cDNA is not subject to RNase degradation, making it more stable than RNA. Reverse transcription PCR (RT-PCR) uses mRNA rather than DNA as the starting template. It measures RNA expression levels. First, the enzyme reverse transcriptase uses the mRNA template to produce a complementary single-stranded DNA strand called cDNA in a process known as reverse transcription. Next, DNA polymerase is used to convert the single-stranded cDNA into double-stranded DNA. These DNA molecules can now be used as templates for a PCR reaction as described above. In RT-PCR, the starting RNA is subsequently degraded, dsDNA is produced, and PCR amplification proceeds in the usual manner.

RT-PCR requires an RNA template, enzyme, nucleotides, buffers and thermocyclers to produce RT-PCR products. Kits are available to simplify and streamline the process. RNA extraction kits for both manual and automated RNA purification exist and, when combined with RT-PCR, make RNA analysis in the clinical laboratory virtually as rapid and equally sensitive as PCR-based DNA amplification. RT-PCR is commonly used in the diagnosis and quantification of RNA virus infections (e.g., human immunodeficiency virus and hepatitis C virus) and the analysis of mRNA transcripts such as those produced by translocations commonly associated with non-Hodgkin's lymphomas, leukemias, and sarcomas.

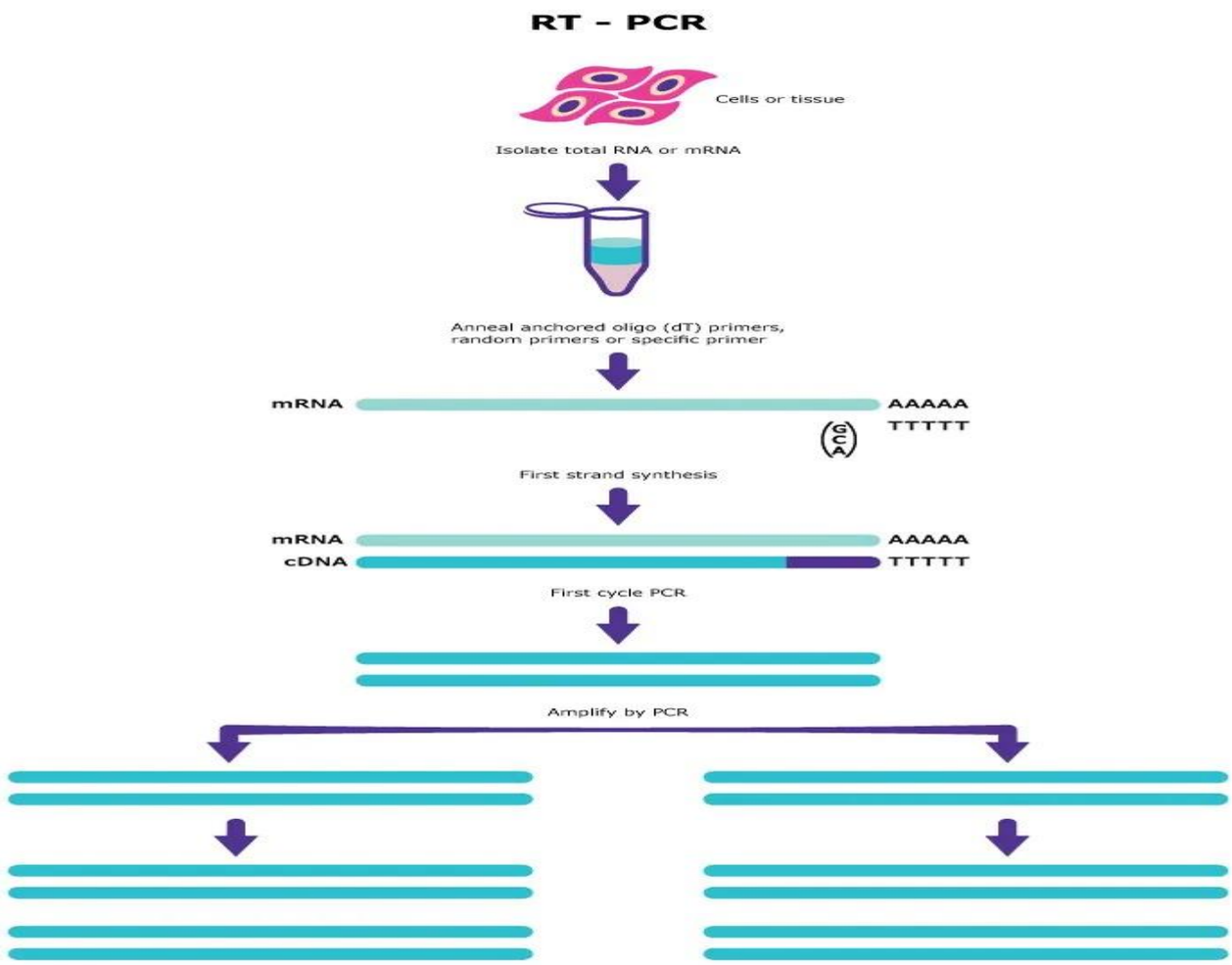


Figure: Steps of RT-PCR

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MODULE# 012 Real-time PCR

Quantitative PCR (qPCR), also called real-time PCR or quantitative real-time PCR, is a PCR-based technique that couples amplification of a target DNA sequence with quantification of the concentration of that DNA species in the reaction. This method enables calculation of the starting template concentration and is therefore a frequently used analytical tool in evaluating DNA copy number, viral load, SNP detection, and allelic discrimination. Conventional PCR is a time-consuming process where the PCR products are analysed through gel electrophoresis. qPCR facilitates the analysis by providing real time detection of products during the exponential phase.

The process is monitored in “real-time”. The reaction is placed in to a real-time PCR machine that watches the reaction occur with a camera or detector. There are a many different techniques that are used to allow the progress of a PCR reaction to be monitored but they all have one thing in common. They all link the amplification of DNA to the generation of fluorescence which can simply be detected with a camera during each PCR cycle. Hence, as the number of gene copies increases during the reaction, so the fluorescence increases.

The principle of real-time PCR depends on the use of fluorescent dye. fluorescent dyes are used to label PCR products during thermal cycling. Real-time PCR instruments measure the accumulation of fluorescent signal during the exponential phase of the reaction for fast, precise quantification of PCR products and objective data analysis. The concentration of the nucleic acid present into the sample is quantified using the fluorescent dye or using the fluorescent labelled oligonucleotides. There are two major detection chemistry options to choose from. Most people use either an intercalating dye (e.g. SYBR® Green) or a hydrolysis probe based detection solution (e.g. TaqMan®). Both technologies are designed to generate fluorescence during the PCR, which allows your real-time PCR machine to monitor the reaction in “real time”. q-PCR is applied in genotyping and quantification of pathogens, microRNA analysis, cancer detection, microbial load testing and GMOs detection.

Figure: Schematic comparing RT-PCR, qPCR and RT-qPCR. (A) RT-PCR workflow. RNA is isolated and cDNA is generated via reverse transcription (RT); PCR is then carried out to amplify areas of interest. (B) qPCR schematic. DNA is isolated and amplified; amplification is quantitated using a probe which fluoresces upon intercalation with double-stranded DNA. (C) RTqPCR procedure. RNA is isolated and cDNA generated before commencing a qPCR procedure.

References:

Dymond, J. S. (2013). Explanatory chapter: quantitative PCR. In *Methods in enzymology* (Vol. 529, pp. 279-289). Academic Press.

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MODULE# 013 Hot Start PCR

Hot start PCR is a novel form of conventional polymerase chain reaction (PCR) that reduces the occurrence of undesired products and formation of primer-dimers due to non-specific DNA amplification at room temperatures. The basic principle of hot-start PCR is the separation of one or more reagents from the reaction mix until the mixture reaches the denaturation temperature upon heating. Hot Start PCR is a technique that inhibits Hot Start Taq

polymerase activity or the incorporation of modified dNTPs during reaction set up until a heat activation step occurs. This technique is based on the principle of initiating the polymerase reaction at or above the primer annealing temperature, thereby preempting possible extensions of the primers nonspecifically annealed at ambient temperature (20°-55°C) at which Taq Pol retains partial activity. For some applications, hot-start PCR may be the only way to amplify successfully the desired product(s) with high yield and specificity.

In practice, a hot-start PCR condition can be attained by adding the polymerase (or any other essential reagent like dNTPs or Mg2+) only when the mixture has reached a temperature higher than Ta and then starting cycles of primer annealing and extension. The use of a wax pellet (e.g., AmpliWax of Perkin-Elmer), which can provide at room temperature a solid-phase barrier between the polymerase and the rest of the reaction components, has made it substantially easier and safer to perform the technique in an uninterrupted PCR operation. Introduction of thermolabile antibodies (For example: TaqStart) which reversibly block the polymerase provides an alternative efficient technique for hot-start PCR: as the thermal cycling starts and temperature of the complete PCR mixture is raised, the inhibitory antibody is selectively denatured (within 30 sec at >85°C), thereby releasing fully active Taq Pol that initiates the polymerization reaction under high stringency priming conditions.

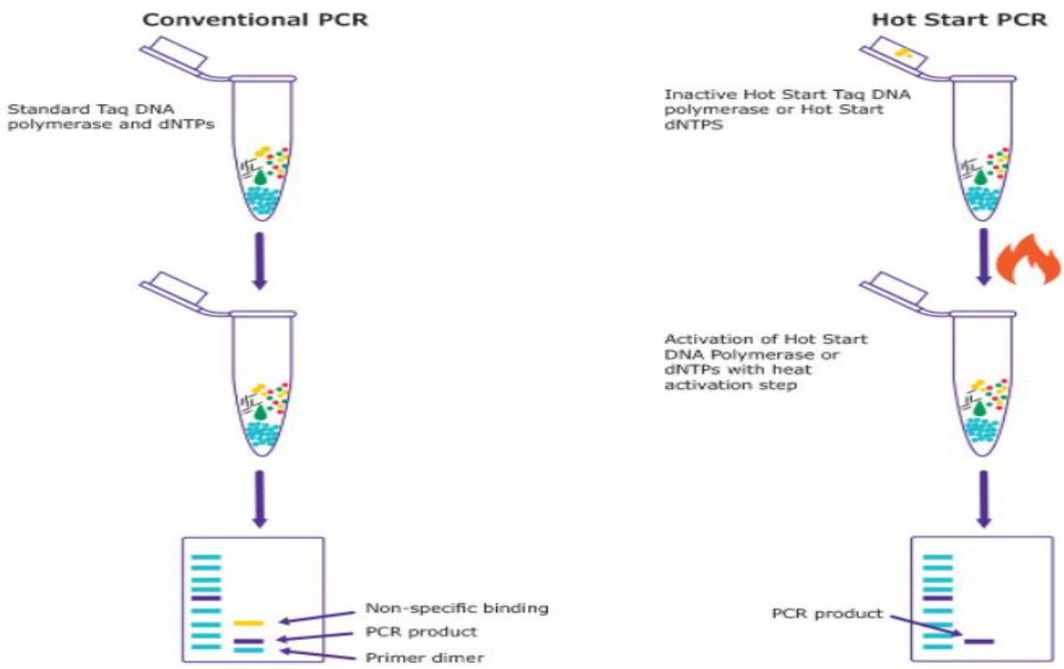


Figure: Comparison of Conventional and Hot Start PCR

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https://microbenotes.com/types-of-pcr/#13-hot-start-pcr

https://www.sigmaaldrich.com/PK/en/technical-documents/technical-article/genomics/pcr/hot-start-pcr

MODULE# 014 Asymmetric PCR

Asymmetric PCR is designed to preferentially amplify one DNA strand. Thus it is useful when amplification of only one of the two complementary strands is needed such as in sequencing, hybridization probing and DNA-aptamer selection. The whole PCR process is similar to regular PCR, except that primers are added at unequal molar ratio to favor the synthesis of the desired DNA strand. Asymmetric PCR differs from regular PCR by the excessive amount

of primers for a chosen strand. As the asymmetric PCR progresses, the lower concentration limiting primer is quantitatively incorporated into newly synthesized double-stranded DNA and used up. Consequently, linear synthesis of the targeted single DNA strand from the excess primer is formed after depletion of the limiting primer. This technique suffers from many limitations such as case-by-case optimization, low reaction efficiency and the frequent generation of non-specific product. Furthermore, limiting the concentration of one primer lowers its melting-temperature below the optimal reaction annealing-temperature which will complicate primer design.

References

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https://microbenotes.com/types-of-pcr/#5-asymmetric-pcr

MODULE# 015 Long PCR

Long range PCR refers to the amplification of DNA targets over 5kb in length which typically cannot be amplified using routine PCR methods or reagents. Traditionally, long range PCR has been performed using a blend of Taq DNA polymerase (for fast elongation) combined with a small amount of proofreading polymerase (for accuracy). The proofreading enzyme repairs DNA mismatches incorporated at the 3' end of the growing strand, allowing Taq polymerase to continue to elongate the DNA much further than it would otherwise, resulting in longer DNA amplification. Long range PCR can also be achieved by using modified high fidelity polymerases with enhanced DNA binding, resulting in highly processive and accurate amplification of long fragments. Whether for cloning, genome mapping or sequencing, PCR Biosystems has a range of solutions enabling the amplification of long fragments with speed and high fidelity. This method allows the amplification of more extended targets within a shorter period and with efficient use of resources.

References:

https://pcrbio.com/row/applications/pcr/long-range-pcr/

https://microbenotes.com/types-of-pcr/#5-asymmetric-pcr

MODULE# 016 Allelic specific PCR

Allele-specific polymerase chain reaction (ASPCR) is an application of the polymerase chain reaction (PCR) that permits the direct detection of any point mutation in human DNA by analyzing the PCR products in an ethidium bromide-stained agarose or polyacrylamide gel. Allele-specific polymerase chain reaction (AS-PCR) is a technique based on allele-specific primers, which can be used to analyze single nucleotide polymorphism. The *allele-specific PCR* is also called the (amplification refractory mutation system) ARMS-PCR corresponding to the use of two different primers for two different alleles. One is the mutant set of primers which are refractory (resistant) to the normal PCR, and the other is the normal set of primers, which are refractory to the mutant PCR reaction. The 3' ends of these primers are modified such that one set of the primer can amplify the normal allele while others amplify the mutant allele. This mismatch allows the primer to amplify a single allele. ASPCR has already been used in DNA-based diagnostic techniques involving the diagnosis of genetic and infectious diseases. It is widely applied in the single gene point mutation detection such as sickle cell anemia and thalassemia. It is also used for the direct determination of ABO blood group genotypes.

References:

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MODULE# 017 Colony PCR

Colony PCR is a method for rapidly screening colonies of yeast or bacteria that have grown up on selective media following a transformation step, to verify that the desired genetic construct is present, or to amplify a portion of the construct. Colony PCR is a method in which, where identification of DNA of interest inserted into the plasmid is obtained by designing the inserted DNA specific primers. The bacterial colony containing the plasmid can directly be amplified using two sets of primers. The first set is of the insert specific primers which amplify the insertion sequence, and the other is of vector-specific flanking primers, which amplifies the plasmid DNA other than the inserted DNA. A bacterial colony is taken and added directly into the master mix containing all other PCR reagents. The main application of colony PCR is in the identification of correct ligation and insertion of inserted DNA into bacteria as well as yeast plasmid. Individual transformants can either be lysed in water with a short heating step or added directly to the PCR reaction and lysed during the initial heating step. This initial heating step causes the release of the plasmid DNA from the cell, so it can serve as template for the amplification reaction. Primers designed to specifically target the insert DNA can be used to determine if the construct contains the DNA fragment of interest. Colony PCR can also be used to determine insert orientation

Procedure of Colony PCR:

- Pick a bacterial colony with an autoclaved toothpick, swirl it into 25 µl of TE autoclaved distilled H2O in a microfuge tube.
- Heat the mix in a boiling water bath (90-100^{0}C) for 2 minutes
- Spin sample for 2 minutes high speed in centrifuge.
- Transfer 20 µl of the supernatant into a new microfuge tube.
- Take 1-2 µl of the supernatant as template in a 25 µl PCR standard PCR reaction.

References:

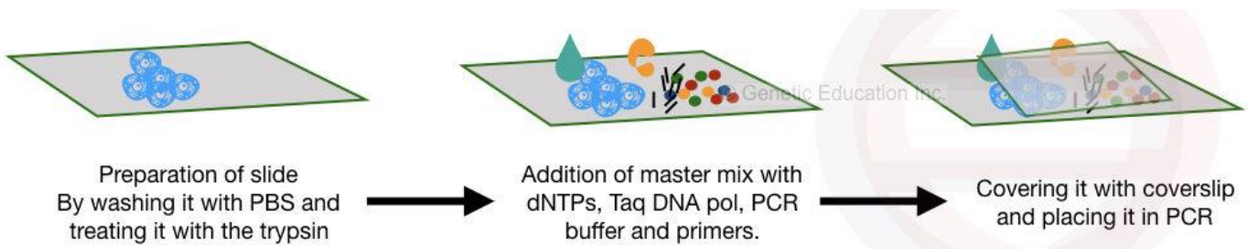
https://international.neb.com/applications/cloning-and-synthetic-biology/dna-analysis/colony-pcr

https://microbenotes.com/types-of-pcr/#5-asymmetric-pcr

MODULE# 018 In situ PCR

- In-Situ Polymerase Chain Reaction (In-situ PCR) is an effective method that detects minute quantities of rare nucleic acid sequences in frozen or paraffin-embedded cells or tissue sections for the compartmentalization of those sequences within the cells.
- This method involves tissue fixing that preserves the cell morphology, which is then followed by the treatment with proteolytic enzymes to provide an entry for the PCR reagents to act on the target DNA.
- In Situ polymerase chain reaction is a histological technique that exploits the advantages of PCR for detection of DNA or mRNA directly in tissue sections. It somehow conjugates together PCR and in situ hybridization that is more traditionally employed for target DNA or mRNA localization in cell organelles, intact cells, or tissue sections.
- In-situ PCR is applicable for the diagnosis of infectious diseases, quantification of DNA, detection of even small amount of DNA and is widely used in the study of organogenesis and embryogenesis

Figure 1: Steps of In situ PCR



References

https://microbenotes.com/types-of-pcr/#5-asymmetric-pcr

https://geneticeducation.co.in/what-is-in-situ-pcr/

https://biotna.net/product/in-situ-pcr/

MODULE# 019 Inverse PCR

Inverse PCR (IPCR) was designed for amplifying anonymous flanking genomic DNA regions. The technique involves the digestion of source DNA, circulation of restriction fragments, and amplification using oligonucleotides that prime the DNA synthesis directed away from the core region of a known sequence, i.e., opposite of the direction of primers used in normal or standard PCR. Prior to the invention of the polymerase chain reaction (PCR), the acquisition of a specific DNA fragment usually entailed the construction and screening of DNA libraries, and the traditional "walking" into flanking DNA fragments involved the successive probing of libraries with clones obtained in the prior screening. These time-consuming procedures could be replaced by IPCR. Because IPCR can be used to efficiently and rapidly amplify regions of unknown sequence flanking any identified segment of cDNA or genomic DNA, researchers do not need to construct and screen DNA libraries to obtain additional unidentified DNA sequence information using this technique. Some recombinant phage or plasmids may be unstable in bacteria and amplified libraries tend to lose them. IPCR eliminates this problem. Inverse PCR is especially useful for the determination of insert locations of various transposons and retroviruses in the host DNA.

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MODULE# 020 AFLP PCR

Amplified fragment length polymorphism (AFLP) is a PCR-based technique that uses selective amplification of a subset of digested DNA fragments to generate and compare unique fingerprints for genomes of interest. It uses selective amplification of a section of digested DNA fragments to generate unique fingerprints for genomes of interest. This technique can quickly generate large numbers of marker fragments for any organism, without prior knowledge of the genomic sequence. AFLP PCR uses restriction enzymes to digest genomic DNA and allows attachment of adaptors to the sticky ends of the fragments. A part of the restriction fragments is then selected to be amplified by using primers that are complementary to the adaptor sequence. The amplified sequences are separated and visualized on denaturing on agarose gel electrophoresis. The power of this method relies mainly in that it does not require prior information regarding the targeted genome, as well as in its high reproducibility and sensitivity for detecting polymorphism at the level of DNA sequence. Widely used for plant and microbial studies, AFLP is employed for a variety of applications, such as: to assess genetic diversity within species or among closely related species, to infer population-level phylogenies and biogeographic patterns, to generate genetic maps and to determine relatedness among cultivars. Variations of standard AFLP methodology have been also developed for targeting additional levels of diversity, like transcriptomic variation and DNA methylation polymorphism.

References

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https://microbenotes.com/types-of-pcr/#5-asymmetric-pcr

MODULE# 021 Assembly PCR

The Assembly PCR method is a two-step PCR approach (assembly and amplification) conventionally used to synthesise genes from multiple overlapping oligonucleotides. In the first step of assembly PCR, multiple oligodeoxynucleotides that contain overlapping regions anneal, and the DNA polymerase extends the primers and fills in the regions between the primers. A range of products of different lengths results from the different possible combinations of annealing that involve less than all the oligodeoxynucleotides. In the second PCR step a pair of

primers is introduced that is specific for the full-length oligodeoxynucleotide and the full-length product is selectively amplified from the mixture. Each cycle of this PCR thus increases the length of various fragments randomly depending on which oligonucleotides find each other. Assembly PCR is used to improve the yield of the desired protein and can also be used to produce large amounts of RNA for structural or biochemical studies**.**

References

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https://microbenotes.com/types-of-pcr/#5-asymmetric-pcr

MODULE# 022 Suicide PCR

Suicide PCR is commonly used studies where avoiding false positives and ensuring the specificity of the amplified fragment is the highest priority. The method requires the use of any primer combination only once in a PCR, which should not have been used in any positive control PCR reaction. These primers should always target a genomic region which has never been amplified before using this particular primer or any other set of primers. This arrangement ensures that no contaminating DNA from previous PCR reactions is present in the lab, which could otherwise generate false positives. Suicide PCR is used in paleogenetics studies which involve an examination of preserved genetic material from the remains of ancient organisms.

References

https://microbenotes.com/types-of-pcr/#5-asymmetric-pcr

MODULE# 023 Methylation Specific PCR

DNA methylation patterns in CpG-rich regions of promoter, CpG islands, are concerned in regulation of gene expression in mammalian cells. Excessive methylation of CpG dinucleotides in promoter represses the gene expression. In cancer, especially, gene silencing is occurred through aberrant methylation in promoter of tumor suppressor genes. Defining DNA methylation patterns in the genome has become essential for understanding diverse biological processes including the regulation of gene expression, imprinted genes, and X chromosome inactivation and how these patterns are deregulated in human diseases. Methylation-specific PCR (MSP) is a method for analysis of DNA methylation patterns in CpG islands. For performing MSP, DNA is modified by and PCR performed with two primer pairs, which are detectable methylated and unmethylated DNA, respectively. This assay requires modification of the genomic DNA by sodium bisulfite and two independent primer sets for PCR amplification, one pair designed to recognize the methylated and the other pair the unmethylated versions of the bisulfite-modified sequence. The amplicons are visualized using ethidium bromide staining following agarose gel electrophoresis. Amplicons of the expected size produced from either primer pair are indicative of the presence of DNA in the original sample with the respective methylation status. Methylation-specific (MS)-PCR is a useful tool for qualitative DNA methylation analysis with multiple advantages, including ease of design and execution, sensitivity in the ability to detect small quantities of methylated DNA, and the ability to rapidly screen a large number of samples without the need for purchase of expensive laboratory equipment.

References

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MODULE# 024 Inter sequence specific PCR (ISSR-PCR)

Inter sequence specific repeat (ISSR)-PCR is a technique that involves the use of microsatellite sequences as primers in a polymerase chain reaction to generate multi-locus markers. It is a simple and quick method that combines most of the advantages of microsatellites (SSRs) and amplified fragment length polymorphism (AFLP) with the universality of random amplified polymorphic DNA (RAPD). ISSR markers are highly polymorphic and are useful in studies on genetic diversity, phylogeny, gene tagging, genome mapping, and evolutionary. Sequences amplified by ISSR-PCR can be used for DNA fingerprinting. Since an ISSR may be a conserved or non-conserved region, this technique is not useful for distinguishing individuals, but rather for phylogeographical analyses or maybe delimiting species; sequence diversity is lower than in SSR-PCR, but still higher than in actual gene sequences. In addition, microsatellite sequencing and ISSR sequencing are mutually assisting, as one produces primers for the other.

Alu PCR is a rapid and easy-to-perform "DNA fingerprinting" technique based on the simultaneous analysis of many genomic loci flanked by Alu repetitive elements, which allows the detection of genetic polymorphisms and mutations in human and primate genomes.

References

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MODULE# O25 Ligation Mediated PCR

Ligation-mediated polymerase chain reaction (LM-PCR) is a genomic analysis technique for determination of (1) primary DNA nucleotide sequences (2) cytosine methylation patterns (3) DNA lesion formation and repair, and (4) in vivo protein–DNA footprints. Ligation-mediated PCR includes three major steps: (i) restriction enzyme digestion of the genomic DNA and ligation of specific adapters; (ii) PCR of restriction fragments using specific primers designed from known sequences and adapter sequences; (iii) analysis of the amplified fragments using gel electrophoresis and sequencing. Owing to the use of specific primers in PCR, ligation-mediated PCR has been one of the preferred approaches to identify flanking sequences with high sensitivity and specificity. However, ligation-mediated PCR requires adapter ligation to generate an artificial priming site at the ends of restriction fragments. There exist common limitations due to the inherent characteristics of this method, such as reduced success rates caused by circularization or multimerization of the restriction fragment, less flexibility, and high expenses, resulting in only one adapter being used with a group of restriction enzymes.

References

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Yu, D., Zhou, T., Sun, X., Sun, Z., Sheng, X., Tan, Y., ... & Yuan, D. (2020). Cyclic Digestion and Ligation-Mediated PCR Used for Flanking Sequence Walking. *Scientific reports*, *10*(1), 1-9.

MODULE# 026 Whole genome amplification PCR

Whole Genome Amplification (WGA) is a method for robust amplification of an entire genome, starting with nanogram quantities of DNA and resulting in microgram quantities of amplified products. WGA has become an invaluable method for preserving limited samples of precious stock material, particularly when using WGA methods that have been developed to amplify material from a single cell. Several methods have been developed for high-fidelity whole genome amplification, including Multiple Displacement Amplification (MDA), Degenerate

Oligonucleotide PCR (DOP-PCR) and Primer Extension Preamplification (PEP). While DOP-PCR and PEP are based on standard PCR techniques, MDA can be achieved with an isothermal reaction setup.

References

https://international.neb.com/applications/dna-amplification-pcr-and-qpcr/whole-genome-amplification

Module# 027 Mini Primer PCR

A PCR method using an engineered polymerase and 10-nucleotide "miniprimers" is termed as Miniprimer PCR. The S-Tbr and *Taq* DNA polymerases are used for PCR (S-Tbr is an engineered polymerase). This method is found to reveal novel 16S rRNA gene sequences that would not have been detected with standard primers. Miniprimer PCR uses a thermostable polymerase enzyme that can extend from short primers (9 or 10 nucleotides). This method allows PCR targeting to smaller primer binding regions, and is used to amplify highly conserved DNA sequences, such as the 16S (or eukaryotic 18S) rRNA gene.

References

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https://microbenotes.com/types-of-pcr/#21-miniprimer-pcr

Module# 028 Advantages and Limitation of PCR

Advantages of PCR	**Disadvantages of PCR**
Highly sensitive	Potentially lower specificity compared to culture and staining
Ability to test for anti-microbial resistance	Need for narrow list of causative agents to use specific primers
Quickly performed in 4-8 hours	Possibility of amplifying normal flora from corneal scrapings
More cost-effective	Becomes less cost-effective when performed with a multi-organism PCR approach
Increased ability to detect less common organisms such as viruses	Supply costs, machinery fees, training expenses

Table: **Advantages and Disadvantages of PCR.**

	Conventional PCR	Real Time PCR
Advantage	• Alternate gold standard for isolation in absence of live agent • Early confirmatory diagnosis • Multiplexing potential • Widely used molecular diagnostic format	• Simultaneous amplification and detection during exponential amplification • Real-time monitoring of amplification as it happens • Quantitative, thus useful for monitoring the microbial load • Lower carry over contamination due to closed tube operation • Increased sensitivity due to fluorescent chemistry • High throughput analysis due to software driven operation
Disadvantage	• Qualitative (yes or no) • End-point detection in plateau phase with non spurious amplification • Post-PCR handling leading to carry over contaminations • Less sensitive, thereby missing borderline cases with low gene copy numbers • Time consuming (3–4 h) • Requirement for thermal cycler and gel documentation system • Prone to inhibitors, nontarget DNA, physicochemical parameter modifications, exposure of reagents to ambient temperature and prolonging times for preparing the reactions	• Expensive detection equipments and consumables • Requirement for fluorescent probe • Restricted to referral laboratory with good financial support • Prone to inhibitors, nontarget DNA, physicochemical parameter modifications, exposure of reagents to ambient temperature and prolonging times for preparing the reactions

Module# 029 Applications of PCR

During the past 40 years, molecular techniques have been under development, however these have had a rapid and tremendous progress in recent year. Among molecular techniques, PCR and its different variations are highlighted as the most commonly used in laboratories and research institutes. Thus, these have contributed to identification and characterization of several organisms and understanding of physiopathology of diverse diseases in human, animal and plant. Also these have provided clues for future research directions in specific topics with impact in public health such as genetics and biochemistry of antimicrobial resistance. The discovery of Polymerase Chain Reaction (PCR) introduced a technological advancement that is relevant for the detection of microorganisms, increasing the sensitivity, precision and accuracy of the diagnosis;

• In virology, the molecular detection and characterization of viral nucleic acids makes a complete viral characterization possible, thereby providing greater knowledge regarding the behavior of the virus and selected infectious processes. In the diagnosis of viruses, this enhances therapeutic treatment as well as clinical and epidemiological virus studies, thereby avoiding unnecessary treatments and reducing overall costs for patients;

• In mycology and parasitology, PCR technology favors the early identification of microorganisms, thereby enhancing epidemiological studies as well as the diagnosis of fungi and parasites, which is essential to the recognition of pathogens. Molecular tests assist in the decision-making process regarding fungal and parasitological diseases by assessing the effects of these microorganisms;

• In microbiology, PCR permits important, immediate observations for diagnostic tests in the detection of microorganisms. Genotyping allows the study of bacteria such as Mycobacterium tuberculosis. This is of tremendous worth for public health, favoring the early recognition and optimized treatment;

• In dentistry, molecular methods enhance knowledge regarding the diagnosis of infectious agents that lead to maxillofacial infections, thereby favoring the assessment of patients at risk for conditions such as caries, periodontal disease, endodontic infections and oral cancer. The publication of the PCR technique was a revolutionary watershed for medicine and science. It has become a standard diagnostic and research tool in dentistry, permitting the early diagnosis of the above-mentioned diseases.

Figure: Application of PCR in different fields of microbiology

References:

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Module# 30 Applications of PCR in Various Scientific Disciplines

PCR can be applied in many fields · In diagnosis therapy · In nucleic acid detection assays · In medical field · In agricultural sciences · In mycology-parasitology · In dentistry · In virological diagnostics · In insert analysis · In molecular systematic evolution · In cancer therapy · In therapy resistant assessment · PCR-as biomarker · In forensic medicine · In virology · In bacteriology · In phytopathology · In PCR-fingerprinting · In the detection of microbiological gene. Some of the key applications of PCR are briefly discussed below.

Genetic Research

PCR has revolutionized scientific research, ever since it was first presented to the outside world in the 1980s. Some of its specialized applications in genetic research are:

- Rapid amplification of tiny fragments of DNA using PCR enabled several techniques such as southern or northern blot hybridization even when the amount of sample material available was very small.
- Study of gene expression patterns is another common application of PCR, where in cells or tissues are analyzed in different stages to check for expression of a specific gene. qPCR can be used here to quantitate the level of gene expression.
- PCR also assists techniques like DNA sequencing using which segments of DNA from an area of interest can be easily amplified to study genetic mutations and their consequences.
- The Human Genome Project used PCR to indicate the presence of a specific genome segment in a particular clone. This enabled mapping of the clones and collating results from several laboratories.
- Advanced variants of the PCR technique have been found to be useful in chromosomal analysis techniques that can help in early detection of genetic birth defects in children.
- PCR augments the traditional method of DNA cloning by amplifying tiny DNA segments for introduction into a vector. By altering the PCR protocol, site-directed or general mutations can be achieved in the DNA fragment of interest.

Medicine

PCR triggered many valuable developments in several medical disciplines.

Microbiology

PCR is a highly valuable technique in microbiology as it allows crucial observations for organism detection. Organisms such as *Mycobacterium tuberculosis* can be studied effectively with the help of genotyping. This allows early identification and treatment and greatly impacts public health monitoring

Virology

In virology, PCR helped detect and characterize the nucleic acids of viruses, which enabled comprehensive viral characterization and a greater understanding of the virus behavior during infection. This understanding immensely helped clinical treatment and enhanced further research on the viruses. For instance, PCR is used to detect HIV infection at an early phase even before the antibodies are formed. This is also useful for screening blood samples collected for donation.

Mycology and Parasitology

PCR technology has also found applications in mycology and parasitology, by enabling early identification of the microorganisms, thus aiding efficient diagnosis and treatment of fungal and parasitic infections.

Dentistry

The PCR technique has become a standard diagnostic and research tool in the field of dentistry. PCR and other molecular biology techniques enable the diagnosis of infectious microbes that cause maxillofacial infections. This helps in the effective management of conditions such as periodontal disease, caries, oral cancer, and endodontic infections.

HIV-AIDS

Quantitative PCR is used to detect HIV-AIDS. Three combinations of drugs used against HIV and regular monitoring of these 3 drugs level in the body can be done by PCR technique.

Cancer-Therapy

Both qualitative and quantitative PCR techniques can be used in detecting diseased strains. p53 gene is helpful in monitoring in cellular divisions, which can be detected by PCRtechnique, regardless it out of control leads to cancerous cell production. PCR technique used to control the cancer diseases by a process called Promoter Methylation(9). Cellular divisions can't be changed but can be switched off. In the DNA region the cell attaches small molecules (-CH3 groups) to the building blocks of DNA. Thus polymerases normally read the genes and produce working copies are no longer allowed to enclose and start to another region. Hence remains silent and no new cell is formed.\

Forensic Science

With the advent of PCR-based DNA fingerprinting, PCR became an invaluable tool in forensic investigations.

- Using DNA fingerprinting, tiny fragments of DNA can be isolated from a crime scene and compared to a huge database of DNA of convicts or criminals. It is also useful in ruling out suspects as part of an investigation.
- DNA fingerprinting is also used in paternity testing, where the DNA from an individual is matched with that of his possible children, siblings, or parents.

Environmental Microbiology

The PCR technique has been successfully used to explore many issues in environmental microbiology. Some of its environmental applications are listed below:

- Sensitive detection of degrading microorganisms in toxic waste and pollutants can be achieved using PCR, which helps efficient biodegradation and bioremediation at the polluted sites.
- A gene probe-based PCR method has been developed by researchers for the detection of indicator bacteria such as coliforms in water supplies, thus supporting measures that enhance water safety.
- PCR is also used to detect and monitor water-borne microbial pathogens, which pose a major public health hazard.

Agricultural Sciences

Identification of multiple infectious diseases is the most useful application of PCR technique. The following are the milestone achievements in agricultural fields. a. Identification of a mutated gene EDA with ectodermal dysplasia in Holstein cattle b. Identification of polymorphism in ABCB1 gene in phenobarbitol responsive-resistant idiopathic epileptica. c. Deletion of Meq gene (which decreases immunosuppresion in chickens). d. MTM/ mutated gene with X-linked Myotubular myopathy in dog species. e. Insertion mutation in ABCD4 with gall bladder mucocele formation in dogs f. Agricultural techniques can also be used for the identification and charcterisation of specific pathogens of animals. g. Bursal diseases virus-Avian samples h. Bovine respiratory syncytial virus i. Actinobacillus pleuropneumoniae-pgs samples j. Canine parvo-virus type-2-faecal samples of dogs k. Feline immunodeficiency virus l. Feline leukemia virus (FeLV).

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Module# 031 PCR-Medicine

Molecular techniques; especially, PCR have a major impact on medicine. The versatility of molecular techniques has allowed advances and changes in all fields of medicine. The following is an overview of the main impacts generated for molecular biology in medical sciences. The Human Genome Project was one of the main drivers that contributed to the development of PCR technology, and it has allowed us to identify many genes related to certain phenotypes, including genetic disorders. PCR helps focus on the actual segment of DNA that is of interest, rather than the whole genome. From a small genetic sample, the genotypes can now be determined, and as a result, many genetic disorders can be detected, diagnosed and monitored. In some cases, gene therapy is available to address these disorders, and PCR is used to monitor the functioning of the relevant genes and gene segments.

Clinical microbiology has been transformed with the use of molecular technology because it has generated a benefit to the patient affected by infectious diseases. Molecular biology has allowed the development of clinical microbiology because it has been possible to identify microorganisms that are difficult to culture, that have many requirements of laboratory or dangerous for laboratory personnel. Pathogenic microorganisms, including some viruses, bacteria, parasites and fungi, cause infectious diseases and can be identified using PCR, aiding efficient diagnosis and treatment. PCR is also used in molecular diagnostics and biochemical analyses. Among other things, these techniques can be used in drug development, especially in measuring the efficacy of drug therapy and research into cancer detection and treatment.

The implementation of molecular tools has allowed a transformation of pathological studies and has changed the clinical practice. This is how the diagnosis and treatment of complex diseases that require a multidisciplinary clinical team currently has a base of molecular biology due to histopathological evaluation of tissues, which is an important part in the morphological assessment, is insufficient by itself. Thus, the ability to define molecular alterations associated with the disease is increasingly required to clarify the diagnosis and therapeutic guidance. At pathological level, molecular biology has allowed the identification of mutations and carriers of diseases as in diabetes, obesity, neurological, muscular, cardiac, metabolic, and congenital diseases and pathologies associated with sensory organs. At the ocular level, implementation of molecular biology has generated enormous advances in knowledge, diagnosis and treatment of ophthalmic diseases.

Molecular tools have also allowed to perform preimplantation genetic diagnosis (PGD) being used for genetic analysis of embryos before transfer into the uterus. It was first developed in England in 1990, as part of the advances in reproductive medicine, genetics and molecular biology. PGD offers couples at risk of having an affected child the opportunity to have normal child by assisted fertilization. The molecular genetic analysis is performed on one or two blastomeres, and only unaffected embryos are transferred into the uterus. It is important to note that in many countries the using of this reproductive procedure has caused controversy. However, this technique provides an opportunity for couples whose children have shown earlier genetic abnormalities. The availability of personal genome testing – determining an individual's unique genetic sequence – has also led to the growth of an industry in personalised medicine. Nutrigenomics is a specific form of consumer genomics, linking genetic information to information about foods that might be better or worse for particular conditions, like inflammatory bowel disease.

Technologies designed to create new products that meet specific human health needs also often rely on PCR as part of the process. For example, some medicinal products are produced by genetic modification (GM) of existing organisms. This includes GM microorganisms to mass produce pharmaceuticals such as insulin and some antibiotics. Sometimes, genetic modification involves transfer of genes from one organism to another – this is called transgenics.

Transplant technologies can also use PCR to improve the accuracy of tissue typing, thereby reducing the likelihood of transplant rejection.

References:

Polymerase Chain Reaction Edited by Dr. Patricia Hernandez-Rodriguez, Chapter # 8: Polymerase Chain Reaction: Types, Utilities and Limitations, Page 163-164.

https://www.sciencelearn.org.nz/resources/2307-using-pcr-in-medicine

Module# 032

PCR-Forensic sciences

In forensic pathology, classic morphology remains as a basic procedure to investigate deaths, but recent advances in molecular biology have provided a very useful tool to research systemic changes involved in the pathophysiological process of death that cannot be detected by morphology. In addition, genetic basis of diseases with sudden death can also be investigated with molecular methods. Practical application of RNA analysis has not been accepted for postmortem research, due to rapid decomposition after death. However, recent studies using variants of conventional PCR (qPCR and RT-PCR) have suggested that relative quantification of RNA transcripts can be applied in molecular pathology to research deaths (molecular autopsy). In a broad sense, forensic molecular pathology involves application of molecular biology in medical science to investigate the genetic basis of pathophysiology of diseases that lead to death. Therefore, molecular tools support and reinforce the morphological and physiological evidence in research of unexplained death. Molecular methods are used in forensic science to establish the filiations of a person (paternity testing) or to obtain evidence from minimal samples of saliva, semen or other tissue debris.

DNA profiling (DNA typing, genetic fingerprinting, DNA testing) is a technique used by forensic scientists to identify someone based on their DNA profile. PCR can be used as a tool in genetic fingerprinting. This technology can identify any one person from millions of others. For example, tiny samples of DNA isolated from a crime scene can be compared with DNA from suspects, or compared with a DNA database. Such procedures can identify or rule out suspects during a police investigation. PCR-based DNA fingerprinting can also be used in parental testing in which an individual is compared with their close relatives and the actual biological father of a child can be confirmed or ruled out. DNA testing can also confirm the biological parents of an adopted child.

In forensic DNA typing, short tandem repeats (STR) or microsatellites are the most frequently genotyped in order to distinguish between individuals, to tie an individual to a crime or to exonerate the innocent. STRs were discovered in the 1980s and since then, they are the “gold standard” in human identification in forensic investigations. They consist of mono-, di-, tri-, tetra-, penta-, and hexa-nucleotide repeat. An individual can be either homozygous (with the same number of repeats) or heterozygous (different number of repeats) in a certainly locus. Tetra-nucleotide repeats are used for genotyping in forensic DNA analysis.

STR profiles obtained from biological samples found at crime scenes are compared with other profiles of known suspects and are identified by police or are included in a national forensic DNA database. Also, the STR profiling is used in paternity/maternity testing, disaster victim identification (DVI), rape perpetrators identification, and kinship testing. Due to the STR profiles, in numerous cases, the persons have been excluded from involvement in crimes and have been exonerate.

A main advantage of the STRs markers consists in the fact that they can test in a rapid, simple, and simultaneously way more than 10 STR loci by multiplexing. Due to this characteristic, it offers an increased degree in the identification of different biological samples.

Other sources of genetic variations that have been demonstrated to present more specialized uses in forensic identification are as follows: autosomal SNPs, the markers on the Y chromosome and mitochondrial DNA (mtDNA).

In forensic medicine, the Y-chromosome has only one useful property: It is present only in males. Thus, in crime cases, the investigators expect to find Y-chromosome at the scene. Also, when talking about male–female ratio in body fluid mixtures, such as rapes, by analyzing the Y-STR component, the investigators can obtain more

information regarding the male component. It is well known that vasectomized or azoospermic rapists do not leave sperm traces, and it is impossible to find spermatozoa on the microscopic examination. In such cases, the Y-STR profiling is very useful, offering information regarding the identity of the rapist.

Mitochondrial DNA (mtDNA) is inherited on the maternal line; thus, all the members of a matrilineal share the same haplotype. Its advantage relies on the fact that it has a number of 200–1700 copies per cell and has an increased probability of survival compared to the nuclear DNA. Therefore, the forensic applications for mtDNA include analysis of biological samples that are old or severely degraded, and analysis of biological samples which contain a low amount of DNA (e.g., hair shafts). MtDNA has been used to identify the Tsar Nicholas II and his brother Georgij Romanov.

Single-nucleotide polymorphisms (SNPs) have a lower heterozygosis when compared to STRs. The advantage of SNP typing consists in the fact that the DNA template size can be as large as 50 bp, compared to the STRs which need the DNA template size of 300 bp, to obtain a good STR profiling.

References:

Polymerase Chain Reaction Edited by Dr. Patricia Hernandez-Rodriguez, Chapter # 8: Polymerase Chain Reaction: Types, Utilities and Limitations, Page 164-165.

https://www.caister.com/highveld/pcr/pcr-forensic-science.html

Module# 033

PCR-Agricultural sciences and environment

Applications of molecular techniques in research in agricultural sciences and environment have been very numerous and varied. It is possible that one of the most important contributions of the applications of some molecular techniques such as PCR has been the identification and characterization of multiple infectious agents that have great impact on human and animal health. Some applications in agricultural science and environment research are described below. Currently, the genome of most domestic animals and major infectious agents that affect animals is known through the use of molecular tools, facilitating the study of mutations associated with disease.

Molecular techniques such as conventional PCR or qPCR have also facilitated research in detection of pathogens in plants, animals, and the environment; understanding of their epidemiology; and, development of new diagnostic tests, treatments or vaccines. Conventional PCR or PCR based methods are being applied to identification and characterization of specific pathogens of animals, e.g., infectious bursal disease virus in avian samples; bovine respiratory syncytial virus; *Actinobacillus pleuropneumoniae* from samples of pigs; canine parvovirus type 2 (CPV 2) in faecal samples of dogs; feline leukemia virus (FeLV) and feline immunodeficiency virus (FIV); among others. Nucleic acid based detection methods are also important to identification of foodborne pathogens, such as *Listeria monocytogenes*; Campylobacter spp., Salmonella enterica, and Escherichia coli O157:H7.

Despite these important applications of molecular methods, one of the purposes with the greatest impact is the detection and characterization of agents with zoonotic potential, such as pandemic (H1N1) influenza; leptospirosis; Canine visceral leishmaniasis etc.

References:

Polymerase Chain Reaction Edited by Dr. Patricia Hernandez-Rodriguez, Chapter # 8: Polymerase Chain Reaction: Types, Utilities and Limitations, Page 165-166.

Module# 034

PCR-Molecular Paleontology

Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains. The field of molecular paleontology has yielded important insights into evolutionary events, the

discovery and characterization of extinct species. By applying molecular analytical techniques to DNA in fossils, one can quantify the level of relatedness between any two organisms for which DNA has been recovered. Advancements in the field of molecular paleontology have allowed scientists to pursue evolutionary questions on a genetic level rather than relying on phenotypic variation alone. Using various biotechnological techniques such as DNA isolation, amplification, and sequencing scientists have been able to gain expanded new insights into the divergence and evolutionary history of countless organisms."

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https://pubmed.ncbi.nlm.nih.gov/11846037/

Module# 035

Types of restriction and modification (R-M) system

At least four different kinds of R-M system are known: type I, type II, type III and type IIs.

Type I

The type I systems were the first to be characterized and a typical example is that from *E. coli* K12. The active enzyme consists of two restriction subunits, two modification (methylation) subunits and one recognition subunit. The methylation and cutting reactions both require ATP and *S*adenosylmethionine as cofactors. The recognition sequences are quite long with no recognizable features such as symmetry. The enzyme also cuts unmodified DNA at some distance from the recognition sequence. However, because the methylation reaction is performed by the same enzyme which mediates cleavage, the target DNA may be modified before it is cut. These features mean that type I systems are of little value for gene manipulation.

Type II

Most of the useful R-M systems are of type II. They have a number of advantages over type I and III systems. First, restriction and modification are mediated by separate enzymes so it is possible to cleave DNA in the absence of modification. Secondly, the restriction activities do not require cofactors such as ATP or *S*-adenosylmethionine, making them easier to use. Most important of all, type II enzymes recognize a defined, usually symmetrical, sequence and cut within it. Many of them also make a staggered break in the DNA and the usefulness of this will become apparent. Although type IIs systems have similar cofactors and macromolecular structure to those of type II systems, the fact that restriction occurs at a distance from the recognition site limits their usefulness.

Type IIS

Type IIS cleaves at least one strand of the DNA duplex outside of the recognition sequence (i.e. cleavage is shifted relative to the recognition sequence). Note that for some enzymes, such as BsmI (recognition sequence: GAATGC), cleavage of the strand written takes place outside of the recognition sequence, whereas cleavage of the complementary strand takes place within the recognition sequence. This is still considered a Type IIS enzyme. However, in most cases both strands are cleaved away from the recognition sequence, which therefore remains intact.

Type III

These systems are composed of two genes (mod and res) encoding protein subunits that function either in DNA recognition and modification (Mod) or restriction (Res). Both subunits are required for restriction, which also has an absolute requirement for ATP hydrolysis. For DNA cleavage, the enzyme must interact with two copies of a non-palindromic recognition sequence and the sites must be in an inverse orientation in the substrate DNA molecule.

References

Roberts, R. J., Belfort, M., Bestor, T., Bhagwat, A. S., Bickle, T. A., Bitinaite, J., & Xu, S. Y. (2003). A nomenclature for restriction enzymes, DNA methyltransferases, homing endonucleases and their genes. *Nucleic acids research*, *31*(7), 1805-1812.

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MODULE# 036

Nomenclature of restriction endonucleases

The discovery of a large number of restriction and modification systems called for a uniform system of nomenclature. Smith and Nathans (1973) introduced a suitable simplified version of this that included the following key features.

i). The species name of the host organism is identified by the first letter of the genus name and the first two letters of the specific epithet to generate a three letter abbreviation. This abbreviation is always written in italics, e.g. *E. coli* = Eco; *H. influenzae*= Hin

ii). Where a particular strain has been the source then this is identified; e.g. *Eco*K

iii). When a particular host strain has several different R-M systems, these are identified by roman numerals. For instance, in case of *H. influenza* different restriction enzymes can be represented as *HindI, HindII, HindIII*

Table : Examples of restriction endonuclease nomenclature

Enzyme s	Enzyme source	Recognition sequence
SmaI	*Serratia marcescens*, 1st enzyme	CCCGGG
HaeIII	*Haemophilus aegyptius*, 3rd enzyme	GGCC
HindII	*H. influenzae*, strain d, 2nd enzyme	GTPyPuAC
HindIII	*H. influenzae*, strain d, 3rd enzyme	AAGCTT
HamHI	*Bacillus amyloliquefaciens*, strain H, 1st enzyme	GGATCC

References

Primrose, S. B., & Twyman, R. (2013). *Principles of gene manipulation and genomics*. Chapter three. John Wiley & Sons.

MODULE# 37

Target sites of restriction endonucleases

Most, but not all, type II restriction endonucleases recognize and cleave DNA within particular sequences of four to eight nucleotides which have a twofold axis of *rotational symmetry*. Such sequences are often referred to as *palindromes* because of their similarity to words that read the same backwards as forwards. For example, the restriction and modification enzymes R. *Eco*RI and M. *Eco*RI recognize the sequence:

5′-GAA │TTC-3′

3′-CTT │AAG-5′

Axis of symmetry

The position at which the restricting enzyme cuts is usually shown by the symbol „/“ and the nucleotides methylated by the modification enzyme are usually marked with an asterisk. For *Eco*RI these would be represented thus:

5′-G/AA*TTC-3′

3′-CTTA*A/G-5′

For convenience it is usual practice to simplify the description of recognition sequences by showing only one strand of DNA, that which runs in the 5′ to 3′ direction. Thus the *Eco*RI recognition sequence would be shown as G/AATTC. From the information shown above we can see that *Eco*RI makes single-stranded breaks four bases apart in the opposite strands of its target sequence so generating fragments with protruding 5′ termini:

5′-G 5′-AATTC-3′

3′-CTTAA-5′ G-5′

These DNA fragments can associate by hydrogen bonding between overlapping 5′ termini, or the fragments can circularize by intramolecular reaction. For this reason, the fragments are said to have sticky or cohesive ends. In principle, DNA fragments from diverse sources can be joined by means of the cohesive ends and, as we shall see later, the nicks in the molecules can be sealed to form an intact artificially recombinant DNA molecule. Not all type II enzymes cleave their target sites like *Eco*RI. Some, such as *Pst*I (CTGCA/G), produce fragments bearing 3′ overhangs, while others, such as *Sma*I (CCC/GGG), and produce blunt or flush ends.

To date, over 10000 microbes from around the world have been screened for restriction enzymes. From these, over 3000 enzymes have been found representing approximately 200 different sequence specificities. Occasionally enzymes with novel DNA sequence specificities are found but most prove to have the same specificity as enzymes already known. Restriction enzymes with the same sequence specificity and cut site are known as isoschizomers. Enzymes that recognize the same sequence but cleave at different points, for example *Sma*I (CCC/GGG) and *Xma*I C/CCGGG), are sometimes known as neoschizomers.

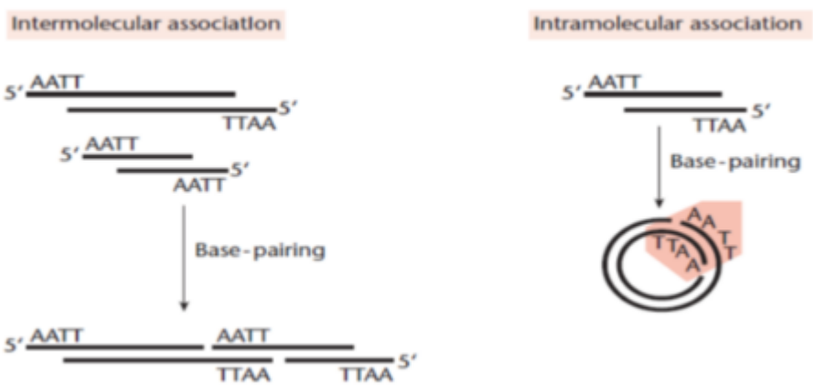


Figure: Cohesive fragments of DNA produced by digestion with EcoR1

References

Primrose, S. B., & Twyman, R. (2013). *Principles of gene manipulation and genomics*. Chapter three. John Wiley & Sons.

MODULE# 038 Number and Size of restriction fragments

The number and size of the fragments generated by a restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut. Assuming a DNA molecule with a 50% G+C content and a random distribution of the four bases, a four base recognition site occurs every 4^4 (256) bp. Similarly, a six-base recognition site occurs every 4^6 (4096) bp and an eight-base recognition sequence every 4^8 (65536) bp. In practice, there is not a random distribution of the four bases and many organisms can be AT- or GC-rich, e.g. the nuclear genome of mammals is 40% G+C and the dinucleotide CG is fivefold less common than statistically expected. Similarly, CCG and CGG are the rarest trinucleotides in most A+T-rich bacterial genomes and CTAG is the rarest tetranucleotide in G+C-rich

bacterial genomes. Thus different restriction endonucleases with six-base recognition sites can produce average fragment sizes significantly different from the expected 4096 bp.

Table: Average fragment size (bp) produced by different enzymes

Enzyme	Target	*Arabidopsis*	*E. coli*	Human
*Apa*I	GGGCCC	25000	15000	2000
*Bam*HI	GGATCC	6000	5000	5000
*Spe*I	ACTAGT	8000	60000	10000

References

Primrose, S. B., & Twyman, R. (2013). *Principles of gene manipulation and genomics*. Chapter three. John Wiley & Sons.

MODULE# 039 Summary of restriction endonucleases

Restriction system allows bacteria to monitor the origin of incoming or foreign DNA continuously. If it recognizes as foreign, then it can by degraded. This cutting ability of the restriction enzymes can be exploited to fragment DNA for gene manipulation purposes. Microbes contain four types of restriction or modification (R-M) system that vary in their specificity. It included type I, II, III or IIs. Type I systems were the first to be characterized from *E. coli* K12. Type I systems are of little value for gene manipulation as they lack specificity. Most of the useful R-M system is Type II that recognizes a defined sequence and cut within it. Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value. Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness. The number and size of fragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut.

References

Primrose, S. B., & Twyman, R. (2013). *Principles of gene manipulation and genomics*. Chapter three. John Wiley & Sons.

Module# 040 Joining DNA Molecules

DNA manipulative enzymes can be grouped into four broad classes, depending on the type of reaction that they catalyze:

- **Nucleases** are enzymes that cut, shorten, or degrade nucleic acid molecules.
- **Ligases** join nucleic acid molecules together.
- **Polymerases** make copies of molecules.
- **Modifying enzymes** remove or add chemical groups.

Nucleases

Nucleases degrade DNA molecules by breaking the phosphodiester bonds that link one nucleotide to the next in a DNA strand. There are two different kinds of nucleases (Figure 1):

Exonucleases: remove nucleotides one at a time from the end of a DNA molecule.

Endonucleases: are able to break internal phosphodiester bonds within a DNA molecule.

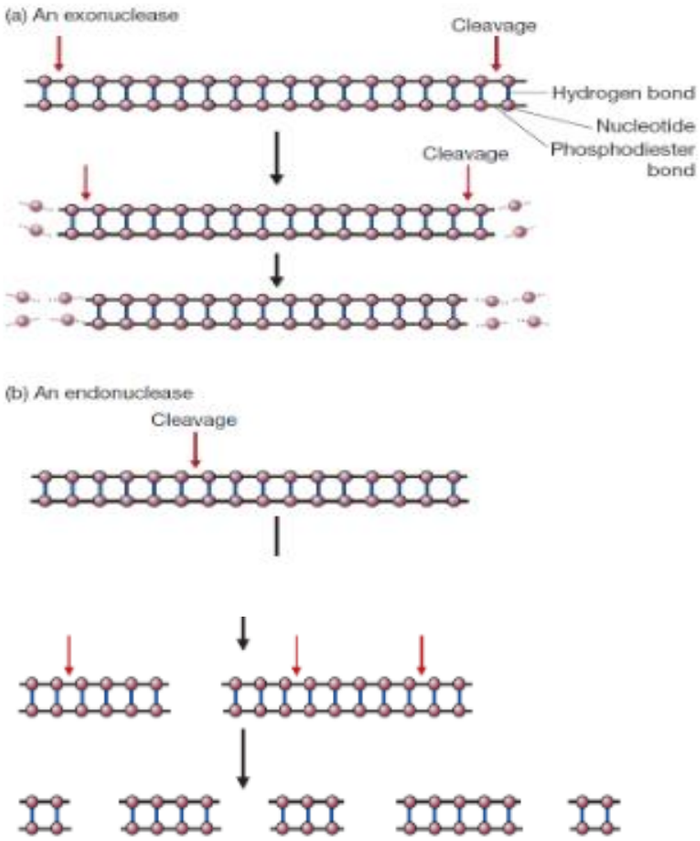


Figure 1: The reactions catalyzed by the two different kinds of nuclease. (a) An exonuclease, which removes nucleotides from the end of a DNA molecule. (b) An endonuclease, which breaks internal phosphodiester bonds.

On the other hand, the special group of enzymes called restriction endonucleases cleave double stranded DNA only at a limited number of specific recognition sites (Figure 2).

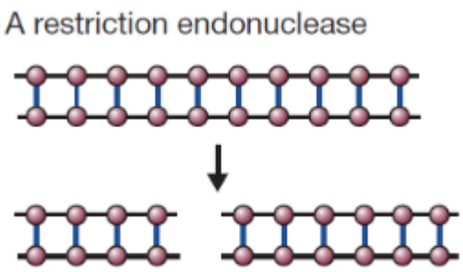


Figure 2: A restriction endonuclease, which cleaves double-stranded DNA, but only at a limited number of sites.

Ligases

In the cell, the function of DNA ligase is to repair single-stranded breaks (“discontinuities”) that arise in double-stranded DNA molecules during, for example, DNA replication. The DNA ligase enzyme family catalyzes the formation of a phosphodiester bond between juxtaposed 5'-phosphate and 3'-hydroxyl termini in duplex DNA. (Figure 3). The role of these enzymes is in the construction of recombinant DNA molecules.

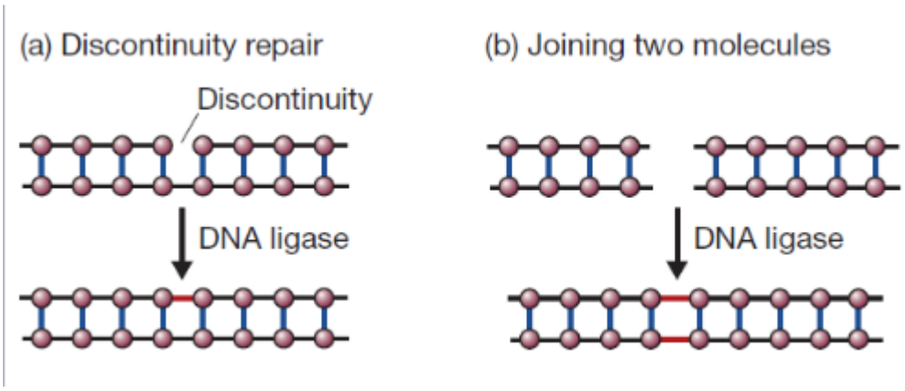


Figure 3: The two reactions catalyzed by DNA ligase. (a) Repair of a discontinuity—a missing phosphodiester bond in one strand of a double stranded molecule. (b) Joining two molecules together.

Polymerases

DNA polymerases are enzymes that synthesize a new strand of DNA complementary to an existing DNA or RNA template (Figure 4.). Most polymerases can function only if the **template** possesses a double-stranded region that acts as a **primer** for initiation of polymerization.

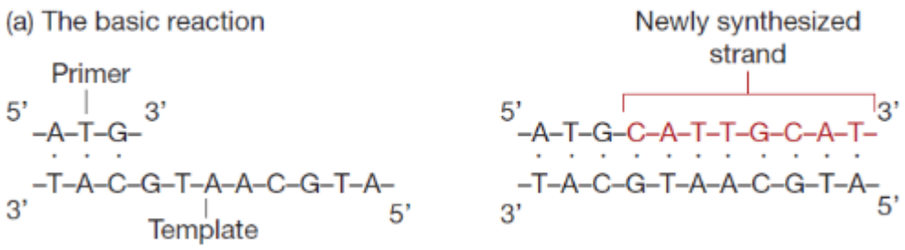


Figure 4: The reactions catalyzed by DNA polymerases. (a) The basic reaction: a new DNA strand is synthesized in the 5′ to 3′ direction.

The *Taq* DNA polymerase used in the polymerase chain reaction (PCR) is the DNA polymerase I enzyme of the bacterium *Thermus aquaticus*. This organism lives in hot springs, and many of its enzymes, including the *Taq* DNA polymerase, are thermostable, meaning that they are resistant to denaturation by heat treatment. This is the special feature of *Taq* DNA polymerase that makes it suitable for PCR, because if it was not thermostable it would be inactivated when the temperature of the reaction is raised to 94°C to denature the DNA.

Reverse Transcriptase

It is a type of DNA polymerase that is important in genetic engineering is **reverse transcriptase**, an enzyme involved in the replication of several kinds of virus. Reverse transcriptase is unique in that it uses as a template not DNA but RNA (Figure 5). The ability of this enzyme to synthesize a DNA strand complementary to an RNA template is central to the technique called complementary DNA (cDNA) cloning.

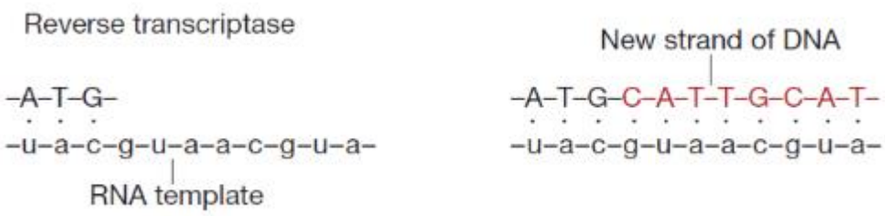


Figure 5: The reactions catalyzed by Reverse transcriptase, which uses a template of RNA.

References

Brown, T. A. (2020). *Gene cloning and DNA analysis: an introduction*. John Wiley & Sons.

Additional Helping Material:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2570007/pdf/12079_2008_Article_26.pdf

https://www.pdffiller.com/jsfiller-desk11/?projectId=466460248#3ec93344c72245649cc4383f05fa0d4b

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Module# 41 Methods of joining DNA Fragments

Mainly three methods are used for joining DNA *in vitro:*

i). Joining covalently annealed cohesive ends by DNA ligase. DNA ligase facilitates the joining of DNA strands together by catalyzing the formation of a phosphodiester bond.

ii). Joining blunt-ended fragments by DNA ligase from phage T4 infected *E. coli*. T4 DNA ligase was purified from *E. coli* cells infected with bacteriophage T4. This enzyme is most efficient when sealing gaps in fragments that are held together by cohesive ends. However, it will also join blunt-ended DNA molecules together under appropriate conditions.

iii). The third utilizes the enzyme terminal deoxynucleotidyltransferase to synthesize homopolymeric 3'single stranded tails at the ends of fragments.

DNA ligase

E. coli and phage T4 encode an enzyme, DNA ligase, which seals single-stranded nicks between adjacent nucleotides in a duplex DNA chain. Although the reactions catalysed by the enzymes of *E. coli* and\ T4-infected *E. coli* are very similar, they differ in their cofactor requirements. The T4 enzyme requires ATP, while the *E. coli* enzyme requires NAD+. In each case the cofactor is split and forms an enzyme– AMP complex. The complex binds to the nick, which must expose a 5′ phosphate and 3′ OH group, and makes a covalent bond in the phosphodiester chain, as shown in the Figure below.

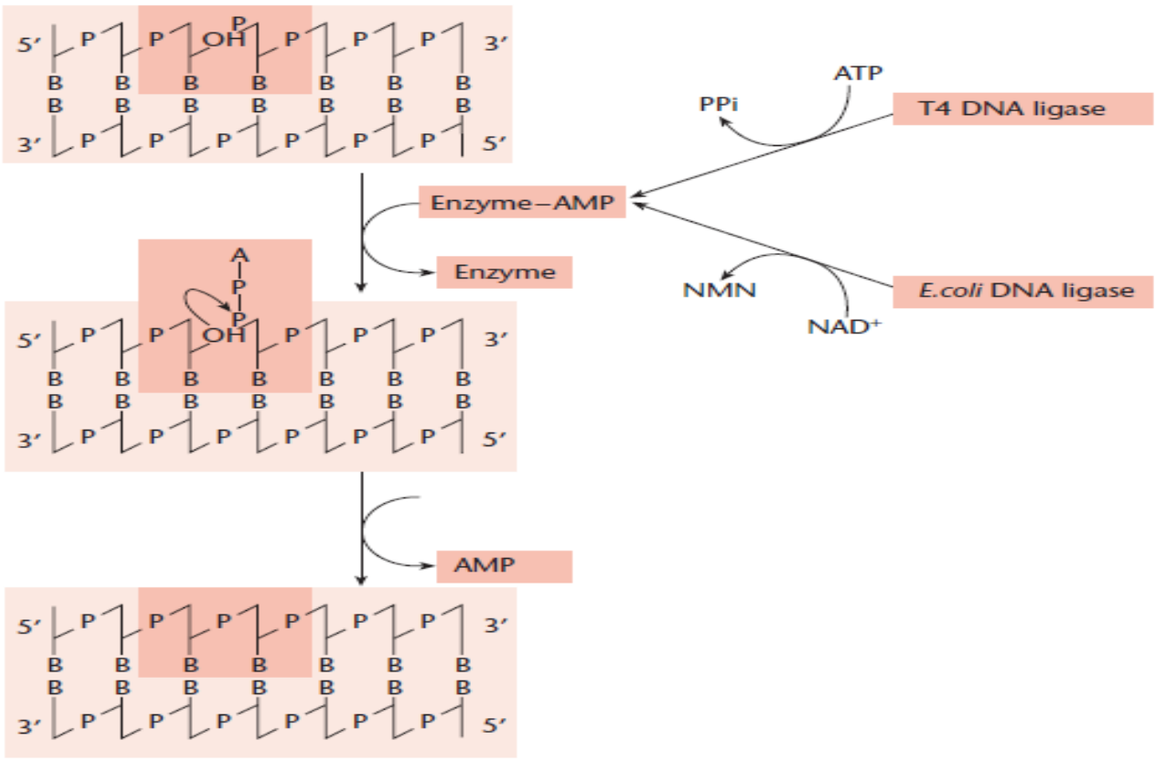


Figure: Action of DNA ligase. An enzyme– AMP complex binds to a nick bearing 3′ OH and 5′ P groups. The AMP reacts with the phosphate group. Attack by the 3′ OH group on this moiety generates a new phosphodiester bond, which seals the nick.

References

Primrose, S. B., Twyman, R. M., & Old, R. W. (2001). *Principles of gene manipulation* (Vol. 6). Chapter No 3. Oxford: Blackwell Science.

Additional Helping Material:

https://www.slideshare.net/shreyamodi100/dna-ligase

https://geneticeducation.co.in/what-is-dna-ligase-and-how-t4-dna-ligase-works/

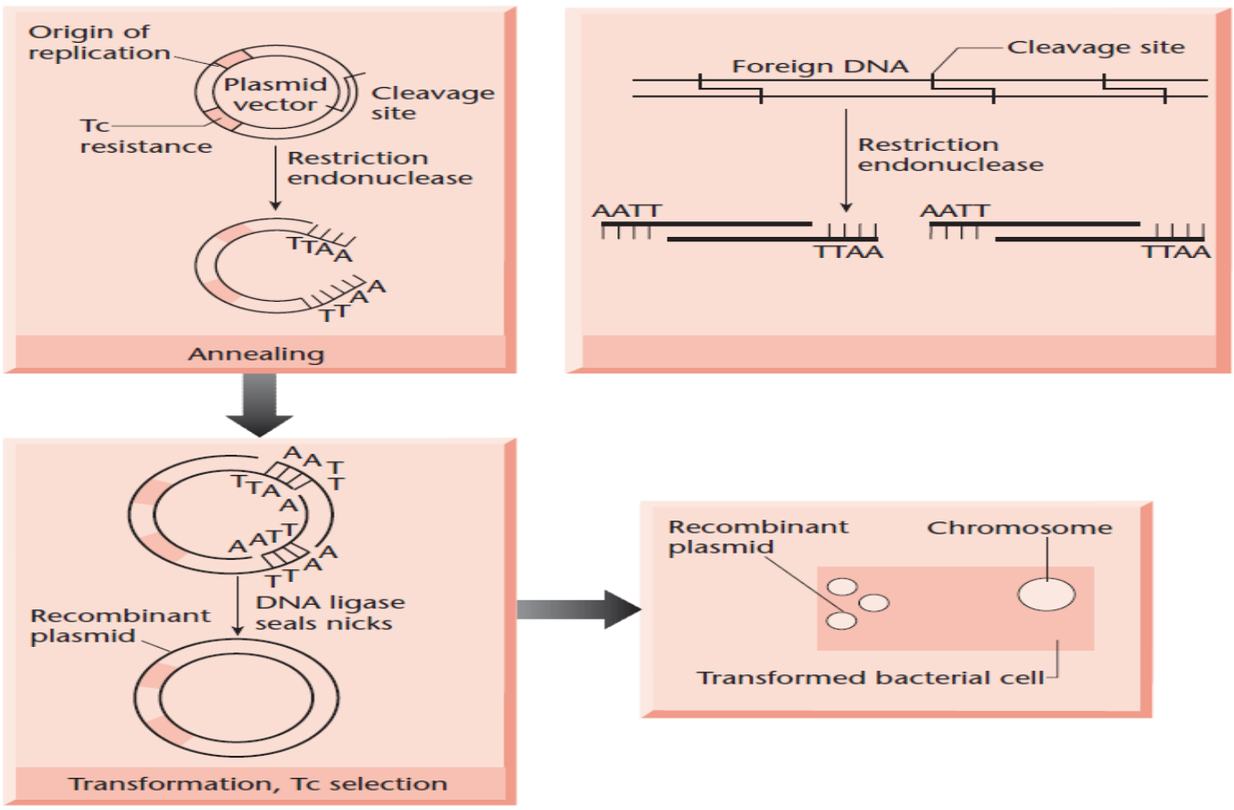
https://www.gbiosciences.com/image/pdfs/protocol/BE-308_protocol.pdf

https://facultystaff.richmond.edu/~lrunyenj/bio554/lectnotes/chapter14.pdf

http://www.afrivip.org/sites/default/files/molecular_gene_cloning_complete.pdf

Module# 42 DNA ligase to create covalent recombinant DNA

When termini created by a restriction endonuclease that creates cohesive ends associate, the joint has nicks a few base pairs apart in opposite strands. DNA ligase can then repair these nicks to form an intact duplex. This reaction, performed *in vitro* with purified DNA ligase, is fundamental to many gene-manipulation procedures, such as that shown in the figure below. optimum temperature for ligation of nicked DNA is 37°C, but at this temperature the hydrogen bonded join between the sticky ends is unstable. *Eco*RI-generated termini associate through only four AT base pairs and these are not sufficient to resist thermal disruption at such a high temperature. The optimum temperature for ligating the cohesive termini is therefore a compromise between the rate of enzyme action and association of the termini, and has been found experimentally to be in the range 4–15°C. The ligation reaction can be performed so as to favour the formation



of recombinants.

Figure: Use of DNA ligase to create a covalent DNA recombinant joined through association of termini generated by *Eco*RI..

References:

Primrose, S. B., Twyman, R. M., & Old, R. W. (2001). *Principles of gene manipulation* (Vol. 6). Chapter No 3. Oxford: Blackwell Science.

Additional Helping Material:

https://www.slideshare.net/SciAmany/dna-ligation

http://vlab.amrita.edu/?sub=3&brch=186&sim=781&cnt=1

https://geneticeducation.co.in/what-is-dna-ligase-and-how-t4-dna-ligase-works/

https://www.gbiosciences.com/image/pdfs/protocol/BE-308_protocol.pdf

https://facultystaff.richmond.edu/~lrunyenj/bio554/lectnotes/chapter14.pdf

http://www.afrivip.org/sites/default/files/molecular_gene_cloning_complete.pdf

Module# 043 Alkaline Phosphatase

Alkaline Phosphatase is suitable for removal of terminal phosphates from deoxyribo-oligonucleotides. Used for the removal of single phosphate groups from 5´-ends of linear vectors to prevent re-circularization during cloning or to dephosphorylate DNA prior to kinase labelling protocols.

The ligation reaction can be performed so as to favour the formation of recombinants. First, the population of recombinants can be increased by performing the reaction at a high DNA concentration; in dilute solutions *circularization* of linear fragments is relatively favoured because of the reduced frequency of intermolecular reactions. Secondly, by treating linearized plasmid vector DNA with alkaline phosphatase to remove 5′-terminal

phosphate groups, both recircularization and plasmid dimer formation is prevented (Figure below). In this case, circularization of the vector can occur only by insertion of non-phosphatase-treated foreign DNA which provides one 5′-terminal phosphate at each join. One nick at each join remains unligated, but, after transformation of host bacteria, cellular repair mechanisms reconstitute the intact duplex.

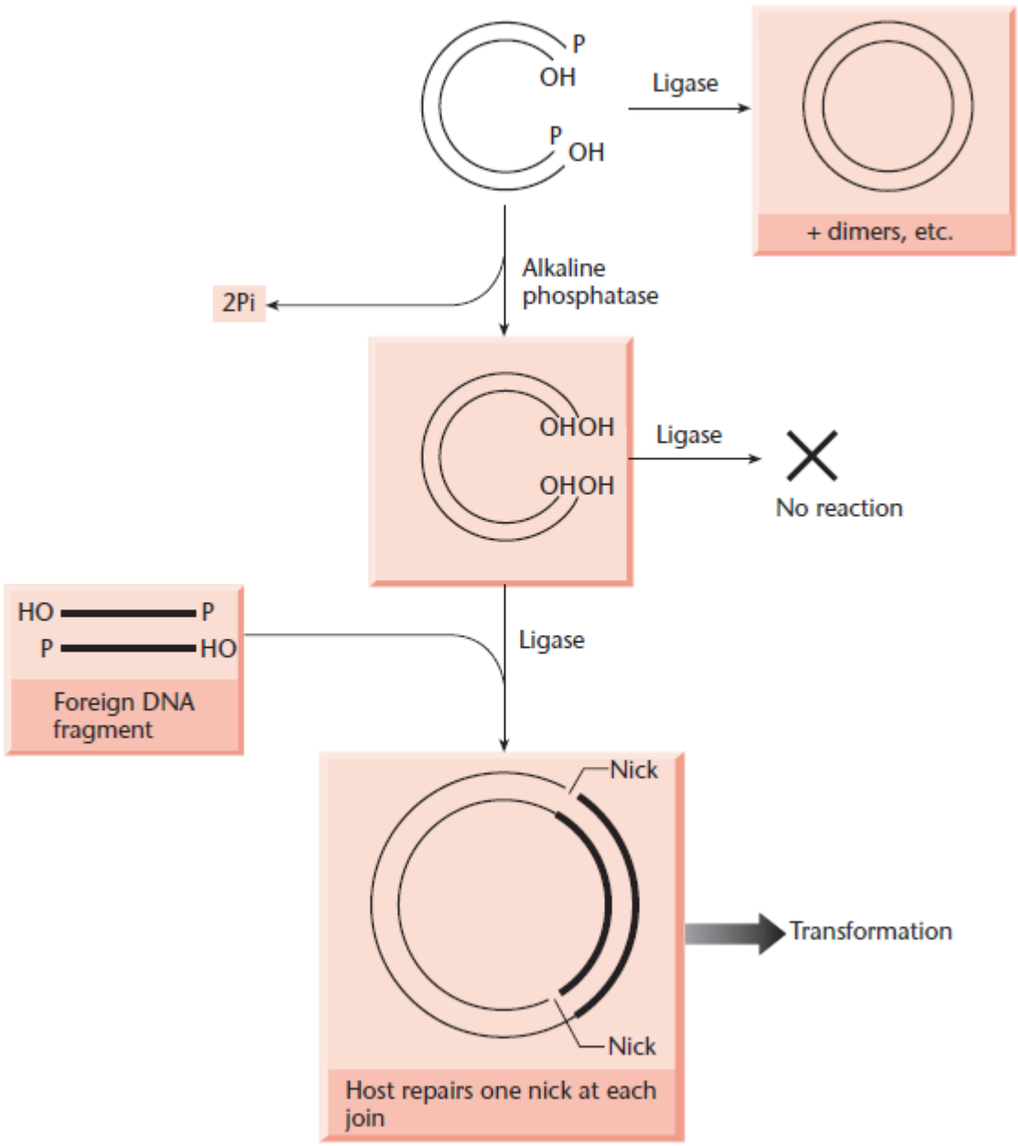


Figure: Application of alkaline phosphatase treatment to prevent recircularization of vector plasmid without insertion of foreign DNA.

References

Primrose, S. B., Twyman, R. M., & Old, R. W. (2001). *Principles of gene manipulation* (Vol. 6). Chapter No 3. Oxford: Blackwell Science.

Additional Helping Material:

https://www.slideshare.net/SmitaShukla1/enzymes-used-in-genetic-engineering

https://www.ncbi.nlm.nih.gov/books/NBK459201/

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4062654/

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2570007/

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Module# 044 Blunt end ligation via Linker Molecules

Joining DNA fragments with cohesive ends by DNA ligase is a relatively efficient process which has been used extensively to create artificial recombinants. A modification of this procedure depends upon the ability of T4 DNA ligase to join blunt-ended DNA molecules (Sgaramella 1972). The *E. coli* DNA ligase will not catalyse blunt ligation except under special reaction conditions of macromolecular crowding (Zimmerman & Pheiffer 1983). Blunt ligation is most usefully applied to joining blunt-ended fragments via *linker* molecules (Linkers are chemically synthesized oligonucleotides that have two blunt ends); in an early example of this, Scheller *et al.* (1977) synthesized self-complementary decameric oligonucleotides, which contain sites for one or more restriction endonucleases. One such molecule is shown in the figure below. The molecule can be ligated to both ends of the foreign DNA to be cloned, and then treated with restriction endonuclease to produce a sticky-ended fragment, which can be incorporated into a vector molecule that has been cut with the same restriction endonuclease. Insertion by means of the linker creates restriction enzyme target sites at each end of the foreign DNA and so enables the foreign DNA to be excised and recovered after cloning and amplification in the host bacterium. one or more restriction endonucleases. One such molecule is shown in the figure below. The molecule can be ligated to both ends of the foreign DNA to be cloned, and then treated with restriction endonuclease to produce a sticky-ended fragment, which can be incorporated into a vector molecule that has been cut with the same restriction endonuclease. Insertion by means of the linker creates restriction enzyme target sites at each end of the foreign DNA and so enables the foreign DNA to be excised and recovered after cloning and amplification in the host bacterium.

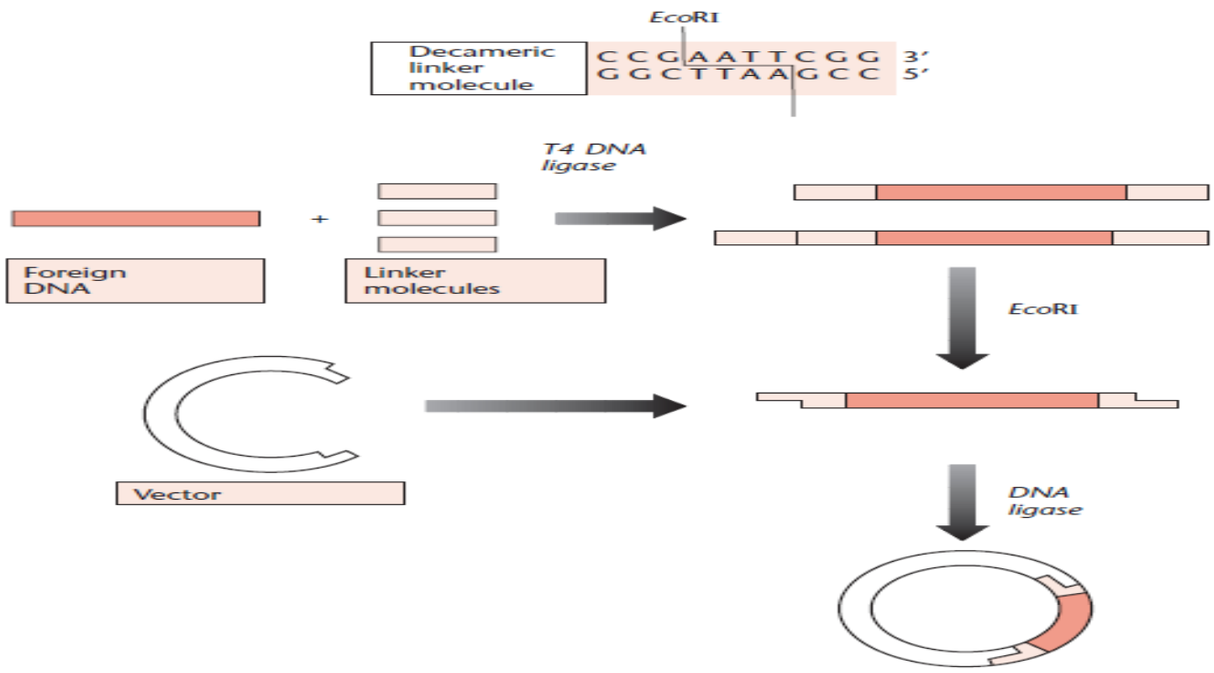


Figure**:** A decameric linker molecule containing an *Eco*RI target site is joined by T4 DNA ligase to both ends of flush ended foreign DNA. Cohesive ends are then generated by *Eco*RI. This DNA can then be incorporated into a vector that has been treated with the same restriction endonuclease.

References

Primrose, S. B., Twyman, R. M., & Old, R. W. (2001). *Principles of gene manipulation* (Vol. 6). Chapter No 3. Oxford: Blackwell Science.

Additional Helping Material:

https://www.slideshare.net/9831302254/linker-adaptor-homopolymeric-tailing-terminal-transferase

https://www.idtdna.com/pages/education/decoded/article/cloning-strategies-part-3-blunt-end-cloning

http://tools.thermofisher.com/content/sfs/brochures/110-dna-insert-ligation-sticky-end-and-blunt-end-into-vector-dna.pdf

http://people.virginia.edu/~owp3a/docs/protocols/Blunt-end%20cloning.pdf

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Module# 045 Adaptors

Blunt ended ligation can be carried out by using another type of synthetic molecules known as adaptors. Adaptors are chemically synthesized oligonucleotides with one blunt and one sticky end. It is double-stranded. It can be single-stranded or double-stranded. During ligation of blunt ended DNA, there can be a case that the restriction enzyme used to generate the cohesive ends in the linker will also cut the foreign DNA at internal sites. In this situation, the foreign DNA will be cloned as two or more sub fragments. One solution to this problem is to choose another restriction enzyme, but there may not be a suitable choice if the foreign DNA is large and has sites for several restriction enzymes. Another solution is to methylate internal restriction sites with the appropriate modification methylase. Alternatively, a general solution to the problem is provided by chemically synthesized adaptor molecules which have a *preformed* cohesive end (Wu *et al.* 1978). Consider a blunt-ended foreign DNA containing an internal *Bam*HI site (Figure below), which is to be cloned in a *Bam*Hl-cut vector. The *Bam* adaptor molecule has one blunt end bearing a 5′ phosphate group and a *Bam* cohesive end which is not phosphorylated. The adaptor can be ligated to the foreign DNA ends. The foreign DNA plus added adaptors is then phosphorylated at the 5′ termini and ligated into the *Bam*HI site of the vector. If the foreign DNA were to be recovered from the recombinant with *Bam*HI, it would be obtained in two fragments. However, the adaptor is designed to contain two other restriction sites (*Sma*I, *Hpa*II), which may enable the foreign DNA to be recovered intact. Note that the only difference between an adaptor and a linker is that the former has cohesive ends and the latter has blunt ends.

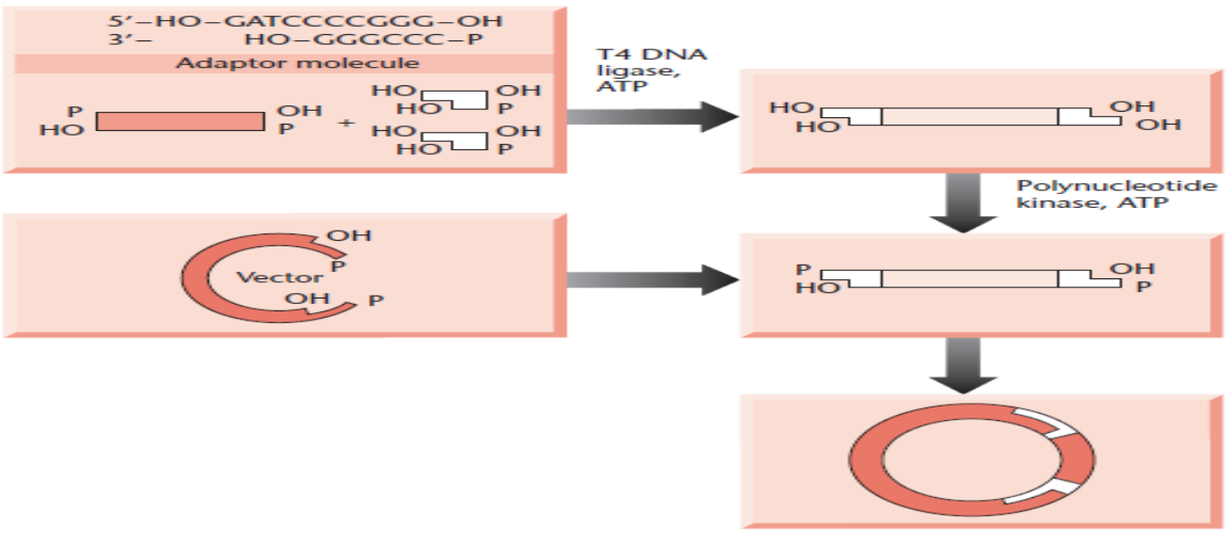


Figure: Use of a *Bam*HI adaptor molecule. A synthetic adaptor molecule is ligated to the foreign DNA. The adaptor is used in the 5′-hydroxyl form to prevent self-polymerization. The foreign DNA plus ligated adaptors is phosphorylated at the 5′-termini and ligated into the vector previously cut with *Bam*HI.

References

Primrose, S. B., Twyman, R. M., & Old, R. W. (2001). *Principles of gene manipulation* (Vol. 6). Chapter No 3. Oxford: Blackwell Science.

Additional Helping Material:

https://www.slideshare.net/9831302254/linker-adaptor-homopolymeric-tailing-terminal-transferase

http://www.himpub.com/documents/Chapter953.pdf

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Module# 046 SOUTHERN BLOTTING

Blotting Techniques

Blotting is the technique in which nucleic acids or proteins are immobilized onto a solid support generally nylon or nitrocellulose membranes. Blotting of nucleic acid is the central technique for hybridization studies. Nucleic acid labeling and hybridization on membranes have formed the basis for a range of experimental techniques involving understanding of gene expression, organization, etc. The main types of blotting techniques are mentioned below.

1. Southern blot is a method used to check for the presence of a DNA sequence in a DNA sample.
2. The northern blot technique is used to study gene expression by detection of RNA (or isolated mRNA) in a sample.
3. The western blot (alternatively, immunoblot) is used to detect specific proteins in a given sample of tissue homogenate or extract.
4. Eastern blotting is a technique to detect protein post translational modification and is an extension of the biochemical technique of western blotting.

Southern Blotting

Southern blot analysis is a laboratory method used to study DNA from different biological samples. The method is named for its creator, British molecular biologist Edwin Southern.

In human tumors, changes in the number of copies or the structure of cancer-causing genes are frequent. Southern blot analysis can be used to investigate whether a gene is amplified, deleted, or structurally rearranged in cancer cells as compared to normal cells. Although this technique is quite labor-intensive, it's particularly useful for detecting large deletions in tumor genomes.

References

http://eagri.org/eagri50/GPBR311/lec25.pdf

https://www.genome.gov/genetics-glossary/Southern-Blot

Additional Helping Material:

https://www.med.upenn.edu/robertsonlab/assets/user-content/documents/Southern%20Blotting%20and%20Hybridization.pdf

https://msu.edu/course/css/451/LabProtocols/09SOUTHERNprocedit.pdf

Module# 047 Southern Blotting procedure

Step I: Restriction digest

- The DNA is extracted from cells, leukocytes and fragmentized by using suitable restriction enzyme. RE cuts the DNA at specific site generating fragments

Step II: Gel electrophoresis

- The resulting fragments are separated according to size by electrophoresis through a standard agarose gel.

Step III: Denaturation

- The gel after electrophoresis is then soaked in alkali (NaOH) or acid (HCl) to denature the double stranded DNA fragments.

Step IV: Blotting

- The separated strands of DNA is then transferred to positively charged membrane nylon membrane (Nitrocellulose paper) by the process of blotting.

Step V: Baking and blocking

- After the DNA of interest bound on the membrane, it is baked on autoclave to fix in the membrane.
- The membrane is then treated with casein or Bovine serum albumin (BSA) which saturates all the binding site of membrane

Step VI: Hybridization with labelled probes

- The DNA bound to membrane is then treated with labelled probe
- The labelled probe contains the complementary sequences to the gene of interest
- The probe bind with complementary DNA on the membrane since all other non-specific binding site on the membrane has been blocked by BSA or casein.

Step VII: Visualization by Autoradiogram

- The membrane bound DNA labelled with probe can be visualized under autoradiogram which give pattern of bands.

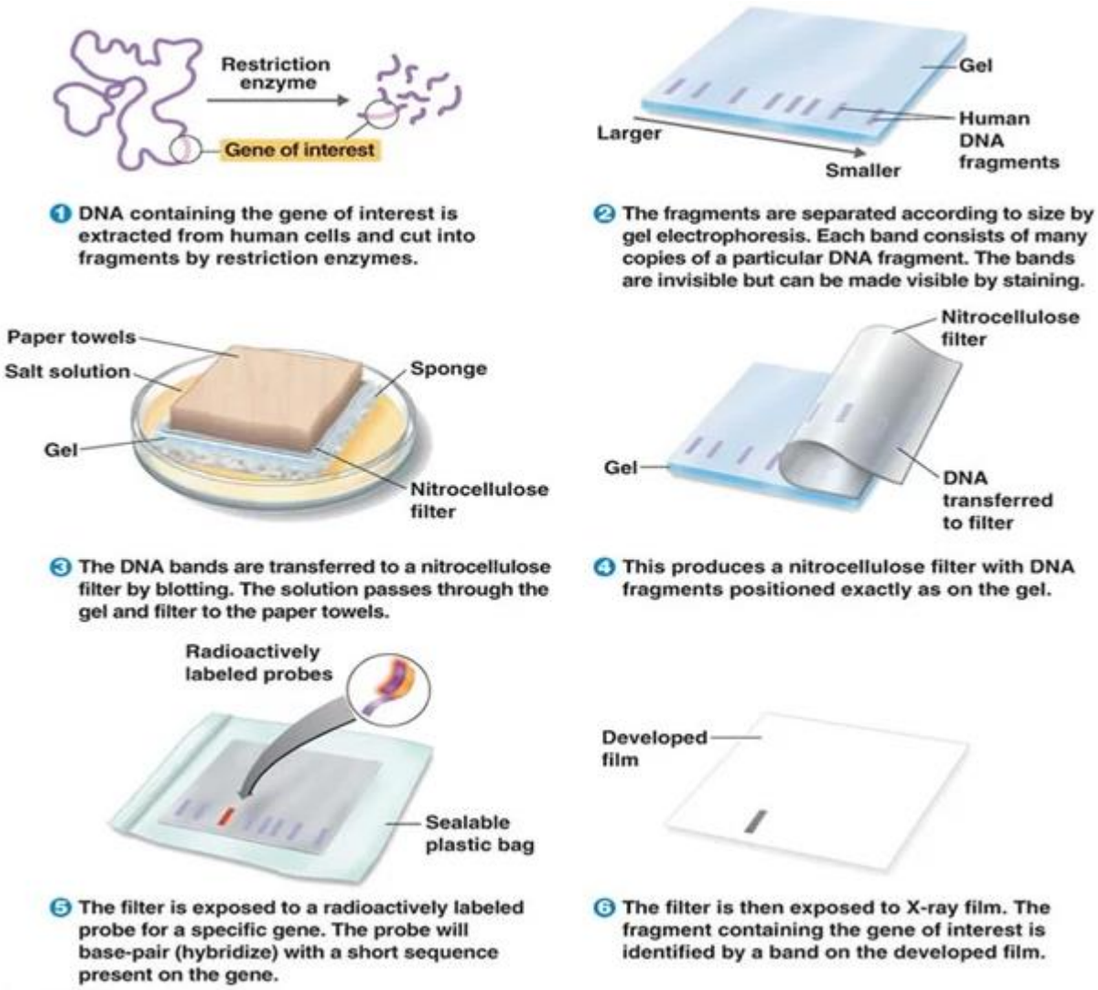


Figure 1: Steps of Southern Blotting

References

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https://www.onlinebiologynotes.com/southern-blotting-principle-procedure-application/

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https://msu.edu/course/css/451/LabProtocols/09SOUTHERNprocedit.pdf

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Module# 048 Southern Blotting Probes

A probe is a fragment of DNA or RNA that is complementary to a sequence of interest. In blotting technique, nucleic acids are fixed to a solid surface and hybridized with a probe. If the target sequence is present in the sample, the probe will hybridize with it and make it detectable. There are two types of probes as radioactive and nonradioactive probes. Therefore, we can tag the probes with a radioactive tag or a fluorescent tag. The key difference between radioactive and nonradioactive probes is that radioactive probes are single-stranded DNA or RNA sequences that are labeled with radioactive isotopes while nonradioactive probes are single-stranded DNA or RNA sequences that are labeled with a chemical tag or a fluorescent tag.

Radioactive Probes

Radioactive probes are the single-stranded DNA or RNA fragments with a radioactive tag. Radioisotopes are used in preparing radioactive probes. Radioisotopes ^{32}P, ^{33}P and ^{35}S are commonly used in the labeling of probes. Moreover, radioisotopes ^{3}H and $^{125}1$ are also used to a lesser extent in the labeling of probes. But they are used for specific applications. Among different radioisotopes, ^{32}P is the most commonly used isotope in labelling radioactive probes.

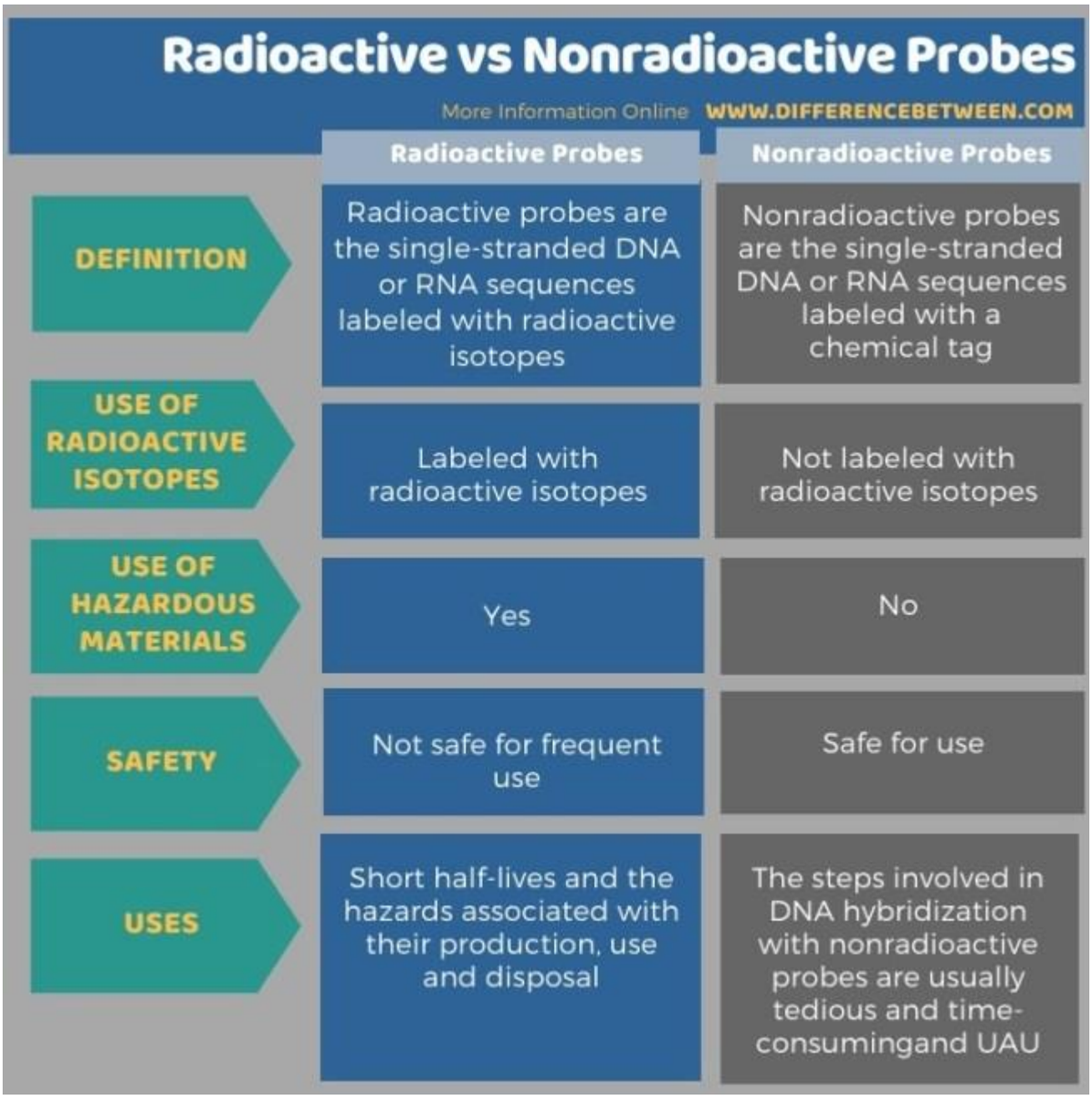
Radioactive probes provide a higher degree of reliability and specificity. Therefore, they provide maximum sensitivity and allow accurate quantification of target sequences. However, there are several disadvantages associated with radioactive probes. They have short half-lives. Moreover, they are hazardous and production, use and disposal are problematic when handling. In addition, radioactive probe preparation is a costly process. Therefore, due to the safety issues and cost, radioactive probes are not used as nonradioactive probes nowadays.

Nonradioactive Probes

Nonradioactive probes are the second type of probes that are chemically labelled. Digoxigenin is a nonradioactive probe, which is an antibody-based marker. Digoxigenin probes are specific and sensitive. Biotin is another label used in nonradioactive probe preparation. Biotin/Streptavidin and Digoxigenin/Antibody-detection systems are the most commonly used nonradioactive probes in hybridization. Furthermore, horseradish peroxidase system is another nonradioactive probe system. Once these nonradioactive probes are hybridized with the target sequences, they can be detected via autoradiography or other imaging techniques.

Nonradioactive probes are used more often in nucleic acid hybridization than radioactive probes. This is because nonradioactive probes are not associated with hazardous materials. Furthermore, nonradioactive detection methods require shorter exposure times to detect the hybridization signal. However, the steps involved in DNA hybridization with nonradioactive probes are usually tedious and time-consuming. Moreover, commercially available solutions are expensive.

Figure 1: Difference between radioactive and non-radioactive probes.



References

https://www.differencebetween.com/difference-between-radioactive-and-nonradioactive-probes/

Additional Reading Material:

https://www.med.upenn.edu/robertsonlab/assets/user-content/documents/Southern%20Blotting%20and%20Hybridization.pdf

https://msu.edu/course/css/451/LabProtocols/09SOUTHERNprocedit.pdf

https://www.csus.edu/indiv/r/rogersa/bio181/probelabeling.pdf

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Take help from additional material/links for topics given in PPT file

Module# 049 Transfer methods in Southern Blotting

Upward capillary transfer

In upward capillary transfer, DNA fragments are carried from the gel in an upward flow of liquid and deposited on the surface of the solid support.

Downward capillary transfer

In downward capillary transfer, DNA fragments are carried in a downward direction in a flow of alkaline buffer and are deposited onto the surface of a charged nylon membrane.

Simultaneous transfer to two membranes

In simultaneous transfer to two membranes, when the target DNA fragments are present in high concentration (e.g., in restriction digests of cloned DNAs), the capillary method can be used to transfer DNA simultaneously and rapidly from a single gel to two nitrocellulose or nylon membranes. The only source of transfer buffer is the liquid trapped in the gel itself, and thus the efficiency of transfer is relatively poor.

Electrophoretic transfer

The electrophoretic transfer method is not practical when nitrocellulose is used as the solid support because the high-ionic-strength buffers required to bind nucleic acids to nitrocellulose conduct electric current very efficiently. In consequence, it is necessary to use large volumes to ensure that the buffering power of the system does not become depleted by electrolysis.

Vacuum transfer

In vacuum transfer, DNA and RNA can be transferred rapidly and quantitatively from gels under vacuum. Several vacuum transfer devices are now commercially available in which the gel is placed in contact with a nitrocellulose or nylon membrane supported on a porous screen over a vacuum chamber.

References

Green, M. R., & Sambrook, J. (2021). Analysis of DNA by Southern blotting. *Cold Spring Harbor Protocols*, *2021*(7), pdb-top100396.

Additional Reading Content

https://www.med.upenn.edu/robertsonlab/assets/user-content/documents/Southern%20Blotting%20and%20Hybridization.pdf

https://msu.edu/course/css/451/LabProtocols/09SOUTHERNprocedit.pdf

Note: To open files, copy and paste link in new browser

Take help from additional material/links for topics given in PPT file

Module# 050 Mutations Identification by Southern Blotting

Southern blot analysis can be used to detect point mutations that disrupt restriction sites, as well as larger structural changes, such as insertions, deletions, and translocations. The presence of a mutation affecting a restriction site causes the pattern of bands to differ from those seen in a normal gene. A change in one nucleotide may alter the nucleotide sequence so that the restriction endonuclease fails to recognize and cleave at that site. Hence, the size & number of the bands generated after digestion of the genomic DNA will be different.

Procedure:

1. Southern blot analysis begins with the isolation of genomic DNA.
2. Digestion of DNA by sequence-specific restriction endonucleases.
3. These genomic fragments are separated based on size using agarose gel electrophoresis.
4. The DNA fragments are denatured in the gel using an alkaline solution, and then they are transferred from the gel to a membrane.
5. Following fixation of the fragments to the membrane, the membrane is hybridized with a labeled probe complementary to the sequence of interest.
6. Detection of the bands complementary to the probe (e.g. by autoradiography).

References

Merker, J. D., & Arber, D. A. (2010). Molecular diagnostics of hematopoietic disorders. In *Molecular Diagnostics* (pp. 243-260). Academic Press.

Module# 051 Northern Blotting

Northern Blotting is a technique used for the study of gene expression. It is done by detection of particular RNA (or isolated mRNA). mRNA is generally represented as 5% of the overall RNA sequence. This method reveals the identity, number, activity, and size of the particular gene.

Procedure

1. The tissue or culture sample collected is first homogenized. The samples may be representative of different types of culture for comparison or it can be for the study of different stages of growth inside the culture.
2. The RNA sequence is separated in the electrophoresis unit an agarose gel is used for the purpose of the nucleic acid separation.
3. Now the separated RNA sequence is transferred to the nylon membrane. This is done by two mechanisms capillary action and the ionic interaction.
4. The transfer operation is done by keeping the gel in the following order. First, the agarose gel is placed on the bottom of the stack, followed by the blotting membrane. On top of these paper towels a mild weight (glass plate) is placed. The entire setup is kept in a beaker containing transfer buffer.
5. RNA transferred to the nylon membrane is then fixed using UV radiation.
6. The fixed nylon membrane is then mixed with probes. The probes are specifically designed for the gene of interest, so that they will hybridize with RNA sequences on the blot corresponding to the sequence of interest.
7. The blot membrane is washed to remove unwanted probe
8. Labeled probe is detected by chemiluminescence or autoradiography. The result will be dark bands in x ray film.

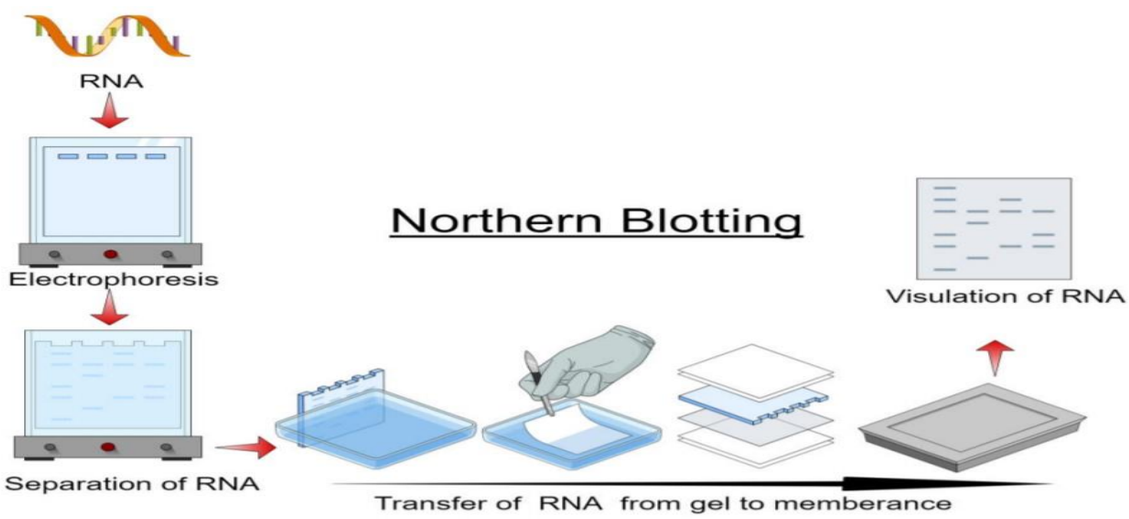


Figure: Procedure of Northern Blotting

References

https://www.thesciencenotes.com/northern-blotting-principle-procedure-and-applications/

https://www.mybiosource.com/learn/testing-procedures/northern-blotting/

Module# 052 Applications of Northern Blotting:

- Standard for studying and inspecting gene manifestation design between tissues, organs, developmental stages, pathogen infection, and over the course of treatment.
- Diagnosis of several diseases (Crohn's diseases) including viral infection.
- Northern blot analysis can be used to investigate whether a gene is overexpressed or underexpressed in cancer cells as compared to normal cells.
- Exhibit overexpression of oncogenes and down regulation of tumor suppressor genes in cancerous cells.
- Study of RNA degradation and splicing.
- Detect specific mRNA molecular weights and contents in a sample.
- Identify mRNA produced by the transgene to protect recombinants

References

https://www.thesciencenotes.com/northern-blotting-principle-procedure-and-applications/

https://www.genome.gov/genetics-glossary/Northern-Blot

Module# 053 Western Blotting

Western blotting is a widely used technique in molecular biology and immuno-genetics for the detection and analyses of proteins, particularly proteins that are of low abundance. It is used to detect the presence of a specific protein in a complex mixture extracted from cells. It is used to detect Size and amount estimate of proteins in the mixture. Also, detects condensed isoforms of proteins as well as tagged proteins. This method is also called "immune blotting" because of its nature to use an antibody for specifically identifying its antigen and also protein blotting. A qualitative and semi quantitative data can be produced using western blotting for the desired protein of interest. It is most widely used as a confirmatory test for the diagnosis of HIV, where this procedure is used to determine whether the patient has antibodies that react with one or more viral proteins or not.

As shown in the figure below, the technique consists of three major processes:

1. Separation of proteins by size (Electrophoresis).
2. Transfer to a solid support (Blotting)
3. Marking target protein using a proper primary and secondary antibody to visualize (Detection).

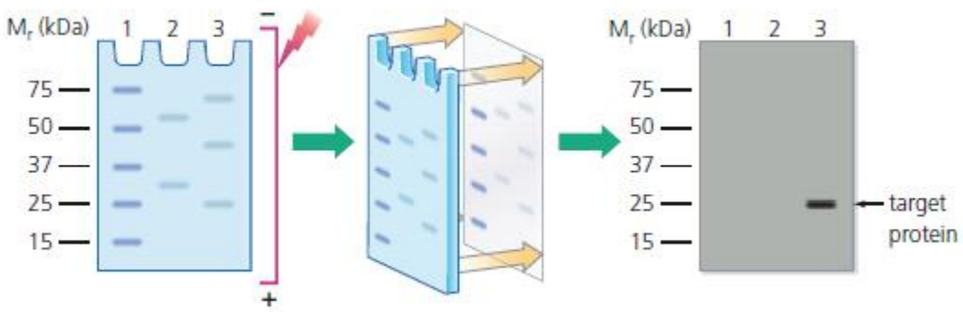


Figure: Separation of protein mixtures by electrophoresis; transfer to a blotting membrane; and detection of target protein, which only becomes visible in the final stage as a band similar to that shown in lane 3. Lane 1: Prestained molecular weight standards. Lanes 2 & 3: Protein mixtures

References

https://www.thesciencenotes.com/western-blotting-principle-procedure-and-applications/

https://www.bio-rad-antibodies.com/western-blotting-immunoblotting-introduction.html

Module# 054 Western Blotting Procedure

Sample Preparation

- Use different samples to extract protein (e.g. tissues or cells)
- Use homogenizer or sonication to breakdown samples.
- Prevent sample digestion at cold temperature through protease and phosphates.
- Finally, observe concentration of proteins and use spectrophotometer for protein concentration.

Gel Electrophoresis

- Proteins are separated on the basis of their size, shape and charge.
- In SDS gel electrophoresis, protein samples are separated according to their molecular weights.
- Load protein samples to polyacrylamide gel (Higher percentage of gels are used for low molecular weight problems and vice versa).

Protein Transfer

- A solid support membrane is placed where we transfer the separated protein for antibody detection.
- The method is called electro blotting where an electric field is directed perpendicular to the surface of the gel maintain their relative position.
- A transfer sandwich is created made up of: a fiber pad (sponge), filter papers, blotting membrane, gel, filter papers and finally a fiber pad.

Protein staining

- The gels must be stained as proteins are not directly visible in the gel.
- Dyes like coomassie blue, silver stain or deep purples are used.
- The gel is imaged with suitable instrument and a permanent record can be made after staining.

Blocking

- Prevents non-specific binding of antibodies by blocking unoccupied sites of membrane with inert protein or non-ionic detergent.
- Bovine Serum Albumin and non-fat dry milk are commonly used typical blockers.
- Blocking agents should possess greater connection towards membrane than the antibodies.

Antibody probing

- Incubate the blot with one or more antibodies
- Primary antibodies are specific depending on the antigens to be detected.
- The secondary antibody (monoclonal or polyclonal) is linked to an enzyme that is used to indicate the location of the protein.

Washing

- Removes unbounded antibodies from the membrane by washing with a suitable buffer.

Protein Detection

- Alkaline phosphates (AP) and horse radish peroxidase (HRP) are widely used which are attached with the secondary antibodies.
- Four types of detection methods: Chromogenic detection, Chemiluminescence detection, Fluorescent detection and radioactive detection.

Analysis and Imaging

- Detection of signals using X-ray film, scanners or CCD.
- Benchmarking with marker protein to estimate the molecular weight of the protein.
- Verification can be done through qualitative and quantitative analysis to show the presence and absence of specific proteins of interests.

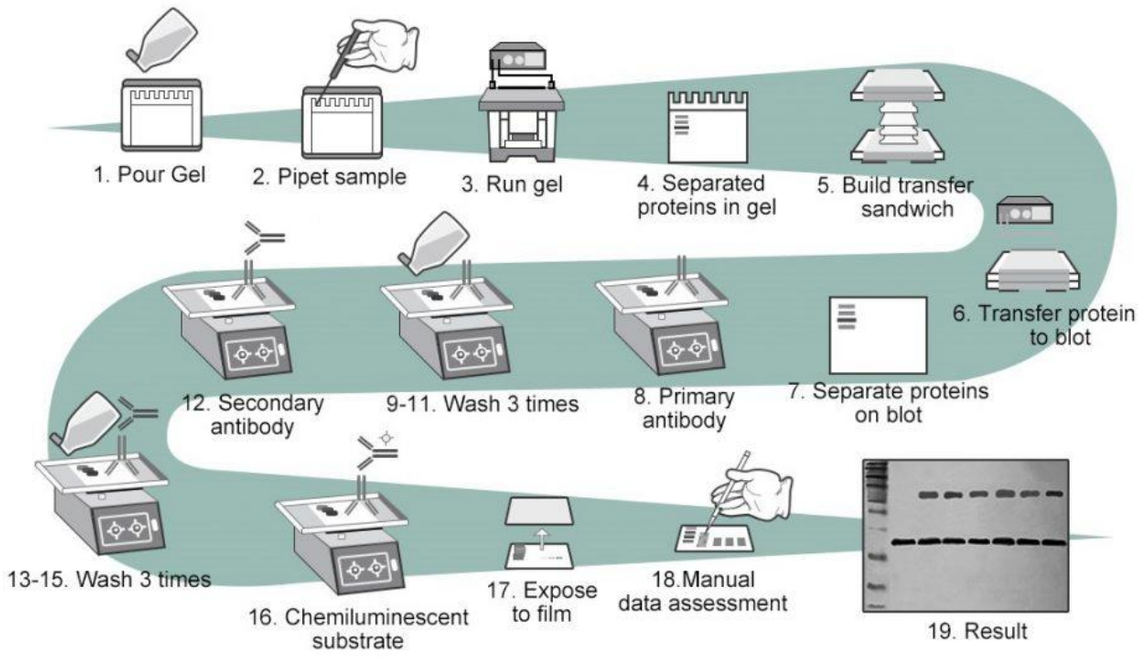


Figure**:** Western Blotting Procedure

References

https://www.thesciencenotes.com/western-blotting-principle-procedure-and-applications/

Module# 055 Applications of western blotting

1. Identification of a specific protein in a complex mixture of proteins and estimation of the size of the protein as well as the amount of protein present in the mixture.
2. Detect condensed isoforms of proteins as well as tagged proteins.
3. It is most widely used as a confirmatory test for the diagnosis of HIV, where this procedure is used to determine whether the patient has antibodies that react with one or more viral proteins or not.
4. Also, used in the diagnosis of Lyme, HIV infection, bovine spongiform encephalopathy, Hepatitis B Virus and Creutzfeldt-Jacob disease.
5. Demonstration of specific antibodies in the serum for diagnosis of neurocysticercosis and tubercular meningitis.
6. It detects post translation modifications, protein-protein and protein-DNA interactions.
7. It is well-suited for evaluating levels of protein expression in cells, and for monitoring fractions during protein purification.
8. Western blots provide confirmation of results both in research and diagnostic testing.

Advantages

- Effective early diagnostic tool.
- Detect minimal immunogenic response form virus or bacteria.
- Requires fewer antibodies for testing.
- Detect specific protein from a large mixture of different proteins. (Even more than 300,000)

Disadvantages

- Requires specific primary antibodies to perform test on desired protein of interest.
- Challenging and hence requires well trained staffs.
- Poorer results as antibodies may revel off-target bindings.
- Detecting and imaging the results can be expensive as equipment cost is high.

References

https://www.thesciencenotes.com/western-blotting-principle-procedure-and-applications/

https://blog.praxilabs.com/2020/05/29/western-blot-concept/#Applications_of_Western_Blotting

Meftahi, G. H., Bahari, Z., Zarei Mahmoudabadi, A., Iman, M., & Jangravi, Z. (2021). Applications of western blot technique: From bench to bedside. *Biochemistry and Molecular Biology Education*, *49*(4), 509-517.

Module# 056 Difference Between Different Blotting Technique

Point of comparison	Southern Blot	Northern Blot	Western Blot
Definition	A procedure used to identify a specific sequence of DNA.	It is an adaptation of the southern blot procedure, which is useful in detecting a specific sequence of DNA through hybridization with complementary DNA.	A blotting procedure used for identifying a specific amino acid sequence in proteins.
Developed by	Edward M. Southern in 1975	James Alwine, David Kemp, and George Stark in 1977	George Stark's group at Stanford University in 1979
Types of molecules detected	Specific DNA sequence	Specific RNA sequence	Specific proteins
Gel electrophoresis	Agarose gel electrophoresis	Denaturing of formaldehyde agarose gel	It involves SDS PAGE/ Polyacrylamide gel
Blotting procedure	Capillary transfer	Capillary transfer	Electric transfer
Probes	DNA probes	cDNA probes	Primary and secondary antibodies
Preparation of sample	The sample needs to be denaturated.	The sample is in its native state.	The sample is in its native state.
Detection system	Autoradiograph, Detection of light, and Changes in color	Autoradiograph, Detection of light, and Changes in color	Detection of light and Changes in color

Applications	• To detect a specific DNA sequence in a given sample. • It is used in paternity testing. • It is used in DNA fingerprinting. • It is used for victim identification. • It is used for identifying criminals. • It is useful in genetic diseases and infectious agents. • It is used in mutation and identification of gene rearrangement.	• Identify the specific RNA sequence in a given sample. • It helps in disease diagnosis. • It is useful in gene expression studies.	• Identifying the number of protein in a given sample. • Detecting the presence of the virus and/or bacteria in the serum. • Detecting the presence of HIV antibody in serum. • Detecting defective proteins. • Used as a definite measure for Lyme disease, herpes, hepatitis B, and Creutzfeldt-Jacob disease.

Module# 057 SINGLE NUCLEOTIDE POLYMORPHISM

Phenotype of living organism is controlled by DNA. Variation in DNA sequence or polymorphism may make individual difference such as the differences in phenotype, risk of various diseases, and response to drugs, vaccine, chemical, and pathogen. Polymorphism or variation in DNA sequence can affect individual phenotypes such as color of skin or eyes, susceptible to diseases, and respond to drug, vaccine, chemical, and pathogen. It occurs more often than mutations (frequency ≥ 1%).

Genetic polymorphism is the occurrence in the same population of two or more alleles at one locus, each with appreciable frequency, where the minimum frequency is typically taken as 1%. An allele is one of the variant forms of a gene at specific locus on a homologous chromosome. The different forms of the polymorphism (alleles) are observed more often in the general population than mutations. The most common polymorphism in the human genome is the single-nucleotide polymorphism (SNP). A single nucleotide polymorphism (SNP) is defined as a single base change in a DNA sequence that occurs in a significant proportion (more than 1 %) of a large population. SNPs occur at a frequency that ranges from 1 in 1000 to 1 in 100 bases.

Around 325 million, SNPs have been identified in the human genome, 15 million of which are present at frequencies of 1% or higher across different populations worldwide. An example for SNP is shown in Figure. It demonstrates that at a specific position of human genome when compared between two individuals and two DNA sequences. The DNA sequence of a person 1 has C nucleotide which is similar to most of the other person (majority group), whereas the DNA sequence of a person 2 has T at this position which is minority group of population. It is said that there is an SNP at this specific position between allele C or T.

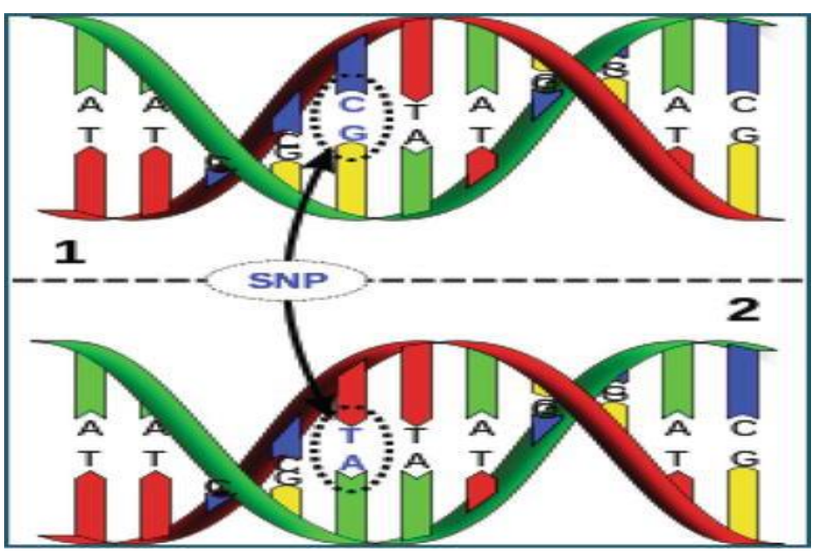


Figure: Single nucleotide polymorphism (SNP)

SNPs are scattered throughout the genome and are found in both coding AND non-coding regions. SNPs can cause silent, harmless, harmful, or latent effects. They occur with a very high frequency, with estimates ranging from about 1 in 1000 bases to 1 in 100 to 300 bases. This means that there could be millions of SNPs in each human genome. The abundance of SNPs and the ease with which they can be measured make these genetic variations significant.

When a SNP occurs, within a coding region, it is possible that there is no effect. There are also SNPs within coding regions that lead to very subtle changes in proteins. For example, if a SNP causes the codon GAU to change to GAG, the amino acid changes from aspartic acid to glutamic acid. There are also SNPs within coding regions that lead to very subtle changes in proteins. For example, if a SNP causes the codon GAU to change to GAG, the amino acid changes from aspartic acid to glutamic acid. If the SNP causes a change in a part of the protein that is not important to its function, the result may be very subtle or totally harmless. When the change in protein lead to disease symptoms in the patient, SNP is harmful and is called a mutation. An example of a SNP causing disease is found in sickle cell anemia.

References

Sukhumsirichart, W. (2018). Polymorphisms. In *Genetic diversity and disease susceptibility*. IntechOpen.

Huang, T., Shu, Y., & Cai, Y. D. (2015). Genetic differences among ethnic groups. *BMC genomics*, *16*, 1-10.

http://www.biostat.umn.edu/~chap/S29-Cancers-SNP.pdf

Module# 058 SNP Maps

SNP maps generate a single map of the human genome containing all possible SNPs by sequencing genomes of a large number of people and comparing the base sequences to discover SNPs. Single-nucleotide polymorphism (SNP) mapping is the easiest and most reliable way to map genes in organisms. SNPs are extremely dense and usually have no associated phenotype, making them ideal markers for mapping. Mapping with SNPs has become a powerful complement (and in some cases outright alternative) to the standard genetic mapping procedures. Moreover, SNP mapping can be routinely used to narrow down the known physical location of mutations to regions smaller than a single cosmid. SNP mapping can theoretically be used to narrow down the implicated region to a single gene.

SNP Profiling

Single nucleotide polymorphisms (SNPs) are the most frequent sequence variations in the human genome, occurring approximately once every 100 to 300 bp. Apart from identical twins, each individual has a unique combination of nucleotides at these positions. Thus, a SNP profile provides a kind of fingerprint. In SNP Profiling, genome of each individual contains distinct SNP pattern. People can be grouped based on the SNP profile. The advantage of SNPs Profiles is important for identifying response to Drug Therapy. A correlation might emerge between certain SNP profiles and specific responses to treatment.

References

http://www.wormbook.org/chapters/www_SNPsintrotwopointmap/SNPsintrotwopointmap.html

Huijsmans, R., Damen, J., van der Linden, H., & Hermans, M. (2007). Single nucleotide polymorphism profiling assay to confirm the identity of human tissues. *The Journal of Molecular Diagnostics*, *9*(2), 205-213.

Module# 059 METHOD FOR DETECTION OF SNPS

SNP genotyping strategies typically involve allele-discrimination and allele-detection. The other methods based on physical properties of DNA.

a.

1. **ALLELE-DISCRIMINATION**

b. **Primer extension**

These approaches involve allele-specific incorporation of nucleotides in primer extension reaction with a DNA template, utilizing enzyme specificity to accomplish allelic discrimination. In the reaction of a common primer extension (CPE) protocol, a designed primer is annealing with its 3′ end near a SNP site and nucleotides are added by polymerase enzyme.

b. **Hybridization**

Hybridization methods use differences in thermal stability of double-stranded DNA to separate between perfectly matched and mismatched target-probe pairs for succeeding allelic discrimination. Hybridization methods have been applied on high-throughput platforms using microarrays. The DNA fragment containing SNPs are amplified from genomic DNA. The PCR products are cleaved, tagged, and hybridized to the probe array under stringent conditions, then wash, and label with fluorescent. The genotypes of SNPs are examined from the fluorescence signal based on probe-target hybridization. Numerous probes that differ at a single base are used to confirm each SNP to increase genotyping accuracy.

c. **Ligation**

Ligation method discriminates the allele by using specificity of ligase enzymes. When two oligonucleotides (allele-specific probes and ligation probes) hybridize to single-stranded template DNA with perfect complementarity, they are nearby to each other, then ligase enzymes join them together to form a single oligonucleotide. Normally, three oligonucleotide probes are used in ligation assays, two probes are allele specific that bind to the template at the SNP site and ligation probe

d. **Enzymatic cleavage**

The method is based on the capacity of enzymes to cleave DNA by recognition of specific sequences and structures. Difference between alleles can be discriminated when SNPs are located in an enzyme recognition site and affect the enzyme activity. The ability of restriction enzymes has been used for detection of genetic variation by the method of restriction fragment length polymorphism (RFLP). These enzymes recognize specific sequences in double-stranded DNA and cleave both strands at a specific site in the sequence or near it to generate smaller DNA fragments. For SNP genotyping, the PCR product containing the SNP is incubated with appropriate restriction enzyme and separated by gel electrophoresis. The SNP genotype is simply determined from sizes of the digested products. This method does not need any probes but it has limited amount and a number of SNPs.

a.

2. **ALLELE DETECTION METHODS**

b. **Mass-based detection**

For mass analysis of oligonucleotides, MALDI-TOF MS is a widely used method.

b. **Fluorescence signal-based detection**

Monitoring fluorescence signal is widely used in genotyping technologies because its operation is simple and detection is fast with high sensitivity. Fluorescence detection is used for direct sequencing (DS) using capillary array electrophoresis.

c. **Chemiluminescence**

Chemiluminescence has several advantages as a detection technique, such as high signal-to-noise ratio, rapid detection, and feasibility for automation. Pyrosequencing™ is an approach that combines sequencing-by-synthesis with chemiluminescence detection. In SNP genotype, it provides sequence information on the region surrounding the SNP site.

3. OTHER METHODS

a. **Single-strand conformation polymorphism (SSCP)**

SSCP discriminates the allele by using secondary structure of single-stranded DNA. The single-stranded DNA molecules which differ at a single base run on a nondenaturing gel electrophoresis display different mobility based on their native conformations

b. **High resolution melting analysis (HRM)**

This method, the fragment cover SNP is amplified by real-time PCR and followed by HRM. The HRM is a technique for the detection mutations and SNPs, which based on analysis of melting curve when double-strand DNA (dsDNA) separate into single-strand DNA (ssDNA) during increased temperature from around 50°C up to around 95°C.

c. **Denaturing high performance liquid chromatography (DHPLC)**

It is a method for screening DNA samples for SNPs and inherited mutations.

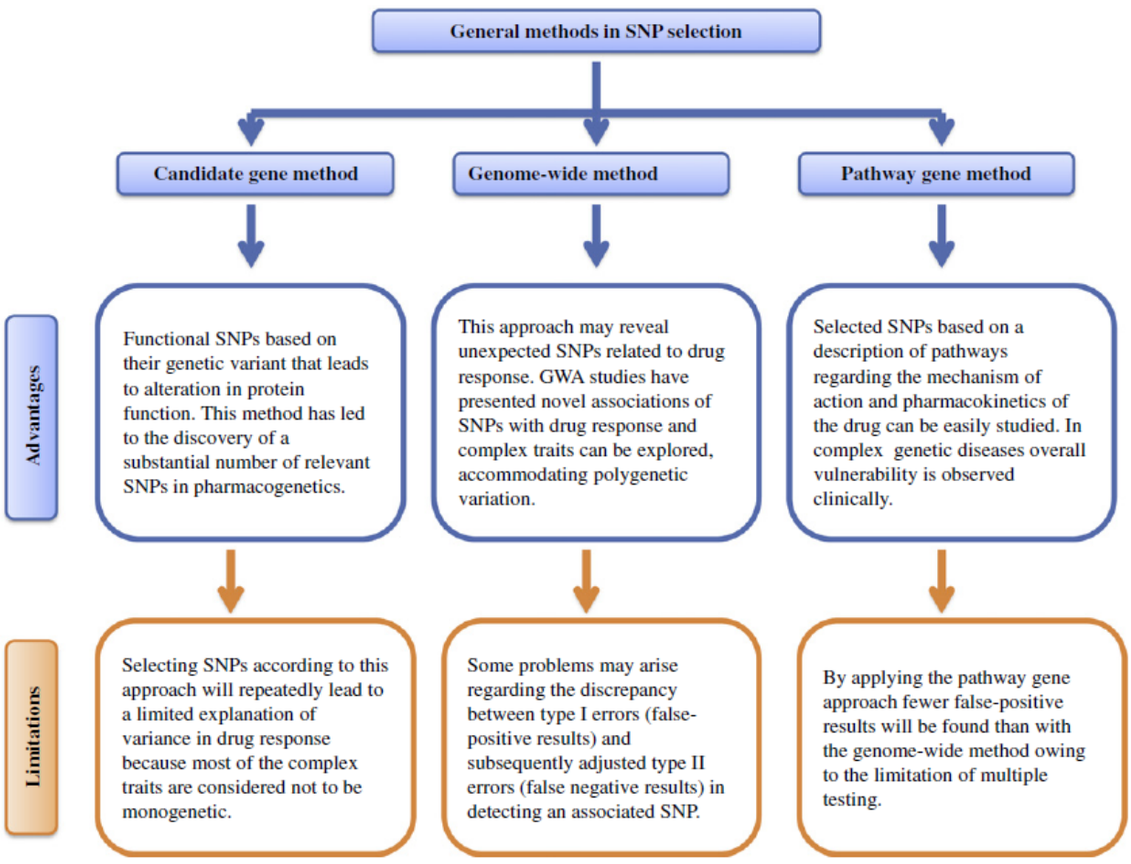


Figure: Selection of SNP made through three different pathways (approaches) exploring the mechanism of genome variation and drug development with some advantages and limitations

References

Sukhumsirichart, W. (2018). Polymorphisms. In *Genetic diversity and disease susceptibility*. IntechOpen.

Module# 060 SNP SCREENING

Two different SNP screening strategies are involved that includes: study of many SNPs in a few individuals or to study a few SNPs in many individuals. Different strategies will require different tools. It is important in determining markers for complex genetic traits. SNP genotyping methods for detecting genes contributing to susceptibility or resistance to multifactorial diseases, adverse drug reactions. For example: in case-control association analysis.

Significance

It helps in disease diagnosis, finding predisposition to diseases, drug discovery & development, drug responses and investigation of migration patterns. All these aspects will help to look for medication & diagnosis at individual level.

Advantages

1. SNPs are the most frequent form of DNA variations
2. They are the disease causing mutations in many genes
3. They are abundant & have slow mutation rates
4. Easy to score
5. May work as the next generation of genetic markers

References

https://studylib.net/doc/9910045/single-nucleotide-polymorphism--snp-

Module# 061 HAPLOTYPE AND LINKAGE DISEQUILIBRIUM

Whole-genome approach to mapping requires the genotyping of thousands of samples. Although there are several high-throughput methods that are available for these studies, they are expensive, laborious and cannot be undertaken by many laboratories. Haplotype refers to the inheritance of a cluster of single nucleotide polymorphisms (SNPs), which are variations at single positions in the DNA sequence among individuals (as shown in figure). A different procedure called haplotype (collection of SNPs on a single chromosome at a locus that is inherited in blocks) analysis has been used to identify common disease genes. Because the genome undergoes recombination involving large stretches of DNA, there may be several SNPs linked together in this large region of DNA. These closely linked SNPs may then be co-transmitted from generation to generation in these large blocks. This phenomenon is called Linkage Disequilibrium. Linkage disequilibrium (LD) is a population-based parameter that describes the degree to which an allele of one genetic variant is inherited or correlated with an allele of a nearby genetic variant within a given population

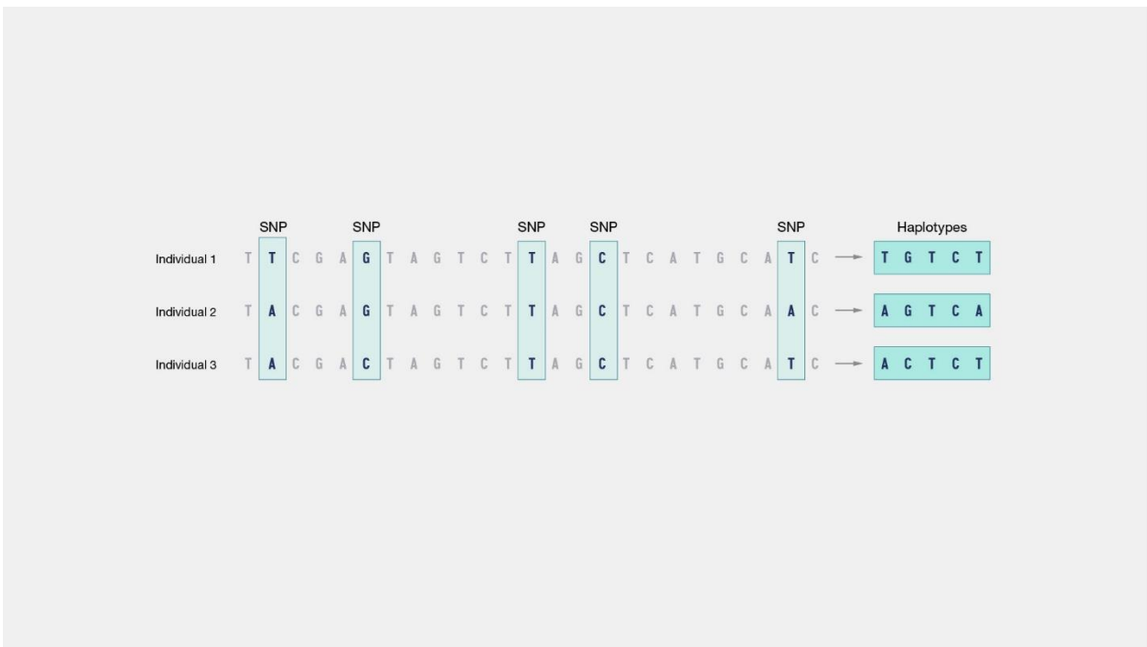


Figure: A haplotype refers to a set of DNA variants along a single chromosome that tend to be inherited together. They tend to be inherited together because they are close to each other on the chromosome, and recombinations between these variants are rare. A haplotype can be limited to a single gene or it can be larger and include multiple genes.

In this type of analysis, multiple genotypes are reduced to haplotypes and hence only a small number of SNPs are required to map the disease gene. Therefore, this method can be more effective in gene mapping and can also provide substantial statistical power in association studies.

HAPLOTYPE CORRELATION WITH PHENOTYPE

The “Haplotype centric” approach combines the information of adjacent SNPs into composite multilocus haplotypes. Haplotypes are not only more informative but also capture the regional LD information, which is assumed to be robust and powerful. Association of haplotype frequencies with the presence of desired phenotypic frequencies in the population will help in utilizing the maximum potential of SNP as a marker.

References

Barkur, S. S. (2007). SNPs in disease gene mapping, medicinal drug development and evolution. *Journal of human genetics/Japan Society of Human Genetics*, *52*(11), 871-880.

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Module# 062 DNA fingerprinting

Background

Almost every cell in our body contains our DNA. On average, about 99.9 per cent of the DNA between two humans is the same. The remaining percentage is what makes us unique. Although this might sound like a small amount, it means that there are around three million base pairs that are different between two people. These differences can be compared and used to help distinguish you from someone else.

Principle of DNA fingerprinting

The human genome consists of innumerable small noncoding sequences which are inheritable and repeatedly present. They can be separated from the bulk DNA as satellite upon performing density gradient centrifugation and thus known as satellite DNA. They can be categorized into either microsatellites or minisatellites depending on the length, base composition and tandemly repetitive units. These satellite DNAs show polymorphism and this polymorphism is the basis of DNA fingerprinting. Variable number tandem repeats (VNTRs) are short nucleotide sequences (20–100 bp) that vary in copy number in bacterial genomes. These repeats act as genetic markers and every individual inherits these repeats from their parents. Thus, every individual has a particular composition of VNTRs and this is the main principle of the DNA fingerprinting technique. There are two principal families of VNTRs: Microsatellites and Minisatellites. Minisatellites are a type of DNA tandem repeat sequence, meaning that the sequences repeat one after another without other sequences or nucleotides in between them. Minisatellites are short sequences (10-60 base pairs long) of repetitive DNA that show greater variation from one person to the next than other parts of the genome. The first minisatellite was discovered in 1980. Microsatellites, also known as simple sequence repeats or short tandem repeats, are widely used genetic markers consisting of nuclear-encoded short sequences of 1–6 base pairs repeated from a few to approximately ~50 times.

DNA fingerprinting is a procedure that shows the hereditary cosmetics of living things. It is a strategy for finding the distinction between the satellite DNA areas in the genome. DNA profiling, DNA testing, DNA examination, Genetic profile, DNA distinguishing proof, genetic fingerprinting, and genetic investigation are a portion of the mainstream names utilized for DNA fingerprinting. DNA fingerprinting was invented in 1984 by Professor Sir Alec Jeffreys after he realized that he could detect variations in human DNA, in the form of these minisatellites.

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MODULE# 63 DNA profiling process

HOW WAS THE FIRST DNA FINGERPRINT PRODUCED?

The process begins with a sample of an individual's DNA (typically called a "reference sample").
The most desirable method of collecting a reference sample is the use of a buccal swab, as this reduces the possibility of contamination. When this is not available may need to be used to collect a sampleof blood, saliva, semen, or other appropriatefluid ortissue from personal items (e.g. toothbrush, razor, etc.) or from stored samples (eg: banked sperm or biopsy tissue).

1. The first step of DNA fingerprinting was to extract DNA from a sample of human material, usually blood.
2. Molecular 'scissors', called restriction enzymes, were used to cut the DNA. This resulted in thousands of pieces of DNA with a variety of different lengths.
3. These pieces of DNA were then separated according to size by a process called gel electrophoresis: The DNA was loaded into wells at one end of a porous gel, which acted a bit like a sieve. An electric current was applied which pulled the negatively-charged DNA through the gel. The shorter pieces of DNA moved through the gel easiest and therefore fastest. It is more difficult for the longer pieces of DNA to move through the gel so they travelled slower. As a result, by the time the electric current was switched off, the DNA pieces had been separated in order of size. The smallest DNA molecules were furthest away from where the original sample was loaded on to the gel.
4. Once the DNA had been sorted, the pieces of DNA were transferred or 'blotted' out of the fragile gel on to a robust piece of nylon membrane and then 'unzipped' to produce single strands of DNA.
5. Next the nylon membrane was incubated with radioactive probes. Probes are small fragments of minisatellite DNA tagged with radioactive phosphorous. The probes only attach to the pieces of DNA that they are complementary to – in this case they attach to the minisatellites in the genome.
6. The minisatellites that the probes have attached to were then visualised by exposing the nylon membrane to X-ray film. When exposed to radioactivity a pattern of more than 30 dark bands appeared on the film where the labelled DNA was. This pattern was the DNA fingerprint. To compare two or more different DNA fingerprints the different DNA samples were run side-by-side on the same electrophoresis gel.

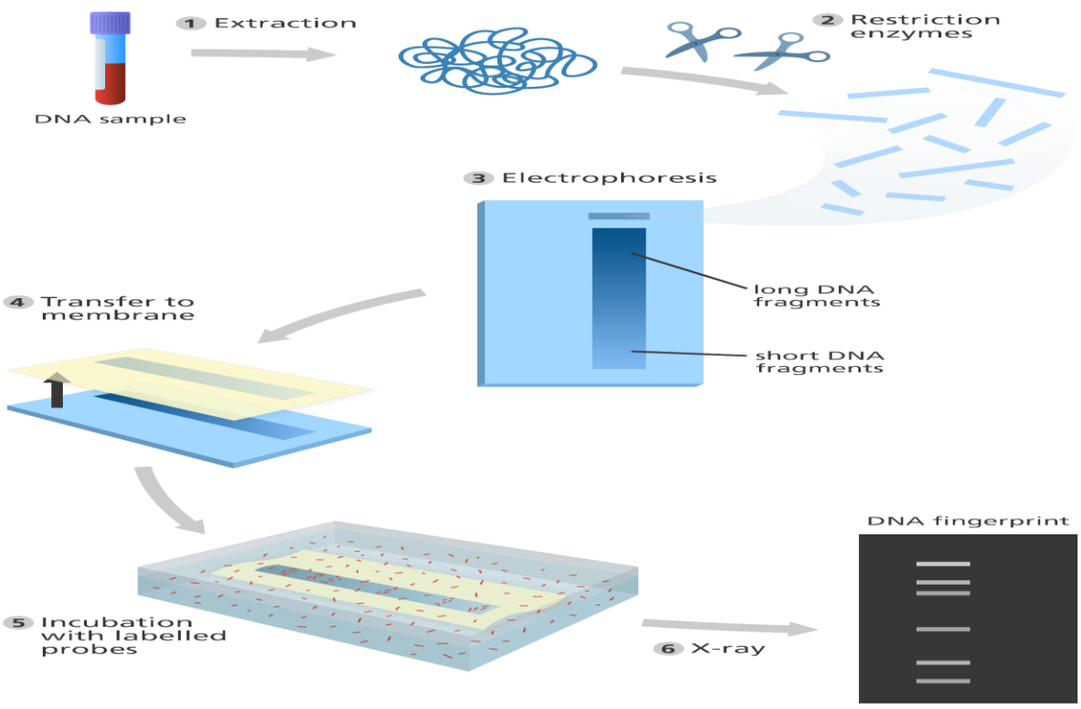


Figure: Illustration showing the steps in DNA fingerprinting.

DNA PROFILING

Modern-day DNA profiling is also called STR analysis and relies on microsatellites rather than the minisatellites used in DNA fingerprinting.

HOW IS A DNA PROFILE PRODUCED TODAY?

1. DNA is extracted from a biological sample.

2. It uses the polymerase chain reaction (PCR) to produce many copies of specific STR sequences.
3. In PCR small bits of DNA called primers bind to complementary sequences of the DNA of interest (here STRs are used) and mark the starting point for the copying of the DNA of interest.
4. The primers for each STR is labelled with a specific coloured fluorescent tag. This makes it easier to identify and record the STR sequences after PCR.
5. Once enough copies of the sequence have been produced by PCR, electrophoresis is used to separate the fragments according to size.
6. Each fragment passes by a laser which causes the fragments with fluorescent tags to glow with a specific colour. The output is displayed as a series of coloured peaks (as shown in the image below) highlighting the colour and length of each STR sequence.

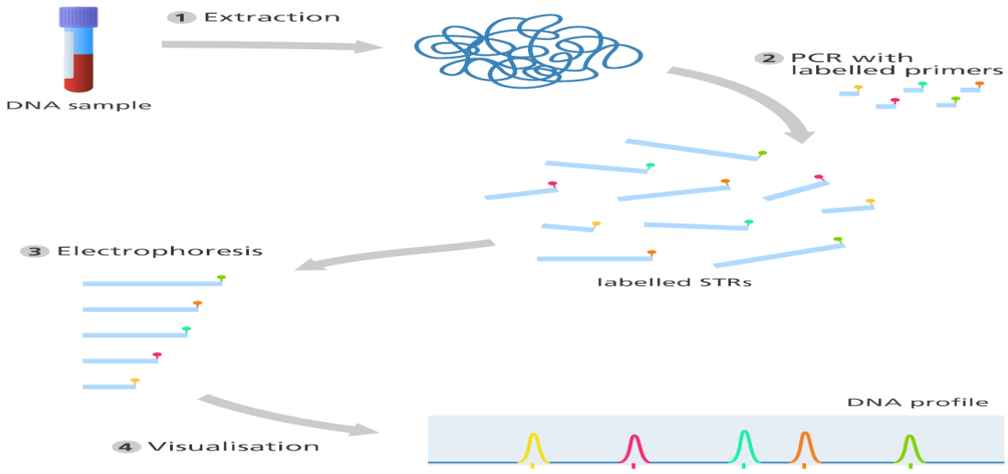


Figure: Illustration showing the steps in DNA profiling.

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https://www.yourgenome.org/facts/what-is-a-dna-fingerprint/

Module# 64 Typs of DNA fingerprinting methods_1

TYPES OF DNA FINGERPRINTING METHODS

Polymerase Chain Reaction

Polymerase chain reaction (PCR) can exponentially amplify a fragment of DNA *in vitro*, and since its invention a series of techniques have emerged in combination with PCR, e.g. PCR—RFLP, AFLP, simple sequences repeats (SSRs), and Single Nucleotide Polymorphisms (SNPs). In PCR, small bits of DNA called primers bind to complementary sequences of the DNA of interest and mark the starting point for the copying of the DNA of interest. In STR analysis the primers used in the PCR are designed to attach to either end of the STR sequence of interest. The primers for each STR is labelled with a specific coloured fluorescent tag. This makes it easier to identify and record the STR sequences after PCR.

Restriction Fragment Length Polymorphism

RFLP is a method established by Grodzicker in 1974, it is used to identify DNA polymorphisms among different individuals. Its basic principle is as follows: first, genomic DNA from different individuals is digested into DNA fragments of varying size, using known restriction enzymes. Second, the digested fragments are separated via electrophoretic analysis. Finally, separated fragments are hybridized with radioactive or chemiluminescent homologous probes and exposed to an X-ray film; the different fragments are visible by autoradiography. The molecular basis of RFLP is that nucleotide base substitutions, insertions, deletions, duplications, and inversions within the whole genome can remove or create new restriction sites.

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https://www.yourgenome.org/facts/what-is-a-dna-fingerprint/

Module# 65 Types of DNA fingerprinting methods -2

Types of DNA Fingerprinting Methods

Rapid Amplified Polymorphic DNA

RAPD was developed by U.S. scientists in 1990. It amplifies the target genomic DNA with short, arbitrary primers (commonly 10 bp) in a PCR reaction, and can be used to produce relatively complicated DNA profiles for detecting amplified fragment length polymorphisms between organisms. Since the arbitrary primers complement different parts of the genomic DNA, PCR products will differ in number and size (polymorphism). RAPD markers have several obvious features as summarized in the literature: 1) no prior sequence knowledge is necessary for designing the specific primers, which can then be used in different templates. 2) The amount of DNA required is very small because it will be amplified by PCR. 3) RAPDs are simple, quick, and cost effective compared to RFLP. However, RAPDs also have some disadvantages, these include 1) the repeatability and reliability of RAPD polymorphic profiles are poor. 2) Some non-specific and therefore non-reproducible binding of primers occurs.

Amplified Fragment Length Polymorphism

AFLP was developed by Zabeau and Vos in 1993; it is a combination of the RFLP and PCR techniques. The AFLP procedure is as follows: first, the genomic DNA is digested with a restriction enzyme, and then the digested fragments are ligated to synthetic adaptors and amplified with specified primers that are complementary to a selective sequence on the adaptors. Subsequent separation of the amplified fragments is obtained by selective primers and visualized using autoradiography. AFLPs overcome the drawbacks of the labor-intensive, time-consuming RFLP method and solve the reliability problem caused by non-specific amplifications in RAPDs.

AFLPs are notable for their genetic stability, they provide an effective, rapid, and economical tool for detecting a large number of polymorphic genetic markers, that can be genotyped automatically.

DNA PROFILING

Modern-day DNA profiling is also called STR analysis and relies on microsatellites rather than the minisatellites used in DNA fingerprinting. Microsatellites, or short tandem repeats (STRs), are the shorter relatives of minisatellites usually two to five base pairs long. Like minisatellites they are repeated many times throughout the human genome, for example ‘TATATATATATA’.

HOW IS A DNA PROFILE PRODUCED TODAY?

1. DNA is extracted from a biological sample. STR analysis is incredibly sensitive so it only needs a tiny amount of someone’s DNA to produce an accurate result. As a result, the DNA can be extracted from a wider range of biological samples, including blood, saliva and hair.
2. Unlike the original DNA fingerprinting method, DNA profiling does not use restriction enzymes to cut the DNA. Instead it uses the polymerase chain reaction (PCR) to produce many copies of specific STR sequences.
3. PCR is an automated procedure that generates lots of copies of a specific sequence of DNA. It only requires small amounts of DNA to start with and can even make copies from a DNA sample that is partially degraded.
4. In PCR small bits of DNA called primers bind to complementary sequences of the DNA of interest and mark the starting point for the copying of the DNA of interest.
5. In STR analysis the primers used in the PCR are designed to attach to either end of the STR sequence of interest.
6. The primers for each STR is labelled with a specific coloured fluorescent tag. This makes it easier to identify and record the STR sequences after PCR.

7. Once enough copies of the sequence have been produced by PCR, electrophoresis is used to separate the fragments according to size.
8. Each fragment passes by a laser which causes the fragments with fluorescent tags to glow with a specific colour. The output is displayed as a series of coloured peaks (as shown in the image below) highlighting the colour and length of each STR sequence.

- The more STR sequences that are tested, the more accurate the test is at identifying someone.
- Other STRs used for forensic purposes are called Y-STRs, which are derived solely from the male Y chromosome. This is useful for identifying a male perpetrator from mixed DNA samples.
- Only one person in every 10 million million (10,000,000,000,000) will have a particular STR profile. With the world human population estimated at only 7,100 million (7,100,000,000) it is therefore extremely unlikely you will share the same profile as someone else, unless you are an identical twin.

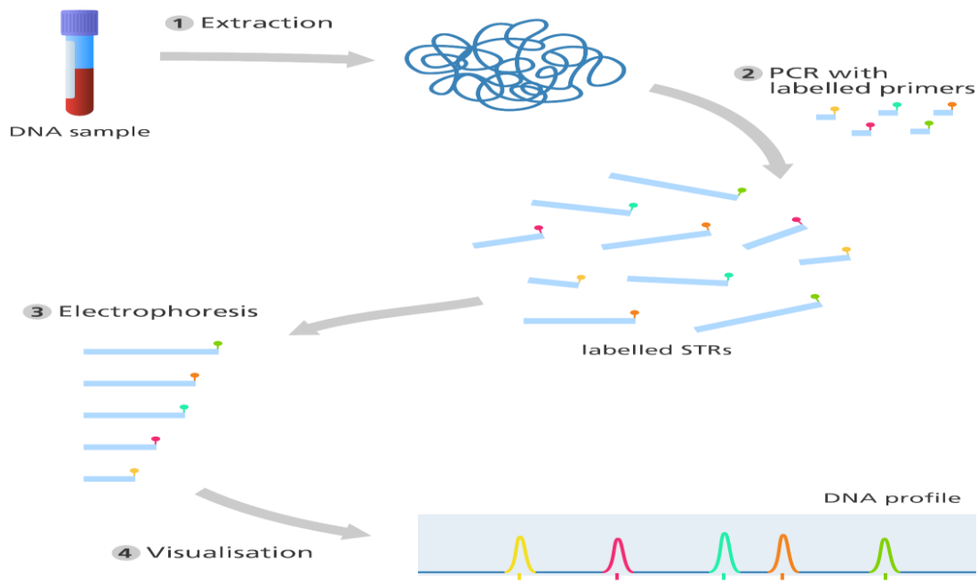


Figure 1: Illustration showing the steps in DNA profiling.

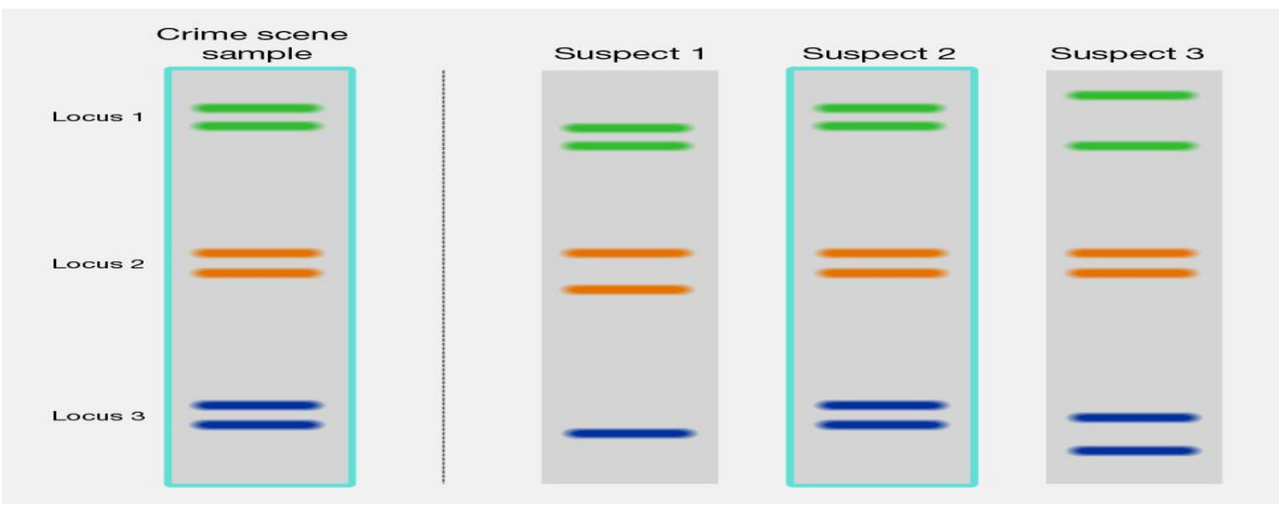


Figure 2: Sample obtained from crime scene is matched with the samples of suspects.

References

Yang, W., Kang, X., Yang, Q., Lin, Y., & Fang, M. (2013). Review on the development of genotyping methods for assessing farm animal diversity. *Journal of animal science and biotechnology*, *4*(1), 1-6.

https://www.yourgenome.org/facts/what-is-a-dna-fingerprint/

Module# 66 Applications of DNA Fingerprinting

Since it was invented in 1984, DNA fingerprinting most often has been used in court cases and legal matters. It can:

- Utilizing the DNA fingerprinting strategy, the natural personality of an individual can be uncovered. For approving one's character, there is no other preferable alternative over DNA fingerprinting.
- Utilizing Blood-Typing in Paternity Tests to show who your parents, siblings, and other relatives may be.
- Gravely harmed dead bodies can be distinguished.
- DNA fingerprinting is extremely accurate. Most countries now keep DNA records on file in much the same way police keep copies of actual fingerprints.

- One of the most significant uses of the current strategy is in the crime scene examination and criminal check. The example is gathered from the crime site which could be salivation, blood, hair follicle, or semen. DNA is removed and investigated against the suspect, utilizing the two markers we clarified previously. By coordinating DNA band designs criminal's connected to wrongdoing can be built up.

It also has medical uses. It can:

- Match tissues of organ donors with those of people who need transplants.
- Identify diseases that are passed down through your family.
- One of the significant downsides of pre-birth determination is maternal cell tainting. The amniotic liquid or CVS test contains the maternal DNA or maternal tissue, once in a while. Contamination expands the opportunity of false-positive outcomes, particularly on account of carrier recognition. Utilizing VNTRs and STRs markers with PCR-gel electrophoresis, maternal cell tainting can be recognized during pregnancy hereditary testing.

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Module# 84 Making Antibodies (Polyclonal Antibodies)

ANTIBODIES:

Antibodies, also known as immunoglobulins, are secreted by B cells (plasma cells) to neutralize antigens such as bacteria and viruses. The classical representation of an antibody is a Y-shaped molecule composed of four polypeptides-two heavy chains and two light chains (Figure 1). Each tip of the "Y" contains a paratope (a structure analogous to a lock) that is specific for one particular epitope (similarly analogous to a key) on an antigen, allowing these two structures to bind together with precision. The ability of binding to an antigen has led to their ubiquitous use in a variety of life science and medical science. Because antibodies can bind antigen epitopes (the region of an antigen where an antibody binds) with great affinity and accuracy, they are essential to host immunity and are valuable in research and medicine for the treatment of disease and health improvement. Antibody affinity and avidity is very important. Antibody affinity is defined as strength of the binding interaction between antigen and antibody. Antibody avidity describes the strength with which an antibody binds to a complex antigen.

These antibodies can be classified into two primary types (monoclonal and polyclonal) by the means in which they are created from lymphocytes. Each of them has important role in the immune system, diagnostic exams, and treatments.

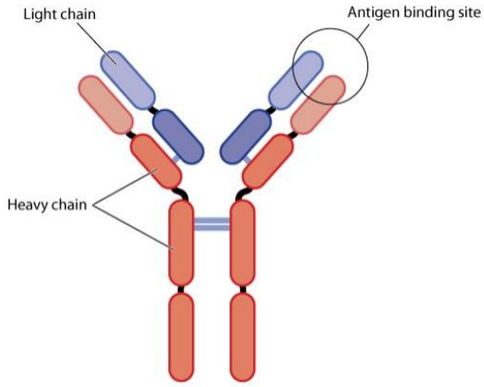


Figure 1: The structure of the antibody

Polyclonal Antibodies Production:

Polyclonal antibodies (pAbs) are mixture of heterogeneous which are usually produced by different B cell clones in the body. They can recognize and bind to many different epitopes of a single antigen.

The schedules and methodologies used to immunize rabbits and produce polyclonal antibodies will continue to vary dependent upon the immunogen (substance that generates B-cell and/or T-cell adaptive immune responses upon exposure to a host organism), the adjuvant (substance that helps create a stronger immune response in people), and the end purpose of the antibody. The choice of adjuvant and immunization schedules are critical components of the polyclonal antibody production process. Polyclonal antibodies (pAbs) are produced by injecting a specific antigen into lab animals, such as rabbits and goats, etc. The animal is immunized repeatedly to obtain higher titers of antibodies specific for the antigen. Within a few weeks, these polyclonal antibodies can be harvested and collected from the antiserum.

Production of polyclonal antibodies is easier and more cost-effective than the production of monoclonal antibodies. Furthermore, polyclonal antisera can be generated in a shorter time (4-8 weeks), whereas it takes about 3 to 6 months to produce mAbs.

Rabbit Polyclonal Antibody Production Process

There are several critical steps involved in the production of polyclonal antibodies in rabbits. A four-step procedure of pAb production is outlined below.

Step 1: Antigen preparation

The production process starts with preparing the protein or peptide antigens. It's important to ensure the quality of the target antigen, in particular when binding to a conformational epitope is desired. The specificity of polyclonal antibody obtained depends on the purity of the antigen applied.

Step 2: Animal immunization

The choice of animals for the production of polyclonal antibodies depends upon the amount of antiserum desired, the evolutionary distance between the species from which the protein of interest has been derived and the species of the animal to be immunized, and prior experience with the immunogens. Rabbits are the usual animal of choice because they are genetically divergent from the human and mouse sources of the proteins most often studied. Rabbits provide as much as 25 ml of serum from each bleed without significant harmful effects.

Adjuvants are used to enhance the immune response for pAb production. Freund's Adjuvant is one of the most commonly used adjuvants in research. In the presence of adjuvant, the protein antigen is injected intramuscularly, intradermally, or subcutaneously into an animal of the chosen species. Booster immunizations are started 4 to 8 weeks after the priming immunization and continued at 2- to 3-week intervals. Prior to the priming immunization, following the primary and each booster immunization, the animal is bled and serum prepared from whole blood. When the antibody titer has reached an acceptable level, the production of polyclonal antibodies should be ended (Figure 2).

Step 3: Antibody purification

Affinity purification is a good choice for purification of polyclonal antibodies.To isolate specific polyclonal antibodies from antiserum, antigen-specific affinity purification is often used. Antigen affinity purification results in

the elimination of the bulk of the non-specific IgG fraction, and enriches the fraction of immunoglobulin that specifically reacts with the target antigen (Figure 2).

Step 4: Quality control

After purification, a series of quality control tests are performed to ensure the quality of polyclonal antibodies.

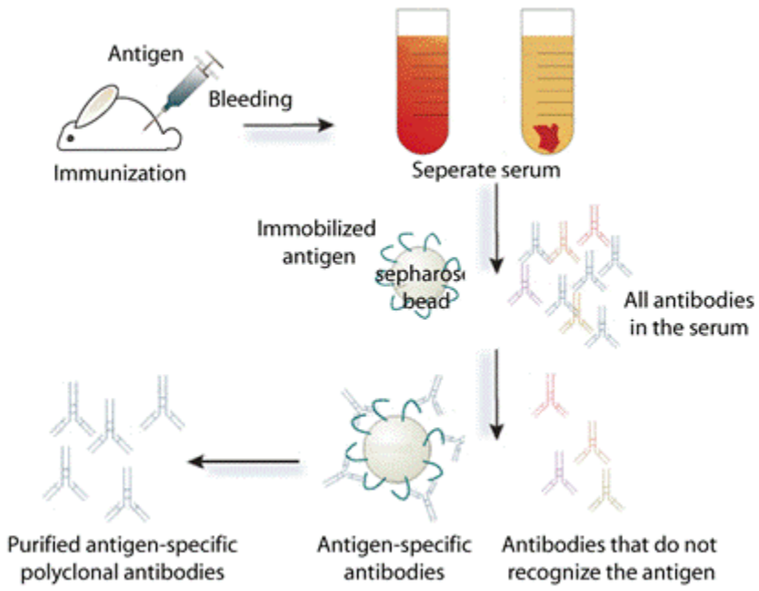


Figure 2: The process to generate the polyclonal antibody

Immunoglubulin Y

The term “IgY technology” was introduced in the literature in the mid 1990s to describe a procedure involving immunization of avian species, mainly laying hens and consequent isolation of the polyclonal IgYs from the “immune” egg yolk (thus avoiding bleeding and animal stress). IgYs have been applied to various fields of medicine and biotechnology. Steps of IgYproduction is shown below in the figure 3. Birds present three classes of immunoglobulins – IgA, IgM and IgY and all of them are incorporated into the egg. The first two is similar to their mammalian counterparts and are transferred into the egg white in trace amounts. However, IgY, the functional equivalent of mammalian IgG, is transferred in large amounts to the egg yolk via specific receptors at the yolk membrane. The IgY concentration in chicken serum, for example, ranges from 8-10mg/ml and can go up to 10-20mg/ml in the egg-yolk making it effectively the sole immunoglobulin isotype in the egg yolk. This has an impact on subsequent antibody isolation and is distinct from what happens in mammalian blood serum.

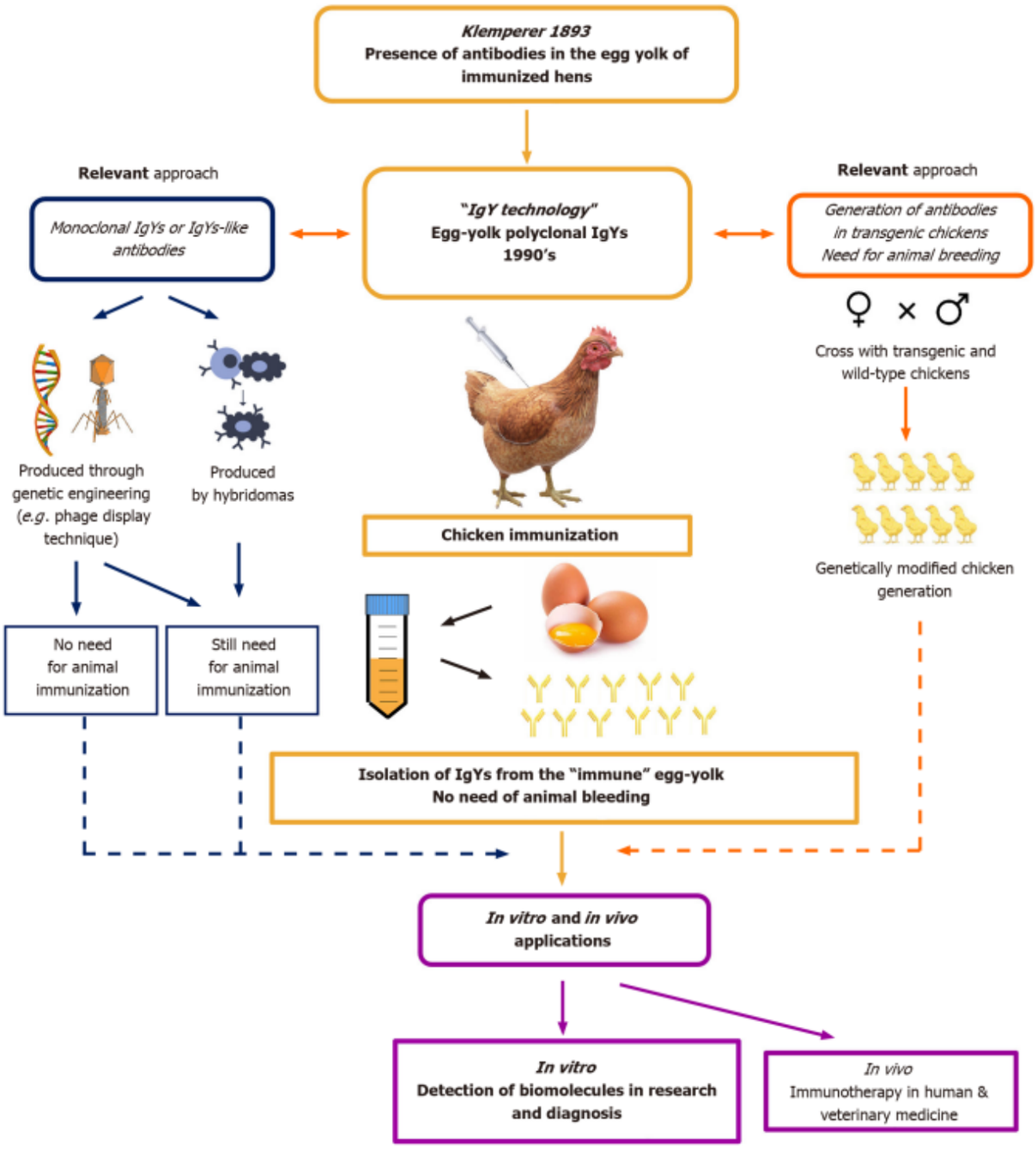


Figure 3: Schematic representation of the main parts comprising the original immunoglobulin Y technology (central axis); promising relevant approaches are also shown (periphery, left and right)

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https://www.drugtargetreview.com/article/33086/antibody-fragment-technology-and-avian-igy-antibodies-a-powerful-combination/

Module# 85 Monoclonal Antibodies Production:

Monoclonal antibodies (mAbs) are generated by identical B cells which are clones from a single parent cell. This means that the monoclonal antibodies have monovalent affinity and only recognize the same epitope of an antigen.

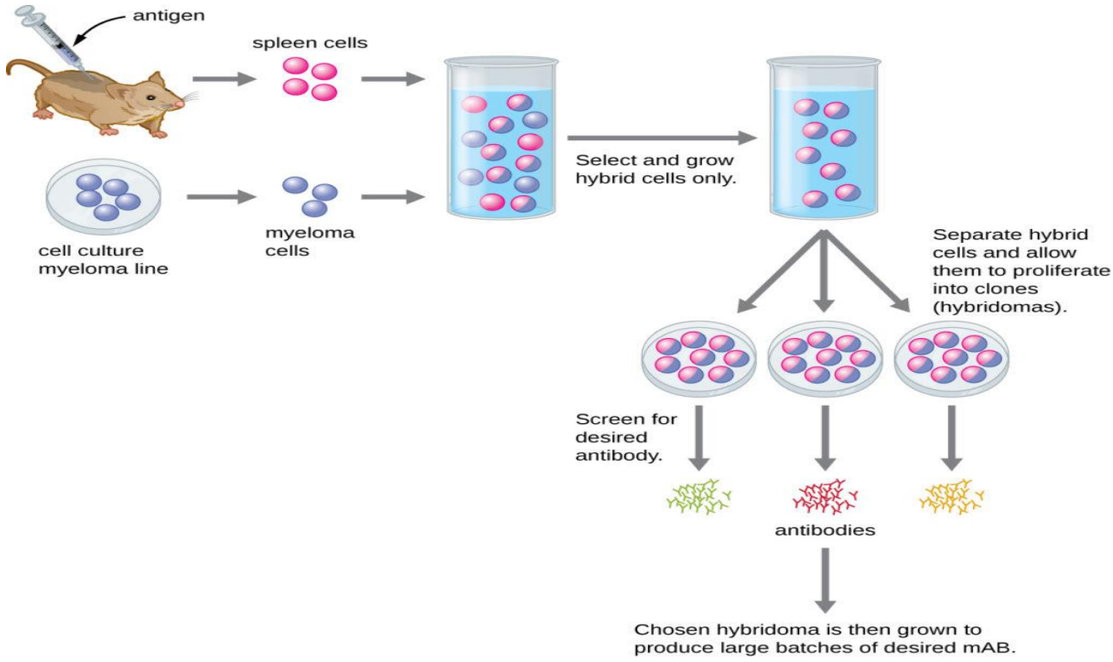


Figure 1: The process to generate the monoclonal antibody

Unlike polyclonal antibodies, which are produced in live animals, monoclonal antibodies are produced *in vitro* using tissue-culture techniques. mAbs are produced by immunizing an animal, often a mouse, multiple times with a specific antigen. B cells from the spleen of the immunized animal are then removed. Since normal B cells are unable to proliferate forever, they are fused with immortal, cancerous B cells called myeloma cells, to yield hybridoma cells. All of the cells are then placed in a selective medium that allows only the hybridomas to grow; unfused myeloma cells cannot grow, and any unfused B cells die off. The hybridomas, which are capable of growing continuously in culture while producing antibodies, are then screened for the desired mAb. Those producing the desired mAb are grown in tissue culture; the culture medium is harvested periodically and mAbs are purified from the medium. This is a very expensive and time-consuming process. It may take weeks of culturing and many liters of media to provide enough mAbs for an experiment or to treat a single patient. mAbs are expensive (Figure 2).

Figure 2: Monoclonal antibodies (mAbs) are produced by introducing an antigen to a mouse and then fusing polyclonal B cells from the mouse's spleen to myeloma cells. The resulting hybridoma cells are cultured and continue to produce antibodies to the antigen. Hybridomas producing the desired mAb are then grown in large numbers on a selective medium that is periodically harvested to obtain the desired mAbs.

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Module# 86 Freezing Cells and Cell Banking

Cell Lines:

Cell lines are cultures of animal cells that can be propagated repeatedly and sometimes indefinitely.They arise from primary cell cultures. Primary cultures are initiated directly from the cells, tissues, or organs of animals and are typically used in experiments within a few days.

Cell Banks:

Cell banks are widely used to preserve cell properties as well as to record and control the use of cell lines in biomedical research. A cell bank is a facility that stores cells of specific genome for the purpose of future use in a product or medicinal needs. They often contain expansive amounts of base cell material that can be utilized for various projects. Cell banks can be used to generate detailed characterizations of cell lines and can also help mitigate cross-contamination of a cell line. Utilizing cell banks also reduces the cost of cell culture processes, providing a cost-efficient alternative to keeping cells in culture constantly. Cell banks are commonly used within

fields including stem cell research and pharmaceuticals, with cryopreservation being the traditional method of keeping cellular material intact.

Cryopreservation

Cryopreservation is a process of using low temperatures to preserve cells and tissues for future use. This technique involves cooling cells to very low temperatures (-80˚C to -196˚C) and suspending their cellular metabolism, which preserves the cells for an indefinite amount of time. When water within cells freezes, the ice formation can cause a solute imbalance and damage the cellular structure. By using proper techniques and a freezing medium containing the right cryoprotectants and additives, researchers can maximize the post-thaw viability of cells for cell culture.

Overview of a General Cryopreservation Protocol

The specific protocol for freezing cells will depend on the cell type and cryopreservation media used. A typical protocol used for storing and freezing down cells involves the following steps (Figure below):

Storage

- Before putting the donated cell lines into storage, they are first proliferated and multiplied into a large number of identical cells before being stored in a number of cryovials.
- Along with the cells, cryoprotection agents are also added to the vials to protect the cells from rupturing from ice crystals during the freezing process.
- 10% DMSO solution is a common cryoprotection agent. The freezers contain nitrogen in either liquid or vapor form, and the cells are frozen at a rate of -1 to -3 degrees Celsius per minute until a temperature of -196 degrees Celsius is reached.
- At a temperature of -196 degrees Celsius, metabolic processes within the cells are significantly slowed to stop all cell growth, thus preserving the cell line, which is especially useful when the cell line has a limited number of cell divisions.
- Cells can be stored for an extended amount of time in this state, reducing the rate of degradation of cellular material.

Freezing

- The general freezing process for mammalian cells involves suspending a small density of the cells of interest in a solution of cryopreservation agents in a cryovial and freezing the cells to a temperature of -196 degrees Celsius. A slow freezing rate is important to maintaining the health of the cell culture.
- Freezing the cells at a rate of -1 to -3 degrees Celsius per minute is generally acceptable in maintaining cell culture health.
- The use of cryopreservation agents is also key to the freezing process. A common cryoprotection agent used is 10% solution of DMSO, which acts to protect the cells from the rupturing caused by ice crystals during freezing and during thawing.
- DMSO has been observed to be toxic to cells, and requires dilution after the cells are thawed.

Note: Short-term storage of cells (< 1 month) at -80°C is acceptable, but should be minimized to ensure maximum stability.

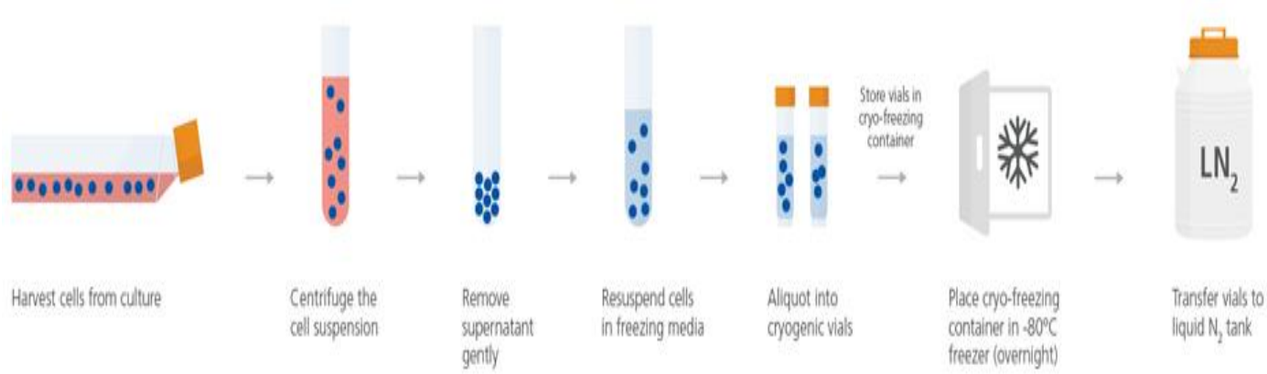


Figure: Typical Step-by-Step Cryopreservation Protocol

Different cells, depending on their biological make-up, may respond differently to a given cryopreservation protocol, which can affect post-thaw viability.

References

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Module# 87

Immunoassay

Immunoassays are bioanalytical methods in which the quantitation of the analyte depends on the reaction of an antigen (analyte) and an antibody. Agglutination is the process that occurs if an antigen is mixed with its corresponding antibody called isoagglutinin.

Enzyme-Linked Immunosorbent Assay

ELISA (which stands for enzyme-linked immunosorbent assay) is a technique to detect the presence of antigens in biological samples. An ELISA, like other types of immunoassays, relies on antibodies to detect a target antigen using highly specific antibody-antigen interactions.

Basic ELISA Principles

In an ELISA assay, the antigen is immobilized to a solid surface. This is done either directly or via the use of a capture antibody itself immobilized on the surface. The antigen is then complexed to a detection antibody conjugated with a molecule amenable for detection such as an enzyme or a fluorophore.

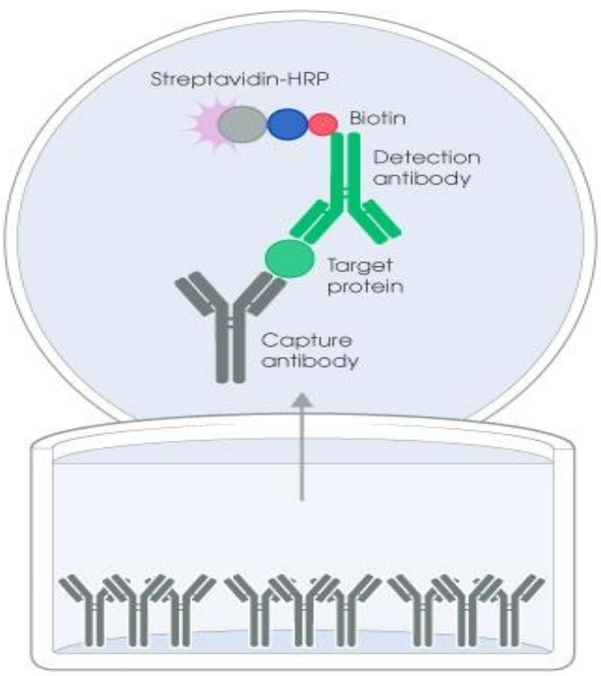


Figure 1: The basic setup of an ELISA assay**.** A capture antibody on a multi-well plate will immobilize the antigen of interest. This antigen will be recognized and bound by a detection antibody conjugated to biotin and streptavidin-HRP.

An ELISA assay is typically performed in a multi-well plate (96- or 384-wells), which provides the solid surface to immobilize the antigen. Immobilization of the analytes facilitates the separation of the antigen from the rest of the components in the sample. This characteristic makes ELISA one of the easiest assays to perform on multiple samples simultaneously.

Advantages

- High sensitivity and specificity: it is common for ELISAs to detect antigens at the picogram level in a very specific manner due to the use of antibodies.

- High throughput: commercial ELISA kits are normally available in a 96-well plate format. But the assay can be easily adapted to 384-well plates.
- Easy to perform: protocols are easy to follow and involve little hands-on time.
- Quantitative: it can determine the concentration of antigen in a sample.
- Possibility to test various sample types: serum, plasma, cellular and tissue extracts, urine, and saliva among others.

Disadvantages

- Temporary readouts: detection is based on enzyme/substrate reactions and therefore readout must be obtained in a short time span.
- Limited antigen information: information limited to the amount or presence of the antigen in the sample.

Types:

There are four main types of ELISA: direct ELISA, indirect ELISA, sandwich ELISA and competitive ELISA. Each has unique advantages, disadvantages and suitability.

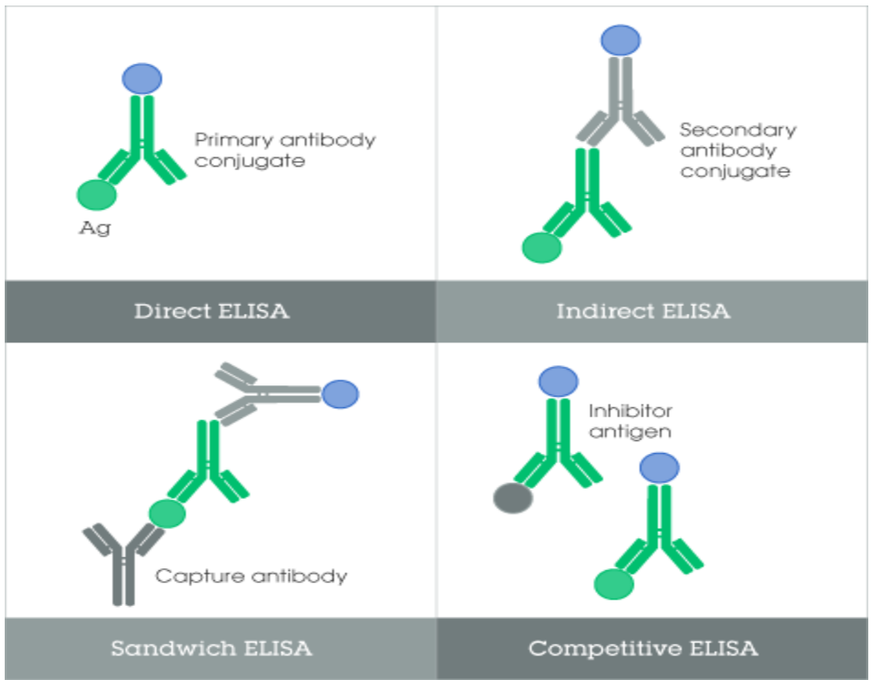


Figure 2: The different types of ELISA (direct, indirect, sandwich, and competitive)

References

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Module# 88

Triple Antibody Sandwich ELISA (Indirect ELISA)

Triple Antibody Sandwich ELISA is also known as Indirect ELISA. Figure below demonstrates how an indirect ELISA is set up; antigen is adsorbed to a well in an ELISA plate. Detection is a two-step process. First, an unlabeled primary antibody binds to the specific antigen. Second, an enzyme conjugated secondary antibody that is directed against the host species of the primary antibody is applied.

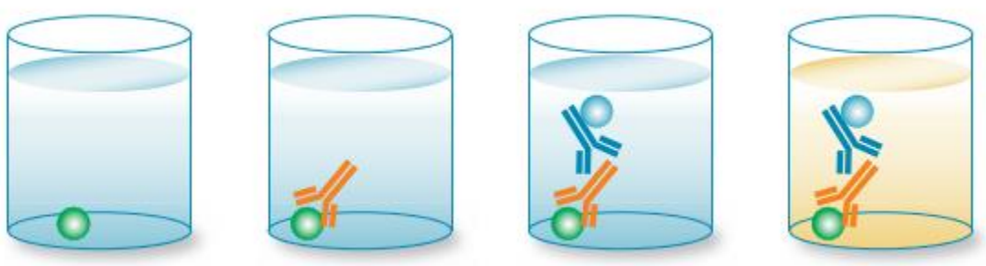


Figure: Overview of indirect ELISA.

The indirect ELISA method has high sensitivity since more than one labeled secondary antibody can bind the primary antibody; it is more economical than the direct ELISA as fewer labeled antibodies are needed. Indirect ELISA delivers greater flexibility since different primary antibodies can be used with a single labeled secondary antibody. Among its disadvantages is the possibility of cross-reactivity of secondary antibody to the adsorbed antigen, which could increase background noise. Also, indirect ELISA assays take longer to run than direct ELISAs since an additional incubation step for the secondary antibody is required. The indirect ELISA is most suitable for determining total antibody concentration in samples.

Advantages	Disadvantages
High sensitivity - more than one labeled secondary antibody can bind the primary antibody	
Economical - fewer labeled antibodies are needed	
Greater flexibility - different primary antibodies can be used with a single labeled secondary antibody	Possibility of background noise - secondary antibody may be cross-reactive Longer procedure than direct ELISA technique - additional incubation step for secondary antibody needed
Best for: determining total antibody concentration in samples.	

References

https://www.bio-rad-antibodies.com/elisa-types-direct-indirect-sandwich-competition-elisa-formats.html

Module# 89 Double Antibody Sandwich ELISA

Sandwich ELISAs require the use of matched antibody pairs (capture and detection antibodies) as shown in the Figure below. Each antibody is therefore specific for a different and non-overlapping region or epitope of the antigen. It is important that matched antibody pairs are tested specifically in sandwich ELISA to ensure that they detect different epitopes, to achieve accurate results. The capture antibody, as its name implies, binds the antigen that can then be detected in a direct ELISA or in an indirect ELISA configuration.

The procedure for a sandwich ELISA firstly requires the well of an ELISA plate to be coated with a capture antibody. The analyte or sample is then added, followed by a detection antibody. The detection antibody can be enzyme conjugated, in which case this is referred to as a direct sandwich ELISA. If the detection antibody used is unlabeled, a secondary enzyme-conjugated detection antibody is required. This is known as an indirect sandwich ELISA. The key advantage of a sandwich ELISA is its high sensitivity; it is 2-5 times more sensitive than direct or indirect ELISAs. Sandwich ELISA also delivers high specificity as two antibodies are used to detect the antigen. It offers flexibility since both direct and indirect methods can be used. The advantages bring with them a few disadvantages; if a standardized ELISA kit or tested antibody pair is not available, antibody optimization has to be worked out since it is important to reduce cross-reactivity between the capture and detection antibodies. Sandwich ELISAs are particularly suited to the analysis of complex samples, since the antigen does not need to be purified prior to the assay yet still delivers high sensitivity and specificity (e.g. measuring cytokine levels in an immune response).

When to Use: Determining analyte concentration in a biological sample.

Advantages:

- Highest specificity and sensitivity
- Compatible with complex sample matrices

Disadvantages:

- Longer protocol
- Challenging to develop

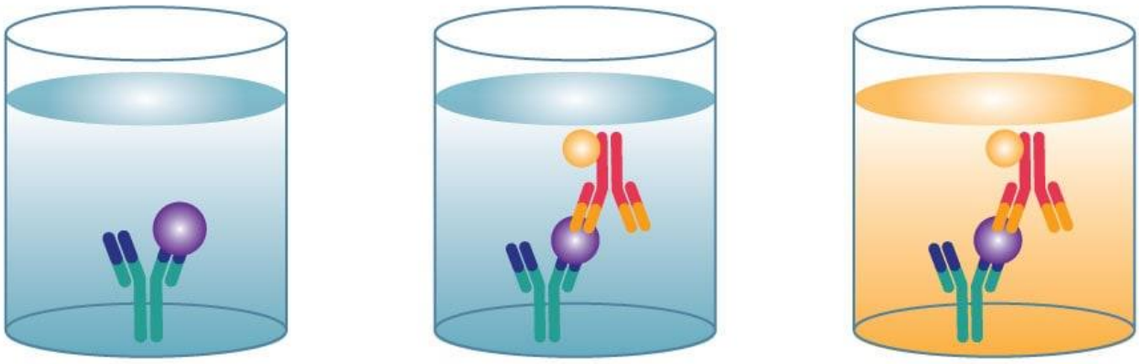


Figure: Sandwich ELISA

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Module# 90 Microscopy

A **microscope** is an instrument that magnifies objects otherwise too small to be seen, producing an image in which the object appears larger. Most photographs of cells are taken using a microscope, and these pictures can also be called **micrographs**. From the definition above, it might sound like a microscope is just a kind of magnifying glass. In fact, magnifying glasses do qualify as microscopes; since they have just one lens, they are called **simple microscopes**. The fancier instruments that we typically think of as microscopes are **compound microscopes**, meaning that they have multiple lenses. Because of the way these lenses are arranged, they can bend light to produce a much more magnified image than that of a magnifying glass. Two parameters are especially important in microscopy: magnification and resolution.

- **Magnification** is a measure of how much larger a microscope (or set of lenses within a microscope) causes an object to appear. For instance, the light microscopes typically used in high schools and colleges magnify up to about 400 times actual size. So, something that was 1 mm wide in real life would be 400 mm wide in the microscope image.
- The **resolution** of a microscope or lens is the smallest distance by which two points can be separated and still be distinguished as separate objects. The smaller this value, the higher the **resolving power** of the microscope and the better the clarity and detail of the image. If two bacterial cells were very close together on a slide, they might look like a single, blurry dot on a microscope with low resolving power, but could be told apart as separate on a microscope with high resolving power.

Parts of a Microscope

There are three structural parts of the microscope i.e. head, base, and arm.

1. **Head** – This is also known as the body. It carries the optical parts in the upper part of the microscope.
2. **Base** – It acts as microscopes support. It also carries microscopic illuminators.
3. **Arms** – This is the part connecting the base and to the head and the eyepiece tube to the base of the microscope. It gives support to the head of the microscope and it is also used when carrying the microscope

The optical parts of the microscope are used to view, magnify, and produce an image from a specimen placed on a slide. These parts include:

1. **Eyepiece** – also known as the ocular. This is the part used to look through the microscope. Its found at the top of the microscope. Its standard magnification is 10x with an optional eyepiece having magnifications from 5X to 30X.

2. **Eyepiece tube** – it's the eyepiece holder. It carries the eyepiece just above the objective lens.
3. **Objective lenses** – These are the major lenses used for specimen visualization. They have a magnification power of 40x-100X. There are about 1- 4 objective lenses placed on one microscope, in that some are rare facing and others face forward. Each lens has its own magnification power.
4. **Nose piece** – also known as the revolving turret. It holds the objective lenses. It is movable hence it can revolve the objective lenses depending on the magnification power of the lens.
5. **The Adjustment knobs** – These are knobs that are used to focus the microscope. There are two types of adjustment knobs i.e fine adjustment knobs and coarse adjustment knobs.
6. **Stage** – This is the section in which the specimen is placed for viewing. They have stage clips that hold the specimen slides in place. The most common stage is the mechanical stage, which allows the control of the slides by moving the slides using the mechanical knobs on the stage instead of moving them manually.
7. **Aperture** – This is a hole on the microscope stage, through which the transmitted light from the source reaches the stage.
8. **Microscopic illuminator** – This is the microscopes light source, located at the base. It is used instead of a mirror. It captures light from an external source of a low voltage of about 100V.
9. **Condenser** – These are lenses that are used to collect and focus light from the illuminator into the specimen. They are found under the stage next to the diaphragm of the microscope. They play a major role in ensuring clear sharp images are produced with a high magnification of 400X and above.
10. **Diaphragm** – it's also known as the iris. Its found under the stage of the microscope and its primary role is to control the amount of light that reaches the specimen.
11. **Condenser focus knob** – this is a knob that moves the condenser up or down thus controlling the focus of light on the specimen.

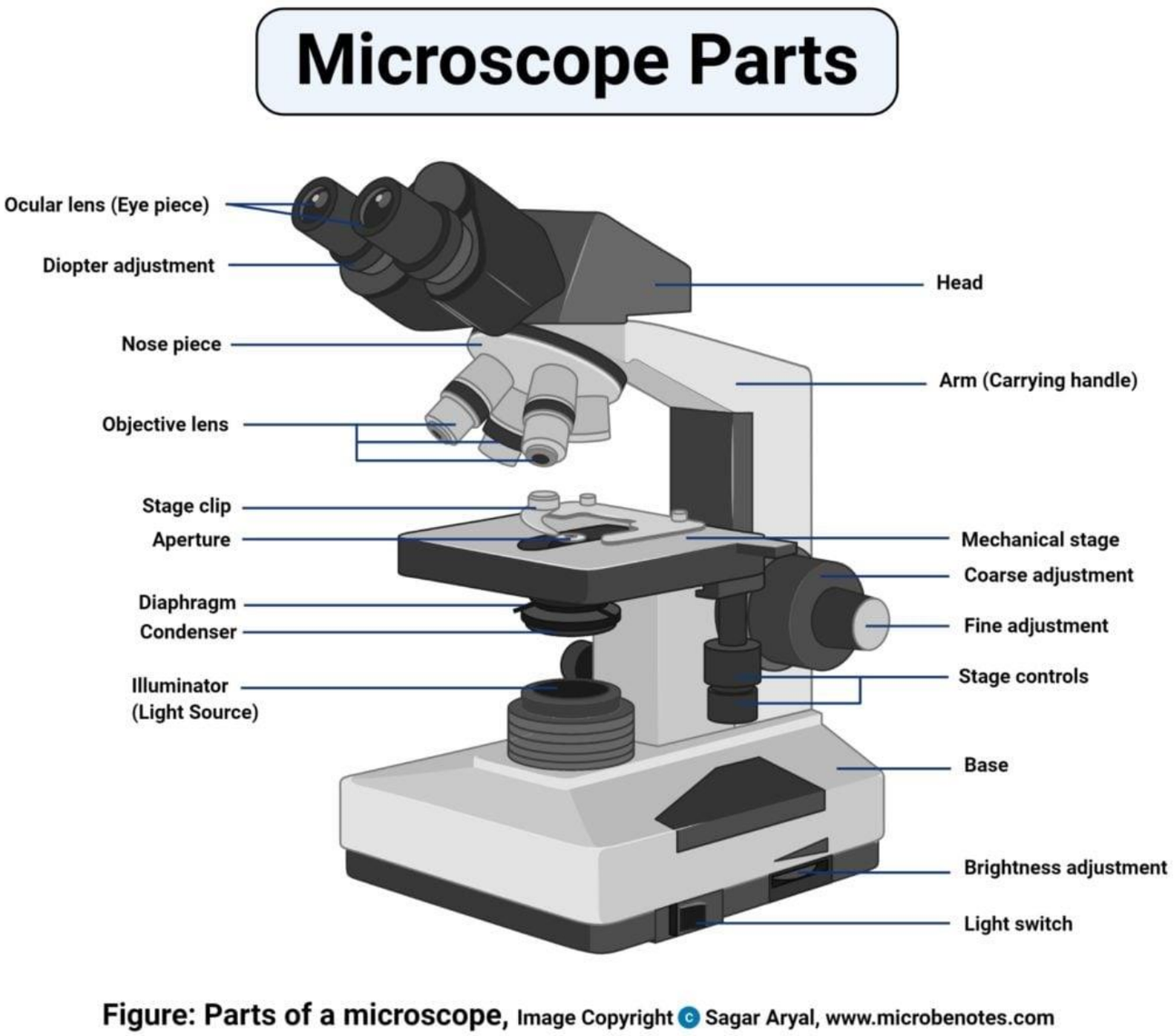


Figure 1: Parts of a microscope.

Principle:

Light from a light source (mirror or electric lamp) passes through a thin transparent object. The objective lens produces a magnified 'real image' (first image) of the object. This image is again magnified by the ocular lens (eyepiece) to obtain a magnified 'virtual image' (final image), which can be seen by eye through the eyepiece. As light passes directly from the source to the eye through the two lenses, the field of vision is brightly illuminated. That is why; it is a bright-field microscope.

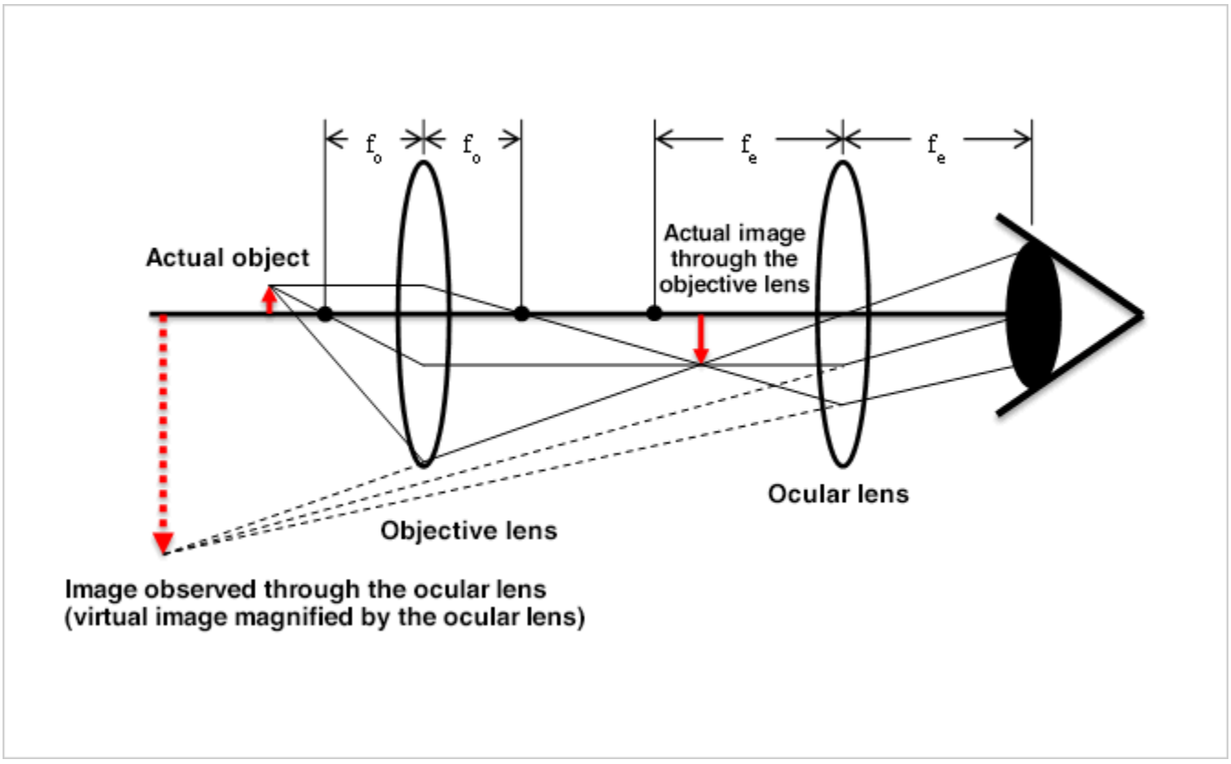


Figure 2: How an image is produced through microscope.

Types of Microscopes:

- Simple Microscope
- Compound Microscope
- Dark Field Microscope
- Phase Contrast Microscope
- Flourescent Microscope
- Interference Microscope
- Electron Microscope
- Atomic Force Microscope
- Polarization Microscope

Phase-Contrast Microscope.

This microscope also contains special condensers that throw light "out of phase" and cause it to pass through the object at different speeds. Live, unstained organisms are seen clearly with this microscope, and internal cell parts such as mitochondria, lysosomes, and the Golgi body can be seen with this instrument.

Fluorescent Microscope

The fluorescent microscope uses ultraviolet light as its light source. When ultraviolet light hits an object, it excites the electrons of the object, and they give off light in various shades of color. Since ultraviolet light is used, the resolution of the object increases. A laboratory technique called the fluorescent-antibody technique employs fluorescent dyes and antibodies to help identify unknown bacteria.

Electron Microscopy

The energy source used in the electron microscope is a beam of electrons. Since the beam has an exceptionally short wavelength, it strikes most objects in its path and increases the resolution of the microscope significantly. Viruses and some large molecules can be seen with this instrument. The electrons travel in a vacuum to avoid contact with deflecting air molecules, and magnets focus the beam on the object to be viewed. An image is created on a monitor and viewed by the technologist.

The more traditional form of electron microscope is the **transmission electron microscope** (TEM). To use this instrument, one places ultrathin slices of microorganisms or viruses on a wire grid and then stains them with gold or palladium before viewing. The densely coated parts of the specimen deflect the electron beam, and both dark and light areas show up on the image.

The **scanning electron microscope** (SEM) is the more contemporary form electron microscope. Although this microscope gives lower magnifications than the TEM, the SEM permits three-dimensional views of microorganisms and other objects. Whole objects are used, and gold or palladium staining is employed.

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Module# 91 Immuno-fluorescent (IF) Microscopy

Immunofluorescence (IF) microscopy is a widely used example of immunostaining and is a form of immunohistochemistry based on the use of fluorophores to visualize the location of bound antibodies. It is a particularly robust and broadly applicable method generally used by researchers to assess both the localization and endogenous expression levels of proteins of interest.

The effective application of this method comprises several considerations, including the nature of the antigen, specificity and sensitivity of the primary antibody, properties of the fluorescent label, permeabilization and fixation technique of the sample, and fluorescence imaging of the cell. Although each protocol will require fine-tuning depending on the cell type, the antibody, and the antigen, there are steps common to nearly all applications. Depending on the type of experiment there are two different Immunofluorescence (IF) variants: In **direct Immunofluorescence (IF)** the first antibody is directly linked to a fluorochrome which visualizes the target structure under the microscope. In **indirect Immunofluorescence (IF)** a fluorescence-coupled secondary antibody is applied during a second incubation step which specifically labels the first antibody. This leads to a greater flexibility in choosing antibodies and fluorochromes and furthermore to a signal amplification, because several secondary antibody molecules can bind to one primary antibody.

Fluorescence detection is possible with use of fluorophore-conjugated antibodies, which emit light upon excitation via light of a shorter wavelength. The most commonly used fluorophores are fluorescein isothiocyanate (FITC) and tetramethylrhodamine isothiocyanate (TRITC). Fluoroscein produces a green fluorescence and rhodamine produces a red fluorescence. Immunofluorescence can be used on tissue sections, cultured cells, or individual cells that are fixed by a variety of methods. Antibodies can be used in this method to analyze the distribution of proteins, glycoproteins, and other antigen targets, including small biological and non-biological molecules.

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Module# 92 Immunosorbent Electron Microscopy

Developments in microscopy, serology and immunology, molecular biology, and laboratory instrumentation have resulted in many new and sophisticated laboratory procedures for the identification of plant pathogens, particularly bacteria, viruses, and viroids. The techniques of traditional scanning microscopy and transmission electron microscopy have been applied to immunosorbent electron microscopy, in which the specimen is subject to an antigen-antibody reaction before observation and scanning tunneling microscopy, which provides information about the surface of a specimen by constructing a three-dimensional image.

Immuno-electron microscopy is used to localize molecules at the ultrastructural level by labeling them with specific antibodies. The antibodies are visualized by electron-opaque markers (colloidal gold particles) attached to them. The effect is to produce an electron-dense label at the site of the antigen-antibody reaction. When the antigen is located inside of the cell, then transmission electron microscopy is required to see it. The labeling can be done pre-embedding or post-embedding (on thin sections).

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Module# 93 ELECTROPHORESIS OF PROTEIN

SDS PAGE or Sodium Dodecyl Sulphate-Polyacrylamide Gel Electrophoresis is a technique used for the separation of proteins based on their molecular weight. It is a technique widely used in forensics, genetics, biotechnology and molecular biology to separate the protein molecules based on their electrophoretic mobility. Polyacrylamide gels are formed by the reaction of acrylamide and bis-acrylamide (*N*,*N*'-methylenebisacrylamide) that results in highly cross-linked gel matrix. The gel acts as a sieve through which the proteins move in response to the electric field. Proteins contain an overall positive or negative charge; this enables the movement of a protein molecule towards the isoelectric point at which the molecule has no net charge. By denaturing the proteins and giving them a uniform negative charge, it is possible to separate them based on the size as they migrate towards the positive electrode (as shown in the figure below). The use of polyacrylamide gels that have a gradient of increasing acrylamide concentration (and hence decreasing pore size) can sometimes have advantages over fixed-concentration acrylamide gels. During electrophoresis in gradient gels, proteins migrate until the decreasing pore size impedes further progress. Once the "pore limit" is reached, the protein banding pattern does not change appreciably with time, although migration does not cease completely.

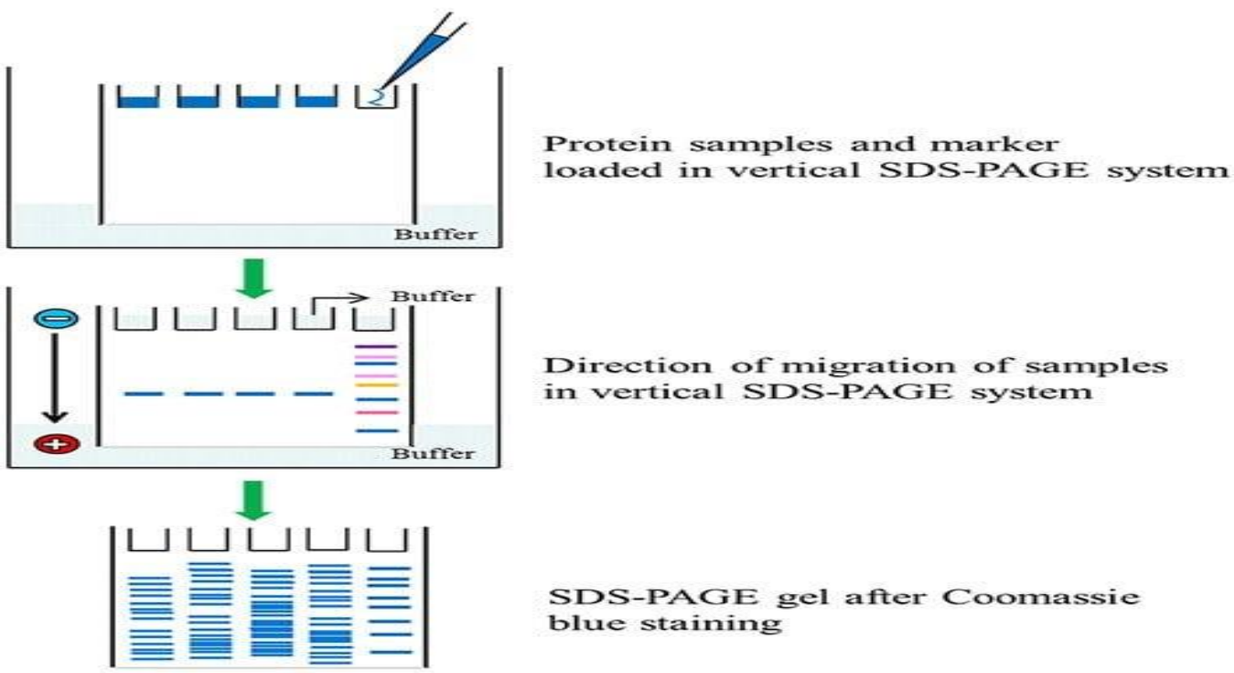


Figure: SDSPAGE of Protein Sample

Principle of SDS-PAGE

The principle of SDS-PAGE states that a charged molecule migrates to the electrode with the opposite sign when placed in an electric field. The separation of the charged molecules depends upon the relative mobility of charged species. The smaller molecules migrate faster due to less resistance during electrophoresis. The structure and the charge of the proteins also influence the rate of migration. Sodium dodecyl sulphate and polyacrylamide eliminate the influence of structure and charge of the proteins, and the proteins are separated based on the length of the polypeptide chain.

Applications of SDS-PAGE

The applications of SDS-PAGE are as follows:

1. It is used to measure the molecular weight of the molecules.
2. It is used to estimate the size of the protein.

3. Used in peptide mapping
4. It is used to compare the polypeptide composition of different structures.
5. It is used to estimate the purity of the proteins.
6. It is used in Western Blotting and protein ubiquitination.
7. It is used in HIV test to separate the HIV proteins.
8. Analyzing the size and number of polypeptide subunits.
9. To analyze post-translational modifications.

Isoelectric focusing (IEF)

Isoelectric focusing (IEF) is an electrophoretic technique for the separation of proteins based on their isoelectric point (pI). The pI is the pH at which a protein has no net charge and thus, does not migrate further in an electric field. In IEF, the separation is carried out in a slab of polyacrylamide or agarose gel that contains a mixture of ampholytes, which migrate in the gel and create a pH gradient when subjected to an electric field.

For IEF analysis, molecules to be focused are distributed over the medium with established pH gradient. An electric current passes through the medium and creates a "positive" anode and "negative" cathode end. Negatively charged molecules migrate through the pH gradient in the medium toward the "positive" end while positively charged molecules move toward the "negative" end. As a particle moves towards the pole opposite of its charge, it is immobilized in the pH gradient as they approach their specific pI. As a result, the proteins become focused into sharp stationary bands with each protein positioned at a point in the pH gradient corresponding to its pI. The IEF technique is capable of extremely high resolution, and even proteins only differing by a single charge could be fractionated into separate bands.

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Module# 94 Electrophoresis of Nucleic Acid

Agarose Gel electrophoresis is a laboratory method used to separate mixtures of DNA or RNA according to molecular size. In gel electrophoresis, the molecules to be separated are pushed by an electrical field through a gel that contains small pores. The molecules travel through the pores in the gel at a speed that is inversely related to their lengths. This means that a small DNA molecule will travel a greater distance through the gel than will a larger DNA molecule. Gel electrophoresis involves an electrical field; in particular, this field is applied such that one end of the gel has a positive charge and the other end has a negative charge. Because DNA and RNA are negatively charged molecules, they will be pulled toward the positively charged end of the gel. All DNA molecules have the same amount of charge per mass. Because of this, gel electrophoresis of DNA fragments separates them based on size only. As the name suggests, gel electrophoresis involves a gel: a slab of Jello-like material. Gels for DNA separation are often made out of a polysaccharide called **agarose**, which comes as dry, powdered flakes. When the agarose is heated in a buffer (water with some salts in it) and allowed to cool, it will form a solid, slightly squishy gel. At the molecular level, the gel is a matrix of agarose molecules that are held together by hydrogen bonds and form tiny pores.

Pulse Field Gel Electrophoresis

Conventional electrophoresis uses a single electrical field to cause biomolecules to migrate through a matrix according to its mass-to-charge ratio; the migration distance of the biomolecule is indicative of its mass or size. Conventional electrophoresis can effectively separate DNA fragments up to ~20 kb. However, larger fragments will co-migrate and appear as a large band at the top of the gel when imaged. In 1984, Schwartz and Cantor invented pulsed field gel electrophoresis (PFGE) to overcome this problem. PFGE resolves DNA by alternating the electrical field between spatially distinct pairs of electrodes. This technique results in the separation of DNA fragments of up to ~10 Mb by their reorientation and movement at different speeds through the pores of an agarose gel.

PFGE stemmed from the observation that DNA molecules elongate upon application of an electric field and return to an unelongated state upon removal of the electric field; this relaxation rate is dependent on the size of the DNA. When the orientation of the electric field is changed during electrophoresis, the DNA molecules must return to their

elongated form prior to reorientation, thus affecting the migration rate. This effect can be used to greatly extend the size range over which electrophoretic DNA separations are possible.

When the electrical field is applied to the gel, the DNA molecules elongate in the direction of the electrical field. The first electrical field is then switched to the second field according to the run specifications. The DNA must change conformation and reorient before it can migrate in the direction of this field. As long as the alternating fields are equal with respect to the voltage and pulse duration, the DNA will migrate in a straight path down the gel (see below).

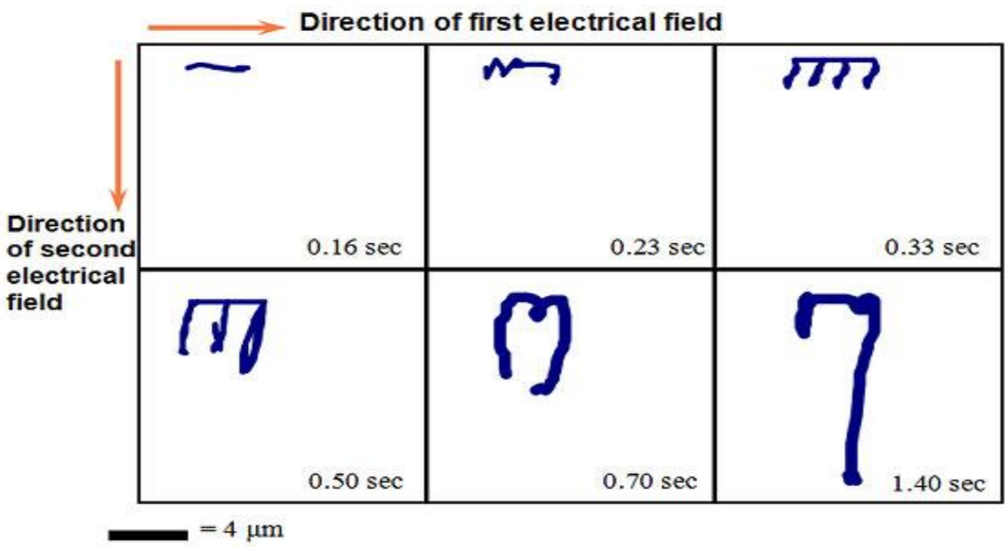


Figure: Time-lapse representation of DNA molecules undergoing PFGE**.**

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Module# 95 CHROMATOGRAPHY

Chromatography is a laboratory technique which is frequently employed for the separation of mixtures. In this technique, the mixture that must be separated is first dissolved into a fluid (commonly referred to as the mobile phase). This fluid carries the mixture through a different structure (commonly referred to as the stationary phase). The fluid carries different components of the mixture at different speeds. Therefore, the components of the mixture are separated in the stationary phase of the chromatography setup.

PURPOSE OF CHROMATOGRAPHY

Analytical: Determine Chemical composition of a sample

Preparative: Used to purify sufficient quantities of a substance

IMPORTANT TERMS OF CHROMATOGRAPHY

Chromatograph: Equipment that enables a sophisticated separation. Example Gas chromatography or Liquid chromatography

Eluent: Fluid entering column/ solvent that carries the analyte.

Eluate: Mobile phase leaving the column.

Retention time: Time takes for a particular analyte to pass through the system (from the column inlet to the detector) under set conditions.

Sample (Anylate): Substance analyzed in chromatography.

Solvent: Any substance capable of solubilizing another substance.

HOW IT WORKS

In all chromatography there is a mobile phase and a stationary phase. The stationary phase is the phase that doesn't move and the mobile phase is the phase that does move. The mobile phase moves through the stationary phase picking up the compounds to be tested. As the mobile phase continues to travel through the stationary phase it takes the compounds with it. At different points in the stationary phase the different components of the compound are going to be absorbed and are going to stop moving with the mobile phase. This is how the results of any chromatography are gotten, from the point at which the different components of the compound stop moving and separate from the other components. In paper and thin-layer chromatography the mobile phase is the solvent. The stationary phase in paper chromatography is the strip or piece of paper that is placed in the solvent. In thin-layer chromatography the stationary phase is the thin-layer cell. Both these kinds of chromatography use capillary action to move the solvent through the stationary phase.

Retention Factor Rf

The retention factor, Rf, is a quantitative indication of how far a particular compound travels in a particular solvent. The Rf value is a good indicator of whether an unknown compound and a known compound are similar, if not identical. If the Rf value for the unknown compound is close or the same as the Rf value for the known compound, then the two compounds are most likely similar or identical.

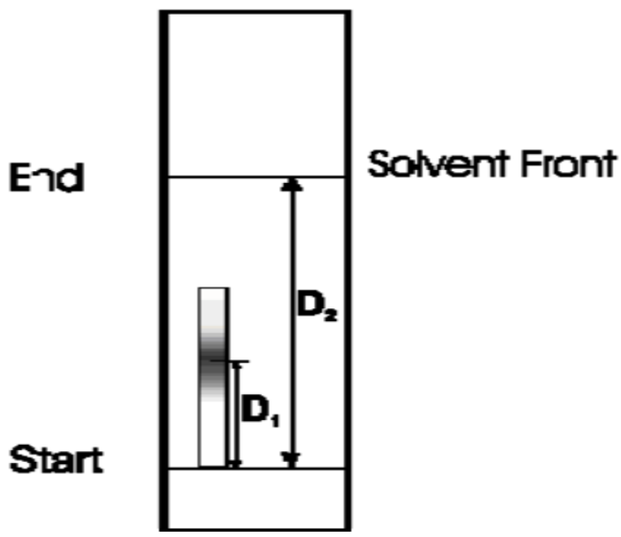


Figure 1: Calculation of Rf value.

The retention factor, Rf, is defined as Rf = distance the solute (D1) moves divided by the distance traveled by the solvent front (D2) Rf = D1 / D2 where D1 = distance that color traveled, measured from center of the band of color to the point where the food color was applied D2 = total distance that solvent traveled (Figure 1).

CLASSIFICATION OF CHROMATOGRAPHY BASED ON THE CHROMATOGRAPHIC BED SHAPE

- **Column Chromatography:** In this type of chromatography, the stationary phase of the setup is placed inside a tube. Then, the particles of the stationary phase (which is in the solid state) are made to fill the inside with the tube. An unrestricted, open path is then prepared for the mobile phase (somewhere along the middle of the tube).
- **Planar Chromatography:** In this type of chromatography, the stationary phase of the apparatus usually has a planar shape. Different subcategories of planar chromatography include paper chromatography (where the stationary phase is a special type of paper) and thin layer chromatography (usually abbreviated as TLC).

CLASSIFICATION OF CHROMATOGRAPHY BASED ON THE PHYSICAL STATE OF THE MOBILE PHASE

- **Gas Chromatography:** In this type of chromatography, the mobile phase is a substance that exists in the gaseous state. It can be noted that gas chromatography is also known as gas-liquid chromatography, and is often abbreviated to GLC. This type of chromatography almost always involves the use of a packed column.

- **Liquid Chromatography:** This type of chromatography involves the use of a mobile phase that exists in the liquid state. Liquid chromatography, often abbreviated to LC, can be carried out either on a plane or in a column. It can be noted that there exist many subcategories under liquid chromatography such as high-performance liquid chromatography and reversed phase liquid chromatography.

CLASSIFICATION OF CHROMATOGRAPHY BASED ON THE MECHANISM OF THE SEPARATION

- **Ion Exchange Chromatography:** This type of chromatography is also known as ion chromatography. Ion exchange chromatography involves the separation of the components of the mixture via an ion exchange mechanism. Differently charged components of the mixture are separated with the help of different ions in this separation technique.
- **Size Exclusion Chromatography:** This type of chromatography involves the separation of different components of the mixture based on their sizes. In size exclusion chromatography, components of the mixture are filtered based on their hydrodynamic volume or hydrodynamic diameters. It can be noted that size exclusion chromatography is also known as gel permeation chromatography or gel filtration chromatography.

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Additional Reading Material:

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Module# 96 Ion Exchange Chromatography

Ion exchange chromatography (or ion chromatography) is a process that allows the separation of ions and polar molecules based on their affinity to ion exchangers. The principle of separation is thus by reversible exchange of ions between the target ions present in the sample solution to the ions present on ion exchangers.

In this process, two types of exchangers i.e., cationic and anionic exchangers can be used.

1. Cationic exchangers possess negatively charged group, and these will attract positively charged cations. These exchangers are also called “Acidic ion exchange” materials, because their negative charges result from the ionization of acidic group.
2. Anionic exchangers have positively charged groups that will attract negatively charged anions. These are also called “Basic ion exchange” materials.

- Ion exchange chromatography is most often performed in the form of column chromatography. However, there are also thin-layer chromatographic methods that work basically based on the principle of ion exchange.

Working Principle of Ion Exchange Chromatography

This form of chromatography relies on the attraction between oppositely charged stationary phase, known as an ion exchanger, and analyte. The ion exchangers basically contain charged groups covalently linked to the surface of an insoluble matrix. The charged groups of the matrix can be positively or negatively charged. When suspended in an aqueous solution, the charged groups of the matrix will be surrounded by ions of the opposite charge. In this “ion cloud”, ions can be reversibly exchanged without changing the nature and the properties of the matrix.

Procedure of Ion Exchange Chromatography

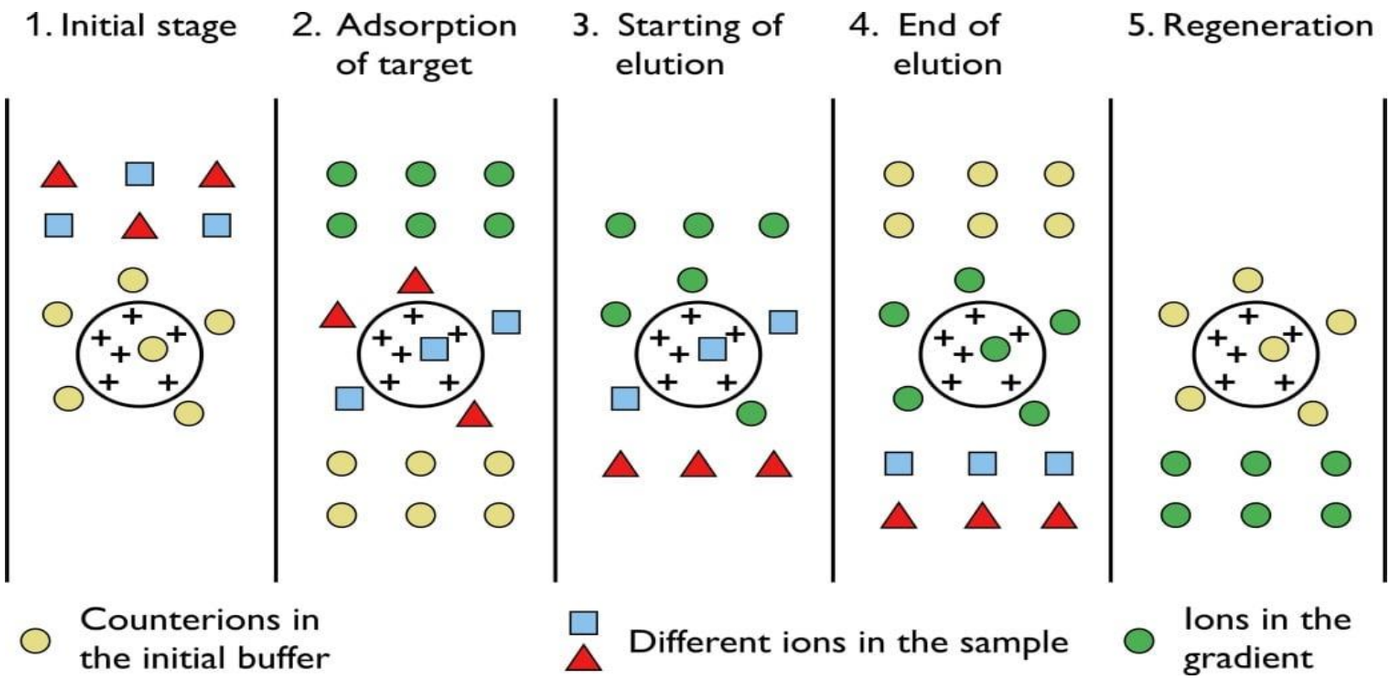


Figure: Procedure of ion exchange chromatography

- Ion exchange separations are carried out mainly in columns packed with an ion-exchanger.
- These ionic exchangers are commercially available. They are made up of styrene and divinyl benzene. Example. DEAE-cellulose is an anionic exchanger, CM-cellulose is a cationic exchanger.
- The choice of the exchanger depends upon the charge of particle to be separated. To separate anions “Anionic exchanger” is used, to separate cations “Cationic exchanger” is used.
- First the column is filled with ion exchanger then the sample is applied followed by the buffer. The tris-buffer, pyridine buffer, acetate buffer, citrate and phosphate buffers are widely used.
- The particles which have high affinity for ion exchanger will come down the column along with buffers.
- In next step using corresponding buffer separates the tightly bound particles.
- Then these particles are analyzed spectroscopically.

Advantages of ion exchange chromatography

1. It is one of the most efficient methods for the separation of charged particles.
2. It can be used for almost any kind of charged molecule including large proteins, small nucleotides and amino acids.
3. Ion exchange is used for both analytical and preparative purposes in the laboratory, the analytical uses being the more common.
4. Inorganic ions also can be separated by ion-exchange chromatography

Limitations of ion exchange chromatography

1. Only charged molecules can be separated.
2. Buffer Requirement

References

https://microbenotes.com/ion-exchange-chromatography/